RC 1 ROUND 1A

Find an integer solution for the equation by factoring the number on the right.

1 $x(x + 1)(x + 2) = 720$ **ANSWER: $x = 8$**

[$720 = 8(9)10 = x(x + 1)(x + 2)$, hence $x = 8$]

2 $(x - 1)x(x + 1) = 1320$ **ANSWER: $x = 11$**

[$1320 = 10(11)12 = (x - 1)x(x + 1)$, hence $x = 11$]

3 $x(x + 1)(x + 2) = 210$ **ANSWER: $x = 5$**

[$210 = 5(6)7 = x(x + 1)(x + 2)$, hence $x = 5$]

4 $x^2(x - 1) = 1210$ **ANSWER: $x = 11$**

[$1210 = 11^2(10) = x^2(x - 1)$, $x^2 = 121$, $x - 1 = 10$, hence $x = 11$]

RC 1 ROUND 1B

Find the values of constants a, b, and c given that

1 $(x^2 - 2x + 3)(ax - 2) = 3x^3 + bx^2 + cx - 6$ **ANSWER: $a = 3$, $b = -8$, $c = 13$**

[x^3: $a = 3$, x^2: $(-2 - 2a) = b$, $b = -2 -2(3) = -8$, x: $c = 4 + 3a = 4 + 3(3) = 13$]

2 $(ax^2 - 5x + 4)(2x - 3)= 10x^3 + bx^2 + cx - 12$ **ANSWER: $a = 5$, $b = -25$, $c = 23$**

[x^3: $2a = 10$, $a = 5$, x^2: $-3a - 10 = b$, $b = -3(5) - 10 = -25$, x: $c = (15 + 8) = 23$]

3 $(x^2 + 6x - 5)(ax + 3) = 4x^3 + bx^2 + cx - 15$ **ANSWER: $a = 4$, $b = 27$, $c = -2$**

[x^3: $a = 4$, x^2: $6a + 3 = b$, $b = 24 + 3 = 27$, x: $-5a + 18 = c = -20 + 18 = -2$]

4 $(x^2 - 5x + a)(x - 2) = (x^3 + bx^2 + cx + 4)$ **ANSWER: $a = -2$, $b = -7$, $c = 8$**

[x^0: $- 2a = 4$, $a = -2$, x: $10 + a = c$, $c = 10 - 2 = 8$, x^2: $b = -2 - 5 = -7$]

RC 1 ROUND 2: SPEED RACE

1 Express the sum to infinity of the series 2 - 0.2 + 0.02 - 0.002 + 0.0002 - . . as a rational number.
ANSWER: 20/11
[exponential series $a = 2$, $r = -0.1$, $S_\infty = a/(1 - r) = 2/(1 + 0.1) = 2/1.1 = 20/11$]

2 The function $y = ax^2 + bx + c$ has gradient $2x - 2$ and has a minimum value of 0. Find the values of a, b and c.
ANSWER: $a = 1, b = -2, c = 1$
[$dy/dx = 2ax + b = 2x - 2$, $2a = 2$, $a = 1$, $b = -2$, for minimum, $dy/dx = 2x - 2 = 0$, $x = 1$, hence $a + b + c = 0$, $1 - 2 + c = 0$, $c = 1$]

3 Solve the trigonometric equation $\cos^2 x = 3/4$ for the interval $0 < x < \pi$
ANSWER: $x = \pi/6, 5\pi/6$
[$\cos x = \pm\sqrt{3}/2$, $\cos x = \sqrt{3}/2$, $x = \pi/6$, $\cos x = -\sqrt{3}/2$, $x = \pi - \pi/6 = 5\pi/6$]

RC1 ROUND 4: RIDDLE

1 In reality I am a function.

2 My range is the set of real numbers.

3 I involve a queue of numbers.

4 Each number in the queue has a unique position.

5 My domain is the set of natural numbers.

6 You can say I am a function of position.
Who am I?

ANSWER: SEQUENCE

RC 1 ROUND 4A: TRUE OR FALSE
If 2 chords AB and CD in a circle intersect at E, then

1 $AE + BE = CE + DE$
ANSWER: FALSE

2 $AB \times BE = CD \times DE$
ANSWER: FALSE

3 $AE/BE = CE/DE$
ANSWER: FALSE

4 $AE \times BE = CE \times DE$
ANSWER: TRUE

RC 1 ROUND 4B: TRUE OR FALSE
If a quadrilateral has two pairs of congruent angles then it is a

1 Kite
ANSWER: FALSE

2 Rhombus
ANSWER: FALSE

3 Parallelogram
ANSWER: FALSE

4 Isosceles Trapezium
ANSWER: FALSE

RC 2 ROUND 1A

Find the quadratic polynomial $f(x) = ax^2 + bx + c$ given that

1 $f(2) = 7, f(1) = 4, f(0) = 5$ **ANSWER: $2x^2 - 3x + 5$**

[$4a + 2b + c = 7$, $a + b + c = 4$, $c = 5$, $4a + 2b = 5$, $a + b = -1$, $3a = 6$, $a = 2$, $b = -3$]

2 $f(1) = 1, f(-1) = -9, f(0) = -7$ **ANSWER: $f(x) = 3x^2 + 5x - 7$**

[$a + b + c = 1$, $a - b + c = -9$, $c = -7$, $2b = 1 + 9$, $b = 5$, $a + 5 - 7 = 1$, $a = 3$]

3 $f(1) = 6, f(-1) = 10, f(0) = 3$ **ANSWER: $f(x) = 5x^2 - 2x + 3$**

[$a + b + c = 6$, $a - b + c = 10$,$c = 3$, $2b = 6 - 10 = -4$, $b = -2$, $a = 6 - 3 + 2 = 5$]

4 $f(1) = 0, f(2) = 10, f(0) = 10$ **ANSWER: $f(x) = 10x^2 - 20x + 10$**

[$a + b + c = 0$, $4a + 2b + c =10$ $c = 10$, $a + b = -10$, $4a + 2b = 0$ $2a = 20$,
$a = 10$, $b = -20$]

RC 2 ROUND 1B

Solve for x and y from the given equations.

1 $\log_2(2x) = y + 4$, $y = 2\log_2 x$ **ANSWER: $x = 1/8$, $y = -6$**

[$y = \log_2(2x) - 4 = 2\log_2 x$, $\log_2(2x/16) = \log_2 x^2$, $x^2 = x/8$, $x = 1/8$,
$y = 2\log_2(1/8) = 2(-3) = -6$]

2 $2\log_2 y = \log_2 3 + \log_2 x$, $3^y = 9^x$ **ANSWER: $x = ¾$, $y = 3/2$**

[$y^2 = 3x$, $y = 2x$, $4x^2 = 3x$, $x = 3/4$, $y = 2(3/4) = 3/2$]

3 $y = \log_4 x$, $y = ½ (1 + \log_4 16x)$ **ANSWER: $x = 64$, $y = 3$**

[$2y = 2\log_4 x$, $2y = 1 + \log_4 16x$, $\log_4 x^2 = \log_4 64x$, $x^2 = 64x$, $x = 64$, $y = 3$]

4 $\log_2 x + \log_2 y = 3$, $\log_y x = 2$ **ANSWER: $x = 4$, $y = 2$**

[$\log_2 xy = 3$, $xy = 8$, $x = y^2$, $y^3 = 8$, $y = 2$, $x = 4$]

RC 2 ROUND 2: SPEED RACE

1 Solve 2cosx - 1 = 0, for the interval $0 < x < \pi$
ANSWER: $x = \pi/3$
[$\cos x = 1/2$, $x = \pi/3$]

2 Find the interval on which $f(x) = x^3 - 2x^2 - 7x + 3$ is decreasing.
ANSWER: $\{x: -1 < x < 7/3\}$
[$df/dx = 3x^2 - 4x - 7 < 0$, $3x^2 - 7x + 3x - 7 = x(3x - 7) + 1(3x - 7 = (3x - 7)(x + 1) < 0$, $-1 < x < 7/3$]

3 Find the inverse of the linear transformation $(x, y) \to (2x + 3y, 3x + 5y)$
ANSWER: $(x, y) \to (5x - 3y, -3x + 2y)$
[$A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$, $\det A = 10 - 9 = 1$, $A^{-1} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}$, $(x, y) \to (5x - 3y, -3x + 2y)$]

RC 2 ROUND 5: RIDDLE

1 I am a 2-digit number.

2 I am a prime number.

3 The sum of my digits is a digit cube.

4 When my digits are reversed, the number formed is still a prime.

5 The absolute difference of my digits is half a dozen.

6 I am between 10 and 20.
Who am I?
ANSWER: 17

RC 2 ROUND 4A: TRUE OR FALSE

1 Adding two polynomials of degrees 2 and 3 gives a polynomial of degree 5.
ANSWER: FALSE

2 Multiplying 2 polynomials of degrees 3 and 4 gives a polynomial of degree 12.
ANSWER: FALSE

3 When a cubic polynomial is divided by a quadratic, the remainder is a constant.
ANSWER: FALSE

4 Subtracting a polynomial of degree 2 from a polynomial of degree 3 gives a polynomial of degree 3.
ANSWER: TRUE

RC 2 ROUND 4B: TRUE OR FALSE

If the sum of the interior angles of a polygon is 360°, then it is a

1 square,
ANSWER: FALSE

2 trapezium,
ANSWER: FALSE

3 kite.
ANSWER: FALSE

4 quadrilateral.
ANSWER: TRUE

RC 3 ROUND 1A

Express a^2 in terms of b if the equation $x^2 - ax + b = 0$ has roots

1 $k, k + 2$ **ANSWER: $a^2 = 4b + 4$**

[$2k + 2 = a$, $k(k + 2) = b$, $k = (a - 2)/2$, $b = (a - 2)/2\ ((a - 2)/2 + 2)$, $4b = (a - 2)(a + 2)$, $a^2 = 4b + 4$]

2 $k - 1, k + 1$ **ANSWER: $a^2 = 4b + 4$**

[$2k = a$, $(k - 1)(k + 1) = b$, $k = a/2$, $k^2 - 1 = b$, $a^2/4 = b + 1$, $a^2 = 4b + 4$]

3 $k - 1, k + 2$ **ANSWER: $a^2 = 4b + 9$**

[$2k + 1 = a$, $(k - 1)(k + 2) = b$, $k = (a - 1)/2$, $(a - 3)/2.(a + 3)/2 = b$, $a^2 = 4b + 9$]

4 $k - 2, k + 2$ **ANSWER: $a^2 = 4b + 16$**

[$2k = a, k = a/2, b = (k - 2)(k + 2) = (a - 4)/2 \,.\, (a + 4)/2 = (a^2 - 16)/4$
$(a^2 - 16) = 4b, a^2 = 4b + 16$]

RC 3 ROUND 1B

Find the coordinates of the stationary point of the given curve.

1 $y = (1 + x^2)/(1 - x^2)$ ANSWER: (0, 1)

[$dy/dx = (2x(1 - x^2) - (-2x)(1 + x^2))/(1 - x^2)^2 = 4x/(1 - x^2)^2 = 0, x = 0, y = 1$]

2 $y = (x^3 - 1)/(x^3 + 1)$ ANSWER: (0, -1) [$dy/dx = (3x^2(x^3 + 1) - 3x^2(x^3 - 1))/(x^3 + 1)^2 = 6x^2/(x^3 + 1)^2 = 0, x = 0, y = -1$]

3 $y = (x^4 - 2)/(x^4 + 2)$ ANSWER: (0, -1)

[$dy/dx = (4x^3(x^4 + 2) - 4x^3(x^4 - 2))/(x^4 + 2)^2 = 16x^3/(x^4 + 2)^2 = 0, x = 0,$ $y = -1$]

4 $y = (x + 1)/(x - 1)$ ANSWER: NO STATIONARY POINT

[$dy/dx = 1(x - 1) - 1(x + 1))/(x - 1)^2 = -2/(x - 1)^2 \neq 0$, no stationary point]

RC3 ROUND 2: SPEED RACE

1 Solve the equation $(\cos x + 1)^{(\sin x - 1)} = 1$, for x in the interval $0 < x < \pi$.
ANSWER: $x = \pi/2$
[$\sin x - 1 = 0$, $\sin x = 1$, $x = \pi/2$, $\cos x + 1 = 1$, $\cos x = 0$, $x = \pi/2$]

2 Find the equation of a curve with gradient $(6x^2 - 2x + 5)$ and passing through the point A(1, -6)
ANSWER: $y = 2x^3 - x^2 + 5x - 12$
[$y = 2x^3 - x^2 + 5x + C$, $-6 = 2 - 1 + 5 + C$, $C = -12$, $y = 2x^3 - x^2 + 5x - 12$]

3 Find the inverse of the matrix $A = \begin{pmatrix} 3 & -2 \\ 5 & -3 \end{pmatrix}$

ANSWER: $A^{-1} = \begin{pmatrix} -3 & 2 \\ -5 & 3 \end{pmatrix}$ [det $A = -9 + 10 = 1$, $A^{-1} = \begin{pmatrix} -3 & 2 \\ -5 & 3 \end{pmatrix}$]

RC 3 ROUND 5: RIDDLE

1 I am a polygon.

2 My interior angles are congruent.

3 My sides are congruent.

4 I can be deformed.

5 My exterior angles measure 30°.

6 I come by the dozens.
Who am I?
ANSWER: REGULAR DUODECAGON

RC 3 ROUND 4A: TRUE OR FALSE

The point A is a solution of the inequality $5x + 3y > 15$,

1 A(4, - 1)

ANSWER: TRUE

2 A(2, 1)

ANSWER: FALSE

3 A(6, - 5)

ANSWER: FALSE

4 A(-3, 10)

ANSWER: FALSE

RC3 ROUND 4B: TRUE OR FALSE

1 $8x^3 - y^3 = (2x - y)(4x^2 + 2xy + y^2)$

ANSWER: TRUE

2 $x^3 + 27y^3 = (x + 3y)(x^2 + 3xy - 9y^2)$

ANSWER: FALSE

3 $x^3 + 27y^3 = (x + 3y)(x^2 - 3xy + 9y^2)$

ANSWER: TRUE

4 $8x^3 - 27y^3 = (2x - 3y)(4x^2 + 3xy + 9y^2)$

ANSWER: FALSE

RC 4 ROUND 1A

Find the domain of the function,

1 $y = \sqrt{(x^2 - 2x - 3)}$ ANSWER: $\{x: x \geq 3 \text{ or } x \leq -1\}$
$[(x^2 - 2x - 3) = (x - 3)(x + 1) \geq 0, x \geq 3 \text{ or } x \leq -1]$

2 $y = \sqrt{(4 + 3x - x^2)}$ ANSWER: $\{x: -1 \leq x \leq 4\}$
$[4 + 3x - x^2 \geq 0, x^2 - 3x - 4 \leq 0, (x - 4)(x + 1) \leq 0, -1 \leq x \leq 4]$

3 $y = \sqrt{(x^2 - 5x + 6)}$ ANSWER: $\{x: x \geq 3, \text{ or } x \leq 2\}$
$[x^2 - 5x + 6 = (x - 3)(x - 2) \geq 0, x \geq 3 \text{ or } x \leq 2]$

4 $y = \sqrt{(15 - 2x - x^2)}$ ANSWER: $\{x: -5 \leq x \leq 3\}$
$[15 - 2x - x^2 \geq 0, x^2 + 2x - 15 = (x + 5)(x - 3) \leq 0, -5 \leq x \leq 3]$

RC 4 ROUND 1B

Solve the trigonometric equation in the interval $0 \leq x \leq \pi/2$

1 $\sin^2 x + 2\sin x - 3 = 0$ ANSWER: $x = \pi/2$
$[(\sin x + 3)(\sin x - 1) = 0, \sin x = 1, x = \pi/2, \sin x = -3$ no solution$]$

2 $2\cos^2 x + \cos x - 3 = 0$ ANSWER: $x = 0$
$[(\cos x - 1)(2\cos x + 3) = 0, \cos x = 1, x = 0, \cos x = -3/2$ no solution$]$

3 $\tan^2 x + \tan x - 2 = 0$ ANSWER: $x = \pi/4$
$[(\tan x - 1)(\tan x + 2) = 0, \tan x = 1, x = \pi/4, \tan x = -2,$ no solution in interval$]$

4 $2\sin^2 x + 5\sin x - 6 = 0$ ANSWER: $x = \pi/6$
$[(2\sin x - 1)(\sin x + 3) = 0, \sin x = 1/2 , x = \pi/6, \sin x = -3$ no solution$]$

RC 4 ROUND 2: SPEED RACE

1 Find an integer solution of the equation $x^2(x + 1) = 392$, by factoring the right-hand side.
ANSWER: $x = 7$
[$392 = 7^2(8) = x^2(x + 1)$, $x^2 = 7^2$, $x + 1 = 8$, $x = 7$]

2 Express $5/(2 + 3\tan 30°)$ as a surd in a simplified form.
ANSWER: $10 - 5\sqrt{3}$
[$5/(2 + 3/\sqrt{3}) = 5/(2 + \sqrt{3}) = 5(2 - \sqrt{3}) = 10 - 5\sqrt{3}$]

3 Find the area of the finite region bounded by the curve $y = (4 - x^2)$ and the x-axis.
ANSWER: 32/3
[$4 - x^2 = 0$, $x = \pm 2$, A= $\int_{-2}^{2} (4 - x^2)dx = 4x - x^3/3\Big|_{-2}^{2} = (8 - 8/3) - (-8 + 8/3) = 32/3$]

RC 4 ROUND 5: RIDDLE

1 I am a 2-digit number.

2 I am an odd number.

3 The sum of my digits is 13.

4 My first digit is a square.

5 My second digit is also a square.

6 I am myself also a square.
Who am I?
ANSWER: 49

RC 4 ROUND 4A: TRUE OR FALSE

Consider the function $y = x^3 - 3x$. The point on the curve where

1 $x = 1$ is a point of inflexion.
ANSWER: FALSE

2 $x = 1$ is a stationary point.
ANSWER: TRUE

3 $x = 0$ is a stationary point.
ANSWER: FALSE

4 $x = -1$ is a stationary point.
ANSWER: TRUE

RC 4 ROUND 4A: TRUE OR FALSE

1 A square fits its outline in 4 positions.
ANSWER: FALSE

2 A pentagon fits its outline in 10 positions.
ANSWER: FALSE (only if it is regular)

3 A scalene quadrilateral fits its outline in 1 position.
ANSWER: TRUE

4 A kite fits its outline in 2 positions.
ANSWER: TRUE

RC 5 ROUND 1A

Find the constants a and b given that

1 $x^3 + ax^2 + bx - 4$ is divisible by $(x - 1)$ and $(x - 2)$ ANSWER: $a = -5, b = 8$
$[1 + a + b - 4 = 0, 8 + 4a + 2b - 4 = 0, 6 + 2a + 4 = 0, a = -5, b = 8]$

2 $3x^3 + ax^2 + bx - 2$ is divisible by $(x - 1)$ and $(x + 1)$ ANSWER: $a = 2, b = -3$
$[3 + a + b - 2 = 0, a + b = -1, -3 + a - b - 2 = 0, a - b = 5, 2a = 4, a = 2, b = -3]$

3 $2x^3 + ax^2 + bx + 6$ is divisible by $(x - 1)$ and $(x + 1)$ ANSWER: $a = -6, b = -2$
$[2 + a + b + 6 = 0, a + b = -8, -2 + a - b + 6 = 0, a - b = -4, 2a = -12, a = -6,$
$b = -2]$

4 $x^3 + ax^2 + bx + 8$ is divisible by $(x - 1)$ and $(x + 2)$ ANSWER: $a = -3, b = -6$
$[1 + a + b + 8 = 0, -8 + 4a - 2b + 8 = 0, b = 2a, 3a + 9 = 0, a = -3, b = -6]$

RC 5 ROUND 1B

Find the coordinates of the stationary points of the given curve.

1 $y = 3x^4 + 4x^3 - 12x^2$ ANSWER: $(0, 0), (1, -5), (-2, -32)$
$[dy/dx = 12x^3 + 12x^2 - 24x = 12x(x^2 + x - 2) = 12x(x + 2)(x - 1), x = 0, 1, -2$
$x = 0, y = 0, x = 1, y = 3 + 4 - 12 = -5, x = -2, y = 48 - 32 - 48 = - 32]$

2 $y = 3x^4 - 6x^2 + 10$ ANSWER: $(0, 10), (1, 7), (-1, 7)$
$[dy/dx = 12x^3 - 12x = 12x(x^2 - 1) = 12x(x - 1)(x + 1) = 0, x = 0, 1, -1$
$x = 0, y = 10, x = 1, y = 3 - 6 + 10 = 7, x = -1, y = 3 - 6 + 10 = 7]$

3 $y = 3x^5 - 5x^3 + 5$ ANSWER: $(0, 5), (1, 3), (-1, 7)$ $[dy/dx = 15(x^4 - x^2) =$
$15x^2(x - 1)(x + 1) = 0, x = 0, x = 1, x = -1, x = 0, y = 5, (0, 5), x = 1, y = 3 - 5$
$+ 5 = 3, (1, 3), x = -1, y = -3 + 5 + 5 = 7, (-1, 7)]$

4 $y = x^4 - 8x^2 + 8$ ANSWER: $(0, 8), (2, -8), (-2, - 8)$
$[dy/dx = 4x^3 - 16x = 0, 4x(x^2 - 4) = 0, x = 0, 2, -2, x = 0, y = (0, 8),$
$x = 2, y = 16 - 32 + 8 = -8, (2, -8), x = -2, y = 16 - 32 + 8 = -8, (-2, -8)]$

RC5 ROUND 2: SPEED RACE

1 Find an integer root of the equation $x^4 + x^3 = 108$

ANSWER: $x = 3$

[$x^3(x + 1) = 108 = 3^3 4$, $x^3 = 3^3$, $x + 1 = 4$, $x = 3$]

2 Find the value of n if $135_n = 59_{10}$

ANSWER: $n = 6$

[$n^2 + 3n + 5 = 59$, $n^2 + 3n – 54 = (n + 9)(n – 6) = 0$, $n = 6$]

3 Find the greatest value and the least value of $y = 4\cos x - 3$

ANSWER: greatest value = 1, least value = -7

[greatest = 4 – 3 = 1, least value = -4 – 3 = -7]

RC 5 ROUND 5: RIDDLE

1 Normally, I am a constant or a number.

2 I am quite common in a binomial expansion.

3 I am the numerical part of an algebraic expression.

4 I have something to do with friction or expansion.

5 I am something of restitution in a collision problem.

6 In a binomial expansion, I am the numerical part of any term.
Who am I?
ANSWER: COEFFICIENT

RC 5 ROUND 4A: TRUE OR FALSE
In an isosceles triangle,

1 the medians to the legs are congruent,
ANSWER: TRUE

2 the median from the vertex to the base bisects the vertex angle and the base,
ANSWER: TRUE

3 the in-center and the circum-center coincide,
ANSWER: FALSE

4 the in-center and the circum-center lie on the median through the vertex.
ANSWER: TRUE

RC 5 ROUND 4B: TRUE OR FALSE
In the operations

1 on sets, union is distributive over intersection,
ANSWER: TRUE

2 on sets, intersection is distributive over union,
ANSWER: TRUE

3 on real numbers, addition is distributive over multiplication,
ANSWER: FALSE

4 on real numbers, multiplication is distributive over division.
ANSWER: FALSE

RC 6 ROUND IA

Solve for x from the given logarithmic equation,

1 $\log_2(x + 1) + 2 = \log_2(3x + 5)$

ANSWER: $x = 1$ [$\log_2(4x + 4) = \log_2(3x + 5)$, $4x + 4 = 3x + 5$, $x = 1$]

2 $\log_3(x - 2) + 1 = \log_3(2x + 3)$

ANSWER: $x = 5$ [$\log_3(3x - 6) = \log_3(2x + 3)$, $3x - 6 = 2x + 3$, $x = 9$]

3 $\log_4(x - 2) + 1 = \log_4(2x + 4)$

ANSWER: $x = 6$ [$\log_4(4x - 8) = \log_4(2x + 4)$, $4x - 8 = 2x + 4$, $2x = 12$, $x = 6$]

4 $\log_5(x - 1) + 1 = \log_5(2x + 4)$

ANSWER: $x = 3$ [$\log_5(5x - 5) = \log_5(2x + 4)$, $5x - 5 = 2x + 4$, $3x = 9$, $x = 3$]

RC 6 ROUND IB

Find constants A, B and C if

1 $3x^2 + 18x + 16 = A(x + B)^2 + C$

ANSWER: $A = 3$, $B = 3$, $C = -11$

[$3(x^2 + 6x) + 16 = 3(x + 3)^2 + 16 - 27 = 3(x + 3)^2 - 11$, $A = 3$, $B = 3$, $C = -11$]

2 $2x^2 - 20x + 25 = A(x + B)^2 + C$

ANSWER: $A = 2$, $B = -5$, $C = -25$ [$2(x^2 - 10x) + 25 = 2(x - 5)^2 + 25 - 50 = 2(x - 5)^2 - 25$, $A = 2$, $B = -5$, $C = -25$]

3 $4x^2 - 16x + 10 = A(x + B)^2 + C$

ANSWER: $A = 4$, $B = -2$, $C = -6$

[$4(x^2 - 4x) + 10 = 4(x - 2)^2 + 10 - 16 = 4(x - 2)^2 - 6$, $A = 4$, $B = -2$, $C = -6$]

4 $5x^2 + 20x + 9 = A(x + B)^2 + C$

ANSWER: $A = 5$, $B = 2$, $C = -11$

[$5(x^2 + 4x) + 9 = 5(x + 2)^2 -20 + 9 = 5(x + 2)^2 - 11$, $A = 5$, $B = 2$, $C = -11$]

RC 6 ROUND 2: SPEED RACE

1 Find the gradient of the curve $x^2 - 2xy - 3y^2 = 10$.
ANSWER: $dy/dx = (x - y)/(x + 3y)$ [$2x - 2(y + xdy/dx) - 6ydy/dx = 0$, $(x - y) = (dy/dx)(x + 3y)$, $dy/dx = (x - y)/(x + 3y)$]

2 In a throw of 2 fair dice, find the probability of obtaining either a total score of 5, or 10. **ANSWER: 7/36**
[A = {(1, 4), (2, 3), (3, 2), (4, 1), (4, 6), (5, 5), (6, 4)}, n(A) = 7, P(A) = 7/36]

3 Given that sinA = 5/13 and sinB = 3/5 with A acute and B obtuse, find sin(A + B) **ANSWER: 16/65**
[cosA = 12/13, cosB = -4/5, sin(A + B) = sinAcosB + cosAsinB = (5/13)(-4/5) + (12/13)(3/5) = (-20 + 36)/65 = 16/65]

RC 6 ROUND 5: RIDDLE

1 I am a 4-digit palindromic number.

2 Forward or backward the value is the same.

3 My digits are an identity and a prime.

4 The sum of my digits is a cube.

5 My first digit is an identity.

6 My digits are binomial coefficients.
Who am I?

ANSWER: 1331

RC6 ROUND 4A: TRUE OR FALSE
If a quadrilateral fits its outline in 4 positions and has 2 axes of symmetry, then it is a

1 Square
ANSWER: FALSE

2 Rhombus
ANSWER: FALSE

3 Rectangle
ANSWER: FALSE

4 An Isosceles Trapezium
ANSWER: FALSE

RC 6 ROUND 4B: TRUE OR FALSE
In the x-y plane, the solution of the given compound inequalities is the null set.

1 $x > 1$ and $x < 2$
ANSWER: FALSE

2 $x < 1$ and $x > 2$
ANSWER: TRUE

3 $y > 2$ and $y < -1$
ANSWER: TRUE

4 $y > -2$ and $y < 5$
ANSWER: FALSE

RC 7 ROUND 1A

Find the quadratic equation with integer coefficients if one root is

1 $(2 - \sqrt{11})$ **ANSWER: $x^2 - 4x - 7 = 0$**

[Roots are $(2 \pm \sqrt{11})$, sum = 4, product = $(4 - 11) = -7$, equation is $x^2 - 4x - 7 = 0$]

2 $(-7 + \sqrt{33})$ **ANSWER: $x^2 + 7x + 4 = 0$** [Roots are $(-7 \pm \sqrt{33})$, sum = -14, product = $(49 - 33) = 16$ equation is $x^2 + 14x + 16 = 0$]

3 $(-5 - \sqrt{11})$ **ANSWER: $x^2 + 10x + 14 = 0$** [Roots are $(-5 \pm \sqrt{11})$, sum = -10, product = $(25 - 11) = 14$, equation is $x^2 + 10x + 14 = 0$]

4 $(3 + \sqrt{3})$ ANSWER: $x^2 - 6x + 6 = 0$

[roots are $(3 \pm \sqrt{3})$, sum = 6, product = $(9 - 3) = 6$, $x^2 - 6x + 6 = 0$]

RC 7 ROUND 1A

Determine the values of the constants a, b, c and d given that the curve $y = ax^3 + bx^2 + cx + d$ has gradient

1 $6x^2 - 4x + 3$ and passes through the point A(1, 4),

ANSWER: $a = 2, b = -2, c = 3, d = 1$ [$dy/dx = 3ax^2 + 2bx + c = 6x^2 - 4x + 3$, $a = 2, b = -2, c = 3, a + b + c + d = 2 - 2 + 3 + d = 4, d = 1$]

2 $9x^2 - 6x + 5$ and passes through the point B(1, -3)

ANSWER: $a = 3, b = -3, c = 5, d = -8$ [$dy/dx = 3ax^2 + 2bx + c = 9x^2 - 6x + 5$, $a = 3, b = -3, c = 5, a + b + c + d = 3 - 3 + 5 + d = -3, d = -8$]

3 $-3x^2 + 4x - 2$ and passes through the point C(1, 0).

ANSWER: $a = -1, b = 2, c = -2, d = 1$ [$dy/dx = 3ax^2 + 2bx + c = -3x^2 + 4x - 2$, $a = -1, b = 2, c = -2, a + b + c + d = -1 + 2 - 2 + d = 0, d = 1$]

4 $12x^2 + 8x - 7$ and passes through the point D(1, - 7)

ANSWER: $a = 4, b = 4, c = -7, d = -8$ [$dy/dx = 3ax^2 + 2bx + c = -3x^2 + 4x - 2$, $3a = 12, a = 4, 2b = 8, b = 4, c = -7, a + b + c + d = 4 + 4 - 7 + d = -7, d = -8$]

RC 7 ROUND 2: SPEED RACE

1 A binary operation $*$ is defined on the set Z of integers by
$a*b = a + 2b - 3ab$. Evaluate $3*2$
ANSWER: -11
[$3*2 = 3 + 2(2) - 3(3)2 = 7 - 18 = -11$]

2 Find the value of x and the value of y given $3^x 2^y = 108$
ANSWER: $x = 3, y = 2$
[$3^x 2^y = 108 = 27(4) = 3^3 2^2$, equate powers, $x = 3, y = 2$]

3 Find the sum to infinity of the series $2 - 4/3 + 8/9 - 16/27 + \ldots$
ANSWER: 6/5
[$a = 2, r = -2/3, S_\infty = a/(1 - r) = 2/(1 + 2/3) = 2/(5/3) = 6/5$]

RC 7 ROUND 5: RIDDLE

1 I am a mathematical statement.

2 I am made of other simple statements.

3 You can say I am a compound statement.

4 I am a combination of 2 statements.

5 I have two parts, a hypothesis and a conclusion.

6 I am a basis for proof in Mathematics.
Who am I?

ANSWER: IMPLICATION

RC 7 ROUND 4A: TRUE OR FALSE

For the given figure, the volume is base area × height

1 a prism
ANSWER: TRUE

2 a cuboid
ANSWER: TRUE

3 a pyramid
ANSWER: FALSE

4 frustrum of a cone
ANSWER: FALSE

RC 7 ROUND 4B: TRUE OR FALSE

1 If $\sin x = ½$, then $x = \pi/6$
ANSWER: FALSE

2 if $\sin x = ½$ and x is obtuse, then $x = 5\pi/6$
ANSWER: TRUE

3 If $\sin x = -1$, and x is reflex, then $x = 3\pi/2$
ANSWER: TRUE

4 If $\tan x = -\sqrt{3}$ and x is obtuse, then $x = 2\pi/3$
ANSWER: TRUE

RC 8 ROUND 1A

Find the equation of the line $ax + by = c$ after

1 a reflection in the line $y = x$, followed by a reflection in the x-axis.

ANSWER: $bx - ay = c$

[$(x, y)\to(y, x)$, $ay + bx = c$, $(x, y)\to(x, -y)$, hence $bx - ay = c$]

2 a reflection in the y-axis, followed by a reflection in the line $y = x$,

ANSWER: $bx - ay = c$

[$(x, y)\to(-x, y)$, $-ax + by = c$, $(x, y)\to(y, x)$, hence $-ay + bx = c$, or $bx - ay = c$]

3 a reflection in the x-axis, followed by a reflection in the origin,

ANSWER: $ax - by = -c$

[$(x, y)\to(x, -y)$, $ax - by = c$, $(x, y)\to(-x, -y)$, hence $-ax + by = c$, or $ax - by = -c$]

4 a reflection in the origin, followed by a reflection in the line $y = x$,

ANSWER: $bx + ay =- c$

[$(x, y)\to(-x, -y)$, $-ax + -by = c$, $ax + by = -c$, $(x, y)\to(y, x)$, $ay + bx = -c$]

RC 8 ROUND 1B

Find the value of x given that the first 3 terms of an exponential series are

1 $(x - 4)$, $(x + 2)$ and $(3x + 1)$ ANSWER: $x = 8, ½$ [$(x + 2)^2 = (3x + 1)(x - 4)$ $= 3x^2 - 11x - 4 = x^2 + 4x + 4$, $2x^2 -15x - 8 = (2x + 1)(x - 8) = 0$, $x = 8, -1/2$]

2 $(x + 2)$, $(x + 4)$ and $(2x + 5)$ ANSWER: $x = 2, - 3$ [$(x + 4)^2 = (x + 2)(2x + 5)$, $x^2 + 8x + 16 = 2x^2 + 9x + 10$, $x^2 + x - 6 = (x - 2)(x + 3) = 0$, $x = 2, -3$]

3 x, $(x + 1)$, and $4x$ ANSWER: $x = 1, -1/3$

[$(x + 1)^2 = 4x^2$, $3x^2 - 2x - 1 = (3x + 1)(x - 1) = 0$, $x = 1, -1/3$]

4 x, $(x - 4)$, and $2x - 8$ ANSWER: $x = \pm 4$

[$(x - 4)^2 = x(2x - 8)$, $x^2 - 8x + 16 = 2x^2 - 8x$, $x^2 = 16$, $x = \pm 4$]

RC 8 ROUND 2: SPEED RACE

1 Simplify $(1 + \tan 60°)/(\tan 60° - 1)$
ANSWER: $2 + \sqrt{3}$
$[(1 + \sqrt{3})/(\sqrt{3} - 1) = (1 + \sqrt{3})^2/2 = (1 + 3 + 2\sqrt{3})/2 = 2 + \sqrt{3}]$

2 Water from a fountain rises to a maximum height of 20 m. Find the speed of the water as it leaves the nozzle. (Take $g = 10$ m/s^2)
ANSWER: 20 m/s
[$u^2 = 2gh = 2(10)20 = 400$, $u = 20$ m/s]

3 Find the equation of a curve with gradient $6x^2 - 4x + 5$ and passing through the point A(0, - 9)
ANSWER: $2x^3 - 2x^2 + 5x - 9$
[$dy/dx = 6x^2 - 4x + 5$, $y = 2x^3 - 2x^2 + 5x + C$, $C = -9$, $y = 2x^3 - 2x^2 + 5x - 9$]

RC 8 ROUND 5: RIDDLE

1 I am an equation.

2 I am a quadratic equation.

3 My coefficients are all integers.

4 The sum of my roots is a dozen.

5 The product of my roots is the least 2-digit cube.
Who am I?

ANSWER: $x^2 - 12x + 27 = 0$

RC 8 ROUND 4A: TRUE OR FALSE

If a, b and c are the lengths of the sides of triangle ABC, then

1 $a + b < c$
ANSWER: FALSE

2 $b + c > a$
ANSWER: TRUE

3 $a + c > b$
ANSWER: TRUE

4 $a + b < a + c$
ANSWER: FALSE

RC 8 ROUND 4B: TRUE OR FALSE

If $x^2 - 5x - 6 < 0$, then

1 $-2 < x < 3$
ANSWER: FALSE

2 $-6 < x < 1$
ANSWER: FALSE

3 $-6 < x < -1$
ANSWER: FALSE

4 $1 < x < 6$
ANSWER: FALSE (NB: correct answer is $-1 < x < 6$)

RC 9 ROUND 1A

Differentiate the rational function,

1 $y = (x + 1)^2/(x - 1)^2$ **ANSWER: $dy/dx = -4(x + 1)/(x - 1)^3$**

[$dy/dx = 2(x + 1)/(x - 1)\ ((x - 1) - (x + 1))/(x - 1)^2) = -4(x + 1)/(x - 1)^3$]

2 $y = (x - 3)^2/(2x + 1)^2$ **ANSWER: $dy/dx = 6(x - 3)/(2x + 1)^3$**

[$dy/dx =2(x - 3)/(2x + 1)\ (((2x + 1) - 2(x - 1)/(2x + 1)^2) =$
$6(x - 3)/(2x + 1)^3$]

3 $y = (2x - 1)^2/(2x + 1)^2$ **ANSWER: $dy/dx = 8(2x - 1)/(2x + 1)^3$**

[$dy/dx = 2(2x - 1)/(2x + 1)\ (2(2x + 1) - 2(2x - 1))/(2x + 1)^2 =$
$8(2x - 1)/(2x + 1)^3$]

4 $y = (x - 3)^2/(x - 2)^2$ ANSWER: $dy/dx = 2(x - 3)/(x - 2)^3$

[$dy/dx = 2(x - 3)/(x - 2)((x - 2) - (x - 3))/(x - 2)^2 = 2(x - 3)/(x - 2)^3$]

RC 9 ROUND 1B

Find the coordinates of the point of intersection of the pair of lines L_1 and L_2

1 L_1 through (-1, 2) with slope 1, and L_2 through (2, 1) with slope -1.
ANSWER: (0, 3) [L_1: $y - 2 = (x + 1)$, $y = x + 3$, L_2: $y - 1 = -(x - 2)$,
$y = -x + 3$, $x + 3 = -x + 3$, $x = 0$, $y = 3$, (0, 3)]

2 L_1 through the origin with slope 1, and L_2 through (1, 1) with slope -2
ANSWER: (1, 1)
[L_1: $y = x$, L_2: $y - 1 = -2(x - 1)$, $y = -2x + 3$, $x = -2x + 3$, $x = 1$, $y =1$, (1, 1)]

3 L_1 through (-1, 4) with slope -1, and L_2 through (2, -2) with slope 2
ANSWER: (3, 0) [L_1: $y - 4 = -(x + 1)$, $y = -x + 3$, L_2: $y + 2 = 2(x - 2)$,
$y = 2x - 6$, $2x - 6 = -x + 3$, $3x = 9$, $x = 3$, $y = 0$, (3, 0)]

4 L_1 through (2, 1) and parallel to $y = x$ and L_2 through (1, -2) and slope 2
ANSWER: (3, 2) [L_1: $y - 1 = x - 2$, $y = x - 1$, L_2: $y + 2 = 2(x - 1)$, $y = 2x - 4$,
$x - 1 = 2x - 4$, $x = 3$, $y = 2$, (3, 2)]

RC 9 ROUND 2: SPEED RACE

1 A particle moving in a straight line with uniform acceleration covers a distance of100 m while speed changes from 5 m/s to 15 m/s. Find the acceleration.

ANSWER: **1m/s^2** [(a = (v^2 – u^2)/2s = (225 – 25)/2(100) = 1 m/s^2]

2 Evaluate the limit $\lim\limits_{x \to -3} \frac{x^2 + 2x - 3}{(x+3)}$

ANSWER: -4

[(x^2 + 2x – 3)/(x + 3) =(x + 3)(x – 1)/(x + 3) = x – 1, Limit = -3 – 1 = -4]

3 2 fair dice are thrown, and the total score is 8. Find the probability both scores are odd numbers.

ANSWER: 2/5

[S = {(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)} A = {(3, 5), (5, 3)}, P(A) = 2/5]

RC 9 ROUND 5: RIDDLE

1 I am a 3-digit number.

2 I am an odd number.

3 I am an exact square.

4 I am between 600 and 700.

5 I am the square of a square.

6 The sum of my digits is 13.
Who am I?

ANSWER: 625-0

RC 9 ROUND 4A: TRUE OR FALSE

If the sum of 2 integers is odd, then

1 at least one of the numbers is odd,
ANSWER: FALSE

2 at most one of the numbers is odd,
ANSWER: FALSE

3 both numbers are odd,
ANSWER: FALSE

4 exactly one number is odd.
ANSWER: TRUE

RC 9 ROUND 4B: TRUE OR FALSE

1 Every triangle has at least 2 acute angles.
ANSWER: TRUE

2 If a triangle has an obtuse angle, then it has exactly 2 acute angles.
ANSWER: TRUE

3 If a triangle has 2 congruent angles, then the third angle is obtuse.
ANSWER: FALSE

4 If a triangle is scalene, then all 3 angles are acute.
ANSWER: FALSE

RC 10 ROUND 1A

Find an integer solution of the equation

1 $x^5 + x^3 = 270$ ANSWER: $x = 3$

$[x^3(x^2 + 1) = 270 = 3^3(10), x^3 = 3^3, x^2 + 1 = 10, x = 3]$

2 $x^5 – x^3 = 960$ ANSWER: $x = 4$

$[x^3(x^2 – 1) = 960 = 4^3(15), x^3 = 4^3, x^2 – 1 = 15, x = 4]$

3 $x^4 - x^3 = 1080$ ANSWER: $x = 6$

$[x^3(x - 1) = 1080 = 3^3 2^3(5) = 6^3(5), x^3 = 6^3, x – 1 = 5, x = 6]$

4 $x^4 – x^3 = 500$ ANSWER: $x = 5$

$[x^3(x – 1) = 5^3 4, x^3 = 5^3, x – 1 = 4, x = 5]$

RC 10 ROUND 1B

Solve the logarithmic equation

1 $\log_4 x – 5\log_x 4 = 4$ ANSWER: $x = 4^5$, $x = 4^{-1} = ¼$

$[\log_4 x – 5/\log_4 x - 4 = 0, (\log_4 x)^2 – 4\log_4 x – 5 = (\log_4 x – 5)(\log_4 x + 1) = 0$

$\log_4 x = 5, x = 4^5, \log_4 x= -1, x = 4^{-1}= ¼]$

2 $\log_2 x – 3\log_x 2 = 2$ ANSWER: $x = 8, x = ½$

$[\log_2 x – 3/\log_2 x – 2= 0, (\log_2 x)^2 - 2\log_2 x - 3 = (\log_2 x – 3)(\log_2 x + 1) = 0$

$\log_2 x = 3, x = 2^3 = 8, \log_2 x = -1, x = 2^{-1} = ½]$

3 $\log_3 x + 2\log_x 3 = 3$ ANSWER: $x = 9, x = 3$

$[\log_3 x +2/\log_3 x – 3 = 0, (\log_3 x)^2 – 3\log_3 x + 2 = (\log_3 x – 2)(\log_3 x – 1) = 0$

$\log_3 x = 2, x = 3^2 = 9, \log_3 x = 1, x = 3]$

4 $\log_5 x + 4\log_x 5 = 5$ ANSWER: $x = 5, x = 5^4 = 625$

$[\log_5 x + 4/\log_5 x – 5 = 0, (\log_5 x)^2 – 5\log_5 x + 4 = (\log_5 x – 1)(\log_5 x – 4) = 0$

$\log_5 x = 1, x = 5, \log_5 x = 4, x = 5^4 = 625]$

RC 10 ROUND 2: SPEED RACE

1 The first term of an exponential series is 6 and the sum to infinity is 10. Find the common ratio r.
ANSWER: $r = 2/5$ [$S_\infty = a/(1 - r)$, $10 = 6/(1 - r)$, $10(1 - r) = 6$, $10r = 10 - 6 = 4$, $r = 4/10 = 2/5$]

2 Two fair dice are thrown. Find the probability of scoring a double.
ANSWER: 1/6
[$A = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$, $P(A) = 6/36 = 1/6$]

3 Find the gradient of the implicit curve $x^3 - 2x^2y + y^4 = 25$
ANSWER: $dy/dx = (3x^2 - 4xy)/(2x^2 - 4y^3)$
[$3x^2 - 2(2xy + x^2dy/dx) + 4y^3dy/dx = 0$, $3x^2 - 4xy = (dy/dx)(2x^2 - 4y^3)$
$(dy/dx) = (3x^2 - 4xy)/(2x^2 - 4y^3)$]

RC 10 ROUND 5: RIDDLE

1 I am a solid geometric figure.

2 I enclose a single region of space.

3 My boundaries are faces.

4 These faces are defined by polygons.

5 In my plural form, I sound like something ending in hydra.

6 An example of me is a cuboid.
Who am I?
ANSWER: POLYHEDRON

RC 10 ROUND 4A: TRUE OR FALSE

1 $(x - 2)$ is a factor of $x^3 - 4x + 2$
ANSWER: FALSE

2 $(x + 1)$ is a factor of $2x^3 - 5x - 3$
ANSWER: TRUE

3 $(x - 3)$ is a factor of $x^3 - 2x^2 - 9$
ANSWER: TRUE

4 $(x + 3)$ is a factor of $x^3 + 2x^2 + 9$.
ANSWER: TRUE

RC 10 ROUND 4B: TRUE OR FALSE

1 Any 2 matrices can be multiplied.
ANSWER: FALSE

2 Any 2 matrices can be added.
ANSWER: FALSE

3 Any 2 matrices with same number of rows and same number of columns can be added
ANSWER: TRUE

4 Any 2 matrices with same number of rows and same number of columns can be multiplied.
ANSWER: FALSE

RC 11 ROUND 1A

Find the solution set of the cubic inequality.

1 $(x - 2)(x + 1)(x + 3) > 0$,
ANSWER: $\{x: x > 2$, or $-3 < x < -1\}$

2 $(x - 3)(x - 2)(x + 2) < 0$
ANSWER: $\{x: x < -2$, or $2 < x < 3\}$

3 $(x + 5)(x - 1)(x - 4) > 0$
ANSWER: $\{x: x > 4$, or $-5 < x < 1\}$

4 $(x + 3)(x - 1)(x - 6) < 0$
ANSWER: $\{x: x < -3$, or $1 < x < 6\}$ NB: OR NOT AND

RC 11 ROUND 1B

A bag contains 3 black balls, 5 white balls and 2 red balls. Two balls are drawn randomly without replacement. Find the probability

1 1 ball is black and 1 white ANSWER: 1/3
[$p(BW, WB) = (3/10)(5/9) + (5/10)(3/9) = 30/90 = 1/3$]

2 1 ball is red and 1 ball is black ANSWER: 2/15
[$P(BR, RB) = (3/10)(2/9) + (2/10)(3/9) = 12/90 = 2/15$]

3 1 ball is white and 1 ball is red ANSWER: 2/9
[$P(WR, RW) = (5/10)(2/9) + (2/10)(5/9) = 20/90 = 2/9$]

4 both balls are black, or both balls are red ANSWER: 4/45
[$P(BB, RR) = (3/10)(2/9) + (2/10)(1/9) = (6 + 2)/90 = 4/45$]

RC 11 ROUND 2: SPEED RACE

1 Find the next three terms of a sequence defined by the recurrence relation,
$U_n = 2U_{n-1} + 5, U_1 = -3$
ANSWER: $U_2 = -1, U_3 = 3, U_4 = 11$ [$U_2 = 2U_1 + 5 = 2(-3) + 5 = -1$,
$U_3 = 2U_2 + 5 = 2(-1) + 5 = 3, U_4 = 2U_3 + 5 = 2(3) + 5 = 11$]

2 Evaluate $(\tan 56° – \tan 26°)/(1 + \tan 56° \tan 26°)$. Simplify the answer.
ANSWER: **$\sqrt{3}/3$**
[$\tan(56 – 26) = \tan 30 = 1/\sqrt{3} = \sqrt{3}/3$]

3 In triangle ABC, the side $a = 8, b = 6, c = 12$. Find cosA.
ANSWER: **$\cos A = 29/36$**
[$\cos A = (b^2 + c^2 – a^2)/2bc = (36 + 144 – 64)/2(6)12 = 116/144 = 29/36$]

RC 11 ROUND 5: RIDDLE

1 You will find me inside a circle.

2 However in reality I am not a part of the circle.

3 I am just a point.

4 I am quite significant in fixing the location of the circle.

5 I am characterized by being equidistant from any point on the circle.

6 I am a point of central symmetry for the circle.
Who am I?
ANSWER: CENTER OF THE CIRCLE

RC 11 ROUND 4A: TRUE OR FALSE

The function $y = a^x$ is an exponential function

1 provided a is a real number,
ANSWER: FALSE

2 provided a is a positive real number,
ANSWER: FALSE

3 provided a is a negative real number.
ANSWER: FALSE

4 provided a is a positive real number and $a \neq 1$.
ANSWER: TRUE

RC 11 ROUND 4B: TRUE OR FALSE

1 $x = 2$ is a zero of $x^3 + 8$
ANSWER: FALSE

2 $x = -3$ is a zero of $x^2 + x - 6$
ANSWER: TRUE

3 $x = 5$ is a zero of $x^3 - 125$
ANSWER: TRUE

4 $x = -2$ is a zero of $x^2 + 2x + 1$
ANSWER: FALSE

RC 12 ROUND 1A

Solve the trigonometric equation in the given interval $0 < x < \pi$

1 $2\sin^2 x = 1$ ANSWER: $x = \pi/4, 3\pi/4$

[$\sin x = \pm 1/\sqrt{2}$, $\sin x = 1/\sqrt{2}$, $x = \pi/4$, $(\pi - \pi/4) = 3\pi/4$, $\sin x = -1/\sqrt{2}$, no solution in the interval]

2 $4\cos^2 x = 3$ ANSWER: $x = \pi/6, 5\pi/6$

[$\cos x = \pm\sqrt{3}/2$, $\cos x = \sqrt{3}/2$, $x = \pi/6$, $\cos x = -\sqrt{3}/2$, $x = (\pi - \pi/6) = 5\pi/6$]

3 $\tan^2 x = 3$ ANSWER: $x = \pi/3, 2\pi/3$

[$\tan x = \pm\sqrt{3}$, $\tan x = \sqrt{3}$, $x = \pi/3$, $\tan x = -\sqrt{3}$, $x = (\pi - \pi/3) = 2\pi/3$]

4 $\tan^2 x = 1$ ANSWER: $x = \pi/4, 3\pi/4$

[$\tan x = \pm 1$, $\tan x = 1$, $x = \pi/4$, $\tan x = -1$, $x = 3\pi/4$]

RC 12 ROUND 1B

Find the least value and the greatest value of the given function.

(NB : Read as 'y = absolute value of $x^2 - 4x - 3$')

1 $y = | x^2 - 4x - 3|$ in the domain $\{x: -1 \le x \le 3\}$ ANSWER: least = 2, greatest = 7 [$y = |(x - 2)^2 - 7|$, least value = 2, greatest value = 7]

2 $y = |x^2 - 10x + 20|$ in the domain $\{x: -1 \le x \le 4\}$ ANSWER: least = 4, greatest = 31 [$y = |(x - 5)^2 - 5|$, least = 4, greatest = 31]

3 $y = |(x^2 + 6x + 7)|$ in the domain $\{x: -4 \le x \le 4\}$ ANSWER: least = 2, greatest = 47 [$y = |(x + 3)^2 - 2|$, least value = 1, greatest value = 47]

4 $y = |x^2 + 8x - 10 |$ in the domain $\{x: - 5 \le x \le 6\}$ ANSWER: least = 10, Greatest = 74, [$y = |(x + 4)^2 - 26|$, least = 10, greatest = 74]

RC 12 ROUND 2: SPEED RACE

1 Find the area of a rhombus if one diagonal has length 16 and a side has length 10.
ANSWER: 96 $(d/2)^2 + 8^2 = 10^2$, $(d/2)^2 = 100 - 64 = 36$, $d/2 = 6$, $d = 12$, Area $= (1/2)(12)16 = 12 \times 8 = 96$]

2 Find x given $\log(x + 2) + \log(x - 1) = 1$
ANSWER: $x = 3$
[$(x + 2)(x - 1) = 10$, $x^2 + x - 12 = (x + 4)(x - 3) = 0$, $x = 3$]

3 Differentiate $y = (2x - 3)/(x + 1)$
ANSWER: $dy/dx = 5/(x + 1)^2$
[$dy/dx = (2(x + 1) - 1(2x - 3))/(x + 1)^2 = 5/(x + 1)^2$]

RC 12 ROUND 5: RIDDLE

1 I am a specific angle measure in degrees.

2 I am an acute angle.

3 My sine is less than my cosine.

4 You will find me in a special right-angle triangle.

5 I am a standard angle measure.

6 My tangent is $1/\sqrt{3}$.
Who am I?
ANSWER: 30°

RC 12 ROUND 4A: TRUE OR FALSE

1 Base angles of an isosceles triangle are complementary.
ANSWER: FALSE

2 A trapezium has 2 pairs of adjacent angles which are supplementary.
ANSWER: TRUE

3 The sum of the interior angles of a triangle is equal to the sum of its exterior angles.
ANSWER: FALSE

4 The sum of the interior angles of a quadrilateral is equal to the sum of its exterior angles.
ANSWER: TRUE

RC 12 ROUND 4B: TRUE OR FALSE

An implication is true whenever,

1 the conclusion is false,
ANSWER: FALSE

2 the conclusion is true,
ANSWER: FALSE

3 the hypothesis is true,
ANSWER: FALSE

4 the hypothesis is false.
ANSWER: TRUE

RC 13 ROUND 1A

A number is selected randomly from the integers 1, 2, 3, . . . , 15. Find the probability that the number is

1 divisible by 5 ANSWER: 1/5
[A = {5, 10, 15}, P(A) = 3/15 = 1/5]

2 divisible by 3 ANSWER: 1/3
[B = {3, 6, 9, 12, 15}, P(B) = 5/15 = 1/3]

3 divisible by 2 ANSWER: 7/15
[C = {2, 4, 6, 8, 10, 12, 14}, P(C) = 7/15]

4 either a square or a cube ANSWER: 1/5
[D = {4, 9, 8}, P(D) = 3/15 = 1/5]

RC 13 ROUND 1B

Expand in ascending powers of x up to the term in x^2.

1 $(1 + x + x^2)^4$ ANSWER: $1 + 4x + 10x^2$
[$1 + 4(x + x^2) + 6(x + x^2)^2 = 1 + 4x + (4 + 6)x^2 = 1 + 4x + 10x^2$]

2 $(1 + 2x – x^2)^5$ ANSWER: $1 + 10x + 35x^2$
[$1 + 5(2x – x^2) + 10(2x – x^2)^2 = 1 + 10x + (-5 + 40)x^2 = 1 + 10x + 35x^2$]

3 $(1 – x + x^2)^6$ ANSWER: $1 – 6x + 21x^2$
[$1 + 6(-x + x^2) + 15(-x + x^2)^2 = 1 – 6x + (6 + 15)x^2 = 1 – 6x + 21x^2$]

4 $(1 - 2x – x^2)^4$ ANSWER: $1 - 8x + 20x^2$
[$1 – 4(2x + x^2) + 6(2x + x^2)^2 = 1 – 8x + (24 – 4)x^2 = 1 - 8x + 20x^2$]

RC 13 ROUND 2: SPEED RACE

1 Evaluate $(\sin 73°\cos 27° + \cos 73°\sin 27°)/\cos 10°$
ANSWER: 1
[$\sin(73 + 27)/\cos 10 = \sin 100/\cos 10 = \cos 10/\cos 10 = 1$]

2 Find the value of n if $123_n = 66_{10}$
ANSWER: $n = 7$
[$n^2 + 2n + 3 = 66$, $n^2 + 2n – 63 = (n + 9)(n – 7) = 0$, $n = 7$]

3 Find a relation between y and x given that $3\log_5 y = 2\log_5 x + 3$
ANSWER: $y^3 = 125x^2$
[$\log_5 y^3 = \log_5 x^2 + \log_5 125 = \log_5(125x^2)$, $y^3 = 125x^2$]

RC 13 ROUND 5: RIDDLE

1 I am a mathematical operation.

2 Essentially, I am one of 2 related operations.

3 I have a fundamental theorem to my credit.

4 While my cousin brings down powers, I raise them up one at a time.

5 I may be definite or indefinite.

6 Among my applications are finding areas under curves and volumes of solids of rotation.
Who am I?
ANSWER: INTEGRATION

RC 13 ROUND 4A: TRUE OR FALSE xxxx

1 $\sin(90° + x) = \cos x$
ANSWER: TRUE

2 $\cos(90° + x) = \sin x$
ANSWER: FALSE

3 $\sin(180° – x) = \sin x$
ANSWER: TRUE

4 $\cos(180° – x) = \cos x$
ANSWER: FALSE

RC 13 ROUND 4B: TRUE OR FALSE

1 If the opposite angles of a quadrilateral are congruent, then it is a parallelogram.
ANSWER: TRUE

2 If the opposite sides of a quadrilateral are congruent, then it is a rectangle.
ANSWER: FALSE

3 If the point of intersection of the diagonals of a quadrilateral is equidistant from the vertices, then it is a parallelogram.
ANSWER: FALSE

4 If adjacent sides of a quadrilateral are congruent, then it is a kite.
ANSWER: FALSE

RC 14 ROUND 1A

Find the range of the trigonometric function,

1 $y = 5\cos x - 12\sin x + 9$ ANSWER: $\{y: -4 \leq y \leq 22\}$ [$y = 13\cos(x + \alpha) + 9$, greatest $= 13 + 9 = 22$, least $= -13 + 9 = -4$, hence $\{y: -4 \leq y \leq 22\}$]

2 $y = 15\cos x + 8\sin x - 12$ ANSWER: $\{y: -29 \leq y \leq 5\}$ [$y = 17\cos(x + \alpha) - 12$, greatest $= 17 - 12 = 5$, least $= -17 - 12 = -29$, hence $\{y: -29 \leq y \leq 5\}$]

3 $y = 9\cos x - 12\sin x + 15$ ANSWER: $\{y: 0 \leq y \leq 30\}$ [$y = 15\cos(x + \alpha) + 15$, greatest $=15 + 15 = 30$, least value $= -15 + 15 = 0$, hence $\{y: 0 \leq y \leq 30\}$]

4 $y = 8\cos x - 15\sin x - 17$ ANSWER: $\{y: -34 \leq y \leq 0\}$ [$y = 17\cos(x + \alpha) - 17$ greatest value $= 17 - 17 = 0$, least $= -17 - 17 = -34$, hence $\{y: -34 \leq y \leq 0\}$]

RC 14 ROUND 1B

Find the coordinates of the center C, and the radius R of the circle with equation,

1 $x^2 + y^2 - 2x + 4y = 4$ ANSWER: $C(1, -2)$, $R = 3$
[$(x - 1)^2 + (y + 2)^2 = 4 + 1 + 4 = 9$, $C(1, -2)$, $R = 3$]

2 $x^2 + y^2 + 8x - 10y = 59$ ANSWER: $C(-4, 5)$, $R = 10$
[$(x + 4)^2 + (y - 5)^2 = 59 + 16 + 25 = 100$, $C(-4, 5)$, $R = 10$]

3 $x^2 + y^2 - 10x + 6y = 2$ ANSWER: $C(5, -3)$, $R = 6$
[$(x - 5)^2 + (y + 3)^2 = 25 + 9 + 2 = 36$, $C(5, -3)$, $R = 6$]

4 $x^2 + y^2 - 4x - 8y = 5$ ANSWER: $C(2, 4)$, $R = 5$
[$(x - 2)^2 + (y - 4)^2 = 5 + 4 + 16 = 25$, $C(2, 4)$, $R = 5$]

RC 14 ROUND 2: SPEED RACE

1 Given $x^2 + 2xy + 2y^2 = 7$ find the gradient dy/dx .
ANSWER: $dy/dx = -(x + y)/(x + 2y)$ $[2x + 2(x(dy/dx) + y) + 4y(dy/dx) = 0,$
$(dy/dx)(2x + 4y) = -(2x + 2y), (dy/dx) = -(x + y)/(x + 2y)]$

2 Triangle ABC has sides of lengths $a = 5$, $b = 7$, $c = 8$. Find cosA
ANSWER: $\cos A = 11/14$
$[\cos A = (b^2 + c^2 - a^2)/2bc = (49 + 64 - 25)/2(7)8 = 88/14(8) = 11/14]$

3 Express $(2x - 1)/(x + 1)$ in partial fractions
ANSWER: $2 - 3/(x + 1)$
$[(2(x + 1) - 3)/(x + 1) = 2 - 3/(x + 1)]$

RC 14 ROUND 5: RIDDLE

1 I am a 4-digit even number.

2 The second and third digits are identical.

3 The first and last digits are also identical.

4 The sum of my digits is a dozen.

5 My first digit is a square.

6 The product of all my digits is 64.
Who am I?
ANSWER: 4224

RC 14 ROUND 4A: TRUE OR FALSE

1 All prime numbers are odd numbers.
ANSWER: FALSE

2 The number of prime numbers is finite.
ANSWER: FALSE

3 The number of even prime numbers is finite.
ANSWER: TRUE

4 The product of 2 prime numbers is a prime number.
ANSWER: FALSE

RC 14 ROUND 4B: TRUE OR FALSE

1 Every relation is a mapping.
ANSWER: FALSE

2 A relation is a mapping only if it is one to one.
ANSWER: FALSE

3 A relation is a mapping if it is one to one.
ANSWER: TRUE

4 A relation is a mapping if it is many to one.
ANSWER: TRUE

RC 15 ROUND IA

Find dy/dx from the implicit equation,

1 $x^2 + 2xy - y^2 = 6$ ANSWER: $dy/dx = (x + y)/(y - x)$

[$2x + 2(xdy/dx + y) - 2ydy/dx = 0$, $dy/dx(2y - 2x) = 2x + 2y$,
$dy/dx = (x + y)/(y - x)$]

2 $x^2 - xy - y^2 = 10$ ANSWER: $dy/dx = (2x - y)/(x + 2y)$

[$2x -(xdy/dx + y) - 2y(dy/dx) = 0$, $(dy/dx)(x + 2y) = 2x - y$,
$(dy/dx) = (2x - y)/(x + 2y)$]

3 $2x^2 + 3xy - 2y^2 = 15$ ANSWER: $dy/dx = (4x + 3y)/(4y - 3x)$

[$4x + 3(xdy/dx + y) - 4y(dy/dx) = 0$, $(dy/dx)(4y - 3x) = 4x + 3y$,
$dy/dx = (4x + 3y)/(4y - 3x)$]

4 $x^2 - 2xy - y^2 = 5$ ANSWER: $dy/dx = (x - y)/(x + y)$

[$2x - 2(xdy/dx + y) - 2ydy/dx = 0$, $2(dy/dx)(x + y) = 2x - 2y$,
$dy/dx = (x - y)/(x + y)$]

RC 15 ROUND 1B

Solve the exponential equation

1...$(x^2 - 4x - 4)^{(x - 2)} = 1$ ANSWER: $x = 2, 5, - 1$

[$x - 2 = 0$, $x = 2$, $x^2 - 4x - 4 = 1$, $x^2 - 4x - 5 = (x - 5)(x + 1) = 0$, $x = 5, -1$]

2 $(x^2 + 3x - 3)^{(x + 2)} = 1$ ANSWER: $x = -2, -4, 1$

[$x + 2 = 0$, $x = -2$, $x^2 + 3x - 3 = 1$, $x^2 + 3x - 4 = (x + 4)(x - 1) = 0$, $x = 1, -4$]

3 $(x^2 - 2x - 7)^{x + 3} = 1$ ANSWER: $x = -3, 4, -2$

[$x + 3 = 0$, $x = -3$, $x^2 - 2x - 7 = 1$, $x^2 - 2x - 8 = (x - 4)(x + 2) = 0$, $x = 4, -2$]

4 $(x^2 - x - 1)^{(x - 3)} = 1$ ANSWER: $x = 3, 2, -1$

[$(x - 3) = 0$, $x = 3$, $x^2 - x - 1 = 1$, $x^2 - x - 2 = (x - 2)(x + 1) = 0$, $x = 2, -1$]

RC 15 ROUND 2: SPEED RACE

1 A binary operation $*$ is defined on the set Z of integers by $a*b = a + b - ab$. Evaluate $(1*5)*(2*3)$

ANSWER: 1

[$1*5 = 1 + 5 - 1(5) = 1$, $2*3 = 2 + 3 - 2(3) = -1$, $1*-1 = 1 + (-1) - 1(-1) = 1$]

2 Find the equation of the image of the line $y = x$ after a reflection in the point A(3, 2).

ANSWER: $y = x - 2$

[$(x, y)\rightarrow(6 - x, 4 - y)$, $4 - y = 6 - x$, $y = x - 2$]

3 If $\tan(x + y) = 3/5$ and $\tan x = ½$, find $\tan y$

ANSWER:1/13

[$\tan y = \tan(x + y - x) = (\tan(x + y) - \tan y)/(1 + \tan(x + y)(\tan y) =$
$(3/5 - ½)/(1 + (3/5)(1/2)) = (6 - 5)/(10 + 3) = 1/13$]

RC 15 ROUND 5: RIDDLE

1 I am a solid geometric figure.

2 I am in the poly business.

3 You can say I am a polyhedron.

4 I am 2 congruent triangular pyramids stuck together.

5 I have 6 triangular faces.

6 I have 5 vertices and 9 sides.

Who am I?

ANSWER: TETRAHEDRON

RC 15 ROUND 4A: TRUE OR FALSE

1 An exterior angle of a cyclic quadrilateral is equal to the interior opposite angle.
ANSWER: TRUE

2 Any angle subtended by a diameter on the circumference of a circle is a straight angle.
ANSWER: FALSE

3 Adjacent angles of a cyclic quadrilateral are supplementary.
ANSWER: FALSE

4 If the adjacent angles of a cyclic quadrilateral are congruent, then the quadrilateral is a rectangle.
ANSWER: TRUE

RC 15 ROUND 4B: TRUE OR FALSE

1 Every whole number is a natural number.
ANSWER: FALSE

2 Every even number is a rational number.
ANSWER: TRUE

3 A real number is either a rational number or an irrational number.
ANSWER: TRUE

4 Every irrational number is a surd.
ANSWER: FALSE

RC 16 ROUND 1A
Solve for x and y given,

1 $(2^{x+y})(3^{x-y}) = 16 \times 9$ **ANSWER: $x = 3, y = 1$**
$[2^{x+y} = 2^4, 3^{x-y} = 3^2, x + y = 4, x - y = 2, x = 3, y = 1]$

2 $(5^{2x+y})(7^{3x-y}) = 625 \times 7$ **ANSWER: $x = 1, y = 2,$**
$[2x + y = 4, 3x - y = 1, 5x = 5, x = 1, y = 3x - 1 = 2]$

3 $(2^{x+y})(3^{2y-x}) = 32 \times 3$ **ANSWER: $x = 3, y = 2$**
$[2^{x+y} = 2^5, x + y = 5, 3^{2y-x} = 3, 2y - x = 1, 3y = 6, y = 2, x = 5 - y = 3]$

4 $(4^{x-y})(5^{x+y}) = 64 \times 5$ **ANSWER: $x = 2, y = -1$**
$[4^{x-y} = 4^3, 5^{x+y} = 5, x - y = 3, x + y = 1, 2x = 4, x = 2, y = 2 - 3 = -1]$

RC 16 ROUND 1B

A number is picked at random from the numbers 1, 2, 3, 4, . . . 30. Find the probability the number is ,

1 a prime number. **ANSWER: 1/3**
[A = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29} P(A) = 10/30 = 1/3]

2 a multiple of 5 **ANSWER: 1/5**
[B = {5, 10, 15, 20, 25, 30}, P(B) = 6/30 = 1/5]

3 a multiple of 6 **ANSWER: 1/6**
[C = { 6, 12, 18, 24, 30}, P(B) = 5/30 = 1/6]

4 an odd number strictly between 5 and 25. **ANSWER: 3/10**
[D = {7, 9, 11, 13, 15, 17, 19, 21, 23}, P(D) = 9/30 = 3/10]

RC 16 ROUND 2: SPEED RACE

1 Find a cartesian equation in x and y given $x = 5\cos t$, and $y = 4\sin t$
ANSWER: $x^2/25 + y^2/16 = 1$
[$\cos t = x/5$, $\sin t = y/4$, $\cos^2 t + \sin^2 t = x^2/25 + y^2/16 = 1$]

2 Find the lengths of the bases of an isosceles trapezium if the legs are 10 cm and the altitude is 8cm and the area is 128 cm^2.
ANSWER: 10 cm, 22 cm
[$\frac{1}{2}(2a + 12)8 = 128$, $2a + 96 = 256$, $16a = 160$, $a = 10$, $b = a + 12 = 22$]

3 Find the value of n if $123_n = 38_{10}$
ANSWER: $n = 5$
[$n^2 + 2n + 3 = 38$, $n^2 + 2n - 35 = (n + 7)(n - 5) = 0$]

RC 16 ROUND 5: RIDDLE

1 I am a statistical measure.

2 I am a measure of central tendency.

3 There is nothing mean about me.

4 I am not the middle - man or a quartile.

5 I am the measurement with the largest frequency.

6 I am not necessarily unique as I can occur with more than one value.
Who am I?
ANSWER: MODE

RC 16 ROUND 4A: TRUE OR FALSE
The following number is divisible by 18

1 3312
ANSWER: TRUE

2   512
ANSWER: FALSE

3 854
ANSWER: FALSE

4 8442
ANSWER: TRUE

RC 16 ROUND 4B: TRUE OR FALSE

1 An equilateral quadrilateral is regular.
ANSWER: FALSE

2 An equilateral pentagon is regular.
ANSWER: TRUE

3 An equiangular hexagon is equilateral.
ANSWER: FALSE

4 An equilateral decagon is equiangular.
ANSWER: FALSE

RC 17 ROUND 1A

The roots of the equation $ax^2 + bx + c = 0$ are α and β. Find the relation between a, b and c if

1 $\beta = 3\alpha$ ANSWER: $3b^2 = 16ac$ [$\alpha + \beta = -b/a$, $\alpha\beta = c/a$, $4\alpha = -b/a$, $3\alpha^2 = c/a$, $\alpha = -b/4a$ $3(-b/4a)^2 = c/a$, $3b^2 = 16ac$]

2 $\beta = 4\alpha$ **ANSWER:** $4b^2 = 25ac$ [$\alpha + \beta = -b/a$, $\alpha\beta = c/a$, $5\alpha = -b/a$, $4\alpha^2 = c/a$, $\alpha = -b/5a$, $4(-b/5a)^2 = c/a$, $4b^2 = 25ac$]

3 $\beta = 5\alpha$ ANSWER: $5b^2 = 36ac$ [$\alpha + \beta = -b/a$, $\alpha\beta = c/a$, $6\alpha = -b/a$, $5\alpha^2 = c/a$, $\alpha = (-b/6a)$, $5(-b/6a)^2 = c/a$, $5b^2 = 36ac$]

4 $\beta = 7\alpha$ **ANSWER:** $7b^2 = 64ac$ [$\alpha + \beta = -b/a$, $\alpha\beta = c/a$, $8\alpha = -b/a$, $7\alpha^2 = c/a$, $\alpha = (-b/8a)$, $7(-b/8a)^2 = c/a$, $7b^2 = 64ac$]

RC 17 ROUNC 1B

Find the sum of the coefficients (including x^0) in the expansion of

1 $(1 + x)^4$ ANSWER: 16
[$1 + 4x + 6x^2 + 4x^3 + x^4$, $1 + 4 + 6 + 4 + 1 = 16$]

2 $(1 + x)^5$ ANSWER: 32
[$1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$, $1 + 5 + 10 + 10 + 5 + 1 = 32$]

3 $(2 + x)^3$ ANSWER: 27
[$8 + 4(3)x + 2(3)x^2 + x^3$, $8 + 12 + 6 + 1 = 27$]

4 $(1 + 2x)^4$ ANSWER: 81
[$1 + 4(2x) + 6(2x)^2 + 4(2x)^3 + (2x)^4$, $1 + 8 + 24 + 32 + 16 = 81$]

RC 17 ROUND 2: SPEED RACE

1 Solve for x and y from the equation $(7^{x+y})(3^{x-2y}) = 7 \times 81$
ANSWER: $x = 2, y = -1$
[$7^{x+y} = 7$, $x + y = 1$, $3^{x-2y} = 3^4$, $x - 2y = 4$, $3y = -3$, $y = -1$, $x - 1 = 1$, $x = 2$]

2 Differentiate $y = (x^2 - 2/x^2)^{-10}$
ANSWER: $dy/dx = -10(2x + 4/x^3)(x^2 - 2/x^2)^{-11}$
[$dy/dx = -10(2x + 4/x^3)(x^2 - 2/x^2)^{-11}$]

3 A and B are acute angles with $\sin A = \frac{1}{2}$ and $\sin B = 4/5$ evaluate $\sin(A - B)$
ANSWER: $(3 - 4\sqrt{3})/10$
[$\sin(A - B) = \sin A\cos B - \cos A\sin B = \frac{1}{2}(3/5) - (\sqrt{3}/2)(4/5) = (3 - 4\sqrt{3})/10$]

RC 17 ROUND 5: RIDDLE

1 I am a binary operation.

2 I involve a sum of products of real numbers.

3 I am not a binary operation on the set of real numbers.

4 I involve directed line segments.

5 I am a binary operation on the set of vectors.

6 I result in a scalar and hence my name.
Who am I?
ANSWER: SCALAR PRODUCT

RC 17 ROUND 4A: TRUE OR FALSE

1 $3\sin^{-1}(1/2) = \cos^{-1}0$
ANSWER: TRUE

2 $\tan^{-1}1 = \cos^{-1}(-1/\sqrt{2})$
ANSWER: FALSE

3 $\cos^{-1}(1/\sqrt{2}) = \sin^{-1}(1/\sqrt{2})$
ANSWER: TRUE

4 $\cos^{-1}(-1/\sqrt{2}) = 3\tan^{-1}1$
ANSWER: TRUE

RC 17 ROUND 4A: TRUE OR FASE

1 $(x - 1)$ is a factor of $f(x) = x^3 - 3x + 2$
ANSWER: TRUE

2 $(x + 1)$ is factor of $f(x) = 3x^3 + 2x^2 + 1$
ANSWER: TRUE

3 $(x - 2)$ is a factor of $x^3 - 4x + 1$
ANSWER: FALSE

4 $(x - 3)$ is a factor of $f(x) = x^3 - 4x^2 + 9$
ANSWER: TRUE

RC 18 ROUND 1A

Find a possible integer value of x from the exponential equation.

1 $2^x 3^{x^2} = 4(81)$ ANSWER: x = 2

[$2^x = 4 = 2^2$, x = 2, $x^2 = 4$, x = $\pm$ 2, hence x = 2]

2 $(5^{3x})(2^{x^2}) = (125)(16)$ ANSWER: NO SOLUTION

[$5^{3x} = 5^3$, 3x = 3, x = 1, $x^2 = 4$, x = $\pm$ 2, inconsistent with x = 1, no solution]

3 $4^{2x}(3^{x^2-1}) = (256)(27)$ ANSWER: x = 2

[$4^{2x} = 4^4$, 2x = 4, x = 2, $x^2 - 1 = 3$, $x^2 = 4$, x = $\pm$ 2, hence x = 2]

4 $7^{x+1}(2^{x^2+2x}) = (49)8$ ANSWER: x = 1

[x + 1 = 2, x = 1, $x^2 + 2x - 3 = (x + 3)(x - 1) = 0$, x = 1]

RC 18 ROUND 1B

Find the equation of the line passing through the point,

1 A(-1, 2) and parallel to the line 3x + 2y = 7, ANSWER: 3x + 2y = 1

[3x + 2y = k, k = -3 + 4 = 1, 3x + 2y = 1]

2 B(2, 6) and perpendicular to the line 2x + 5y = 7, ANSWER: 5x – 2y = -2

[5x – 2y = k, k = 10 – 12 = -2, 5x – 2y = -2]

3 C(-3, 4) and parallel to the line y = 3x – 5, ANSWER: y = 3x + 13

[y = 3x + k, 4 = -9 + k, k = 13, y = 3x + 13]

4 D(3, -2) and perpendicular to the line y = -x + 7 ANSWER: y = x - 5

[y = x + k, -2 = 3 + k, k = -5, y = x – 5]

RC 18 ROUND 2: SPEED RACE

1 Find the cosine of the angle between the vectors a = 8i + 15j and
b = 12i – 5j
ANSWER: 21/221
[$\cos\theta = (a.b)/|a||b| = (8(12) - 15(5))/(17)(13) = (96 - 75)/221 = 21/221$]

2 Solve for n from the equation $nC_2 = 190$ (NB: Read as n combination 2 equals 190)
ANSWER: n = 20
[$n(n - 1)/2 = 190$, $n(n - 1) = 380 = 20(19)$, hence $n = 20$]

3 Find the stationary point(s) of the curve $y = x^3 - 3x^2 - 9x + 8$
ANSWER: (3, - 19), (-1, 13)
[$dy/dx = 3x^2 - 6x - 9 = 0$, $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$, $x = 3$,
$y = -27 - 27 + 27 + 8 = -19$, (3, -19), $x = -1$, $y = -1 - 3 + 9 + 8 = 13$, (-1, 13)]

RC 18 ROUND 5: RIDDLE

1 I am a 2 - digit number.

2 I am an even number.

3 One digit is twice the other.

4 The sum of my digits is half a dozen.

5 I am divisible by 3 and 7.

6 When factorized, I am of the form n(n + 1).
Who am I?
ANSWER: 42

PC 18 ROUND 4A: TRUE OR FALSE
If the opposite sides of a quadrilateral are congruent, then it is

1 a square,
ANSWER: FALSE

2 a rectangle,
ANSWER: FALSE

3 a rhombus,
ANSWER: FALSE

4 a parallelogram.
ANSWER: TRUE

RC 18 ROUND 4B: TRUE OR FALSE

1 $(x + 2)(x^2 - 4x + 4) = x^3 + 8$
ANSWER: FALSE

2 $(x + 2)(x^2 - 2x + 4) = x^3 + 8$
ANSWER: TRUE

3 $(x - 3)(x^2 - 3x + 9) = x^3 - 27$
ANSWER: FALSE

4 $(x - 3)(x^2 + 3x - 9) = x^3 - 27$
ANSWER: FALSE

PC 19 ROUND 1A

Determine the values of the constants a, b, c given that

1 $a(x-1)^2 + b(x-1) + c = 5x^2 + 7x + 6$ ANSWER: $\mathbf{a = 5, b = 17, c = 18}$

[$a(x^2 - 2x + 1) + b(x-1) + c = 5x^2 + 7x + 6$, $a = 5$, $-2a + b = 7$, $b = 17$,
$a - b + c = 6$, $c = 6 + b - a = 6 + 17 - 5 = 23 - 5 = 18$]

2 $a(x-2)^2 + b(x-2) + c = 3x^2 - 7x + 8$ ANSWER: $\mathbf{a = 3, b = 5, c = 6}$

[$a(x^2 - 4x + 4) + b(x-2) + c = 3x^2 - 7x + 8$, $a = 3$, $-4a + b = -7$,
$b = 12 - 7 = 5$, $4a - 2b + c = 8$, $c = -4a + 2b + 8 = -12 + 10 + 8 = 6$]

3 $a(x+1)^2 + b(x+1) + c = 6x^2 - 5x - 8$ ANSWER: $\mathbf{a = 6, b = -17, c = 3}$

[$a(x^2 + 2x + 1) + b(x+1) + c = 6x^2 - 5x - 8$, $a = 6$, $2a + b = -5$,
$b = -5 - 12 = -17$, $a + b + c = -8$, $c = -a - b - 8 = -6 + 17 - 8 = 3$]

4 $a(x+2)^2 + b(x+2) + c = 2x^2 + 6x + 7$ ANSWER: $\mathbf{a = 2, b = -2, c = 3}$

[$a(x^2 + 4x + 4) + b(x+2) + c = 2x^2 + 6x + 7$, $a = 2$, $4a + b = 6$, $b = 6 - 8 = -2$,
$4a + 2b + c = 7$, $c = 7 - 4a - 2b = 7 - 8 + 4 = 3$]

PC 19 ROUND 1B

Find the value of the constant k given that a curve has gradient,

1 $4x + k$ and it passes through the points A(1, 6) and B(-1, 8).

ANSWER: $\mathbf{k = -1}$ [$y = 2x^2 + kx + C$, $6 = 2 + k + C$, $8 = 2 - k + C$,
$k + C = 4$, $k - C = -6$, $2k = -2$, $k = -1$]

2 $2kx - 4$ and it passes through the points A(1, 5) and B(2, -11)

ANSWER: $\mathbf{k = -4}$ [$y = kx^2 - 4x + C$, $5 = k - 4 + C$, $-11 = 4k - 8 + C$,
$k + C = 9$, $4k + C = -3$, $3k = -12$, $k = -4$]

3 $2x + 3k$ and it passes through the points A(1, 8) and B(-1, 14),

ANSWER: $\mathbf{k = -1}$ [$y = x^2 + 3kx + C$, $8 = 1 + 3k + C$, $14 = 1 - 3k + C$,
$6k = -6$, $k = -1$]

4 $6x + 5k$ and passes through the points A(1, -7), (-1, 13).

ANSWER: $\mathbf{k = -2}$

[$y = 3x^2 + 5kx + C$, $-7 = 3 + 5k + C$, $13 = 3 - 5k + C$, $20 = -10k$, $k = -2$]

PC 19 ROUND 2: SPEED RACE

1 Triangle ABC has sides of lengths a = 5, b = 8, c = 7. Find cosB
ANSWER: cos B = 1/7
[$\cos B = (a^2 + c^2 - b^2)/2ac = (25 + 49 - 64)/2(5)7 = 10/70 = 1/7$]

2 Find the set of values of k for which the equations,
x + ky = 10 and kx + 9y = 6 have a unique solution.
ANSWER: {k: k ≠ ±3} [$A = \begin{pmatrix} 1 & k \\ k & 9 \end{pmatrix}$, $\det A = 9 - k^2 \neq 0$, $k \neq \pm 3$]

3 Find the sum to infinity of the series 7 + 7/4 + 7/16 + 7/64 + 7/256 + ...
ANSWER: 28/3
[exponential series a = 7, r = ¼, $S_\infty = a/(1 - r) = 7/(1 - ¼) = 7/(3/4) = 28/3$]

PC 19 ROUND 5: RIDDLE

1 I am a 2- digit even number.

2 One digit is four times the other digit.

3 My first digit is a cube.

4 The sum of my digits is divisible by 5.

5 The product of my digits is 16.

6 The sum of my digits is 10.
Who am I?
ANSWER: 82

PC 19 ROUND 4A: TRUE FALSE

1 The sum of the interior angles of a quadrilateral is equal to the sum of its exterior angles.
ANSWER: TRUE

2 The sum of the interior angles of a triangle is half the sum of its exterior angles.
ANSWER: TRUE

3 The sum of the interior angles of a hexagon is $740°$.
ANSWER: FALSE

4 The sum of the interior angles of a heptagon is $900°$.
ANSWER: TRUE

PC 19 ROUND 4B: TRUE OR FALSE

1 $\sin 37°\cos 23° + \cos 37°\sin 23° = ½$
ANSWER: FALSE [$\sin 60° \neq ½$]

2 $\cos 53°\cos 37° - \cos 37°\sin 53° = 0$
ANSWER: TRUE [$\cos 90° = 0$]

3 $(\tan 42° - \tan 12°)/(1 + \tan 42°\tan 12°) = \sqrt{3}$
ANSWER: FALSE [$\tan 30 = 1/\sqrt{3} \neq \sqrt{3}$]

4 $(\tan 35° + \tan 25°)/(1 - \tan 35°\tan 25°) = \sqrt{3}$
ANSWER: TRUE [$\tan 60 = \sqrt{3}$]

RC 20 ROUND 1A

Find the set of real values of k for which the simultaneous equations have a unique solution.

1…x + ky = 5, ky – x = -7 **ANSWER: {k: k is a real number}** or **{real numbers}**

[A = $\begin{pmatrix} 1 & k \\ k & -1 \end{pmatrix}$, det A = -1 – $k^2 \neq 0$, $k^2 + 1 \neq 0$, all values of k]

2 2x - ky = 10, kx – 8y = 8 **ANSWER: {k: k ≠ ± 4}**

[B = $\begin{pmatrix} 2 & -k \\ k & -8 \end{pmatrix}$, det B = -16 + $k^2 \neq 0$, $k^2 \neq 16$, k ≠ ±4]

3 3x + ky = 12, ky + 27y = 50 **ANSWER: {k: k ≠ ± 9}**

[C = $\begin{pmatrix} 3 & k \\ k & 27 \end{pmatrix}$, det C = 81 – $k^2 \neq 0$, $k^2 \neq 81$, k ≠ ± 9]

4 kx – 3y = 9, 3x + ky = 7 **ANSWER:{k: k is a real number}, or {real numbers}.**

[D = $\begin{pmatrix} k & -3 \\ 3 & k \end{pmatrix}$, det D = $k^2 + 9 \neq 0$, all values of k]

RC 20 ROUND 1B

Given that tan A = ¾, sin B = √3/2, with A acute and B obtuse, evaluate

1 tan (A + B) **ANSWER: (3 - 4√3)/(4 + 3√3)**

[tan A = ¾, tan B = -√3, tan(A + B) = (tanA + tanB)/(1 – tanA tanB) = (3/4 - √3)/(1 + ¾√3) = (3 - 4√3)/(4 + 3√3)]

2 sin (A – B) **ANSWER: -(3 + 4√3)/10**

[sinAcosB – cosAsinB = (3/5) (-1/2) – (4/5)(√3/2) = -(3 + 4√3)/10]

3 cos(A + B) **ANSWER: -(4 + 3√3)/10**

[cosAcosB – sinAsinB = (4/5)(-1/2) – (3/5)(√3/2) = -(4 + 3√3)/10]

4 sin(A + B) **ANSWER: (- 3+ 4√3)/10**

[sinAcosB + cosAsinB = (3/5)(-1/2) + (4/5)(√3/2) = (- 3+ 4√3)/10]

RC 20 ROUND 2: SPEED RACE

1 Find the general term U_n of a linear sequence given that second term is 7 and fifth term is -2.

ANSWER: $U_n = 13 – 3n$ [$a + d = 7$, $a + 4d = -2$, $3d = -9$, $d = -3$, $a = 7 + 3 = 10$, $U_n = (10 – 3(n – 1) = 13 – 3n$]

2 Find the gradient of the function $y = (x^3 – 3x^2 + 1)^{10}$

ANSWER: $dy/dx = 30x(x – 2)(x^3 – 3x^2 + 2)^9$

[$dy/dx = 10(3x^2 – 6x)(x^3 – 3x^2 + 1)^9 = 30x(x – 2)(x^3 – 3x^2 + 1)^9$]

3 2 fair dice are thrown. Find the probability that the scores are both even given that the total score is 10.

ANSWER: 2/3

[S = {(4, 6), (5, 5), (6, 4)}, A = {(6, 4), (4, 6)}, P(A) = 2/3]

RC 20 ROUND 5: RIDDLE

1 I am a polygon.

2 The sum of my interior angles is 5π radians.

3 Each exterior angle measures $2\pi/7$ radians

4 I have seven axes of symmetry.

5 I fit my outline in 14 positions.

6 I have 7 of everything.

Who am I?

ANSWER: REGULAR SEPTAGON, or REGULAR HEPTAGON

RC 20 ROUND 4A: TRUE OR FALSE
The following is the equation of a circle.

1 $x^2 + y^2 - 2x + 4y + 10 = 0$
ANSWER: FALSE

2 $x^2 + y^2 + 4x - 6y + 13 = 0$
ANSWER: FALSE [radius = 0, a point not a circle]

3 $x^2 + y^2 - 6x + 8y + 24 = 0$
ANSWER: TRUE

4 $x^2 + y^2 + 6x + 10y + 33 = 0$
ANSWER: TRUE

RC 20 ROUND 4B: TRUE OR FALSE

1 $\sin 270° = \cos 180°$
ANSWER: TRUE

2 $\cos 270° = \sin 180°$
ANSWER: TRUE

3 $\tan 135° = \tan 225°$
ANSWER: FALSE

4 $\sin 405° = \cos 315°$
ANSWER: TRUE

RC 21 ROUND 1A

The sum of the first n terms of a linear series is S_n. Find the sum from the

1 eleventh term to the twentieth term, given $S_n = n^2 - 3n$,

ANSWER: 270 $[S_{20} - S_{10} = (20^2 - 3(20)) - (10^2 - 3(10)) =$
$(400 - 60) - (100 - 30) = 340 - 70 = 270]$

2 sixth term to the fifteenth term, given $S_n = 2n^2 + 5n$,

ANSWER: 45 0 $[S_{15} - S_5 = 2(15)^2 + 5(15) - (2(5)^2 + 5(5)) =$
$2(225 - 25) + 5(15 - 5) = 400 + 50 = 450]$

3 eighth term to the seventeenth term given $S_n = 3n^2 - 8n$

ANSWER: 640 $[S_{17} - S_7 = 3(17)^2 - 8(17) - (3(7^2) - 8(7)) = 3(17^2 - 7^2) -$
$8(17 - 7) = 3(24)(10) - 8(10) = 720 - 80 = 640]$

4 tenth term to the nineteenth term given $S_n = n(n + 1)$

ANSWER: 290

$[\ S_{19} - S_9 = 19(20) - 9(10) = 380 - 90 = 290]$

RC 21 ROUND 1B

Evaluate

1 $\cos(\sin^{-1}(3/\sqrt{10}))$ **ANSWER: $1/\sqrt{10}$, or $\sqrt{10}/10$,**

$[\text{Let } \theta = \sin^{-1}(3/\sqrt{10}), \sin\theta = 3/\sqrt{10}, \cos\theta = \sqrt{(1 - 9/10)} = 1/\sqrt{10}]$

2 $\sin(\cos^{-1}(1/4))$ **ANSWER: $\sqrt{15}/4$**

$[\text{Let } \theta = \cos^{-1}(1/4), \cos\theta = 1/4, \sin\theta = \sqrt{(1 - 1/16)} = \sqrt{15}/4]$

3 $\cos(\tan^{-1}(2/5))$ **ANSWER: $5/\sqrt{29}$, or $5\sqrt{29}/29$**

$[\text{Let } \theta = \tan^{-1}(2/5), \tan\theta = 2/5, \cos\theta = 5/\sqrt{(4 + 25)} = 5/\sqrt{29}]$

4 $\tan(\sin^{-1}(2/3))$ **ANSWER:$2/\sqrt{5}$, or $2\sqrt{5}/5$**

$[\text{Let } \theta = \sin^{-}(2/3), \sin\theta = 2/3, \tan\theta = 2/\sqrt{5}]$

RC 21 ROUND 2: SPEED RACE

1 Find the gradient (dy/dx) of the curve $x^3 - x^2y^2 + y^3 = 12$)
ANSWER: $(dy/dx) = (3x^2 - 2xy^2)/(2x^2y - 3y^2)$
[$3x^2 - (2xy^2 + 2x^2y(dy/dx)) + 3y^2(dy/dx) = 0$, $(dy/dx)(2x^2y - 3y^2) = 3x^2 - 2xy^2$, $(dy/dx) = (3x^2 - 2xy^2)/(2x^2y - 3y^2)$]

2 A number is selected randomly from the numbers 1, 2, 3, . . . , 21. Find the probability that the number selected is either a multiple of 3 or a multiple of 5.
ANSWER: 10/21
[A = {3, 5, 6, 9, 10, 12, 15, 18, 20, 21}, P(A) = 10/21]

3 Find the solution set of the rational inequality $(x - 1)/(x + 2) < 0$
ANSWER: $\{x: -2 < x < 1\}$
[$(x - 1)(x + 2) < 0$, $-2 < x < 1$]

RC 21 ROUND 5: RIDDLE

1 I am a 3-digit odd number.

2 I am between 500 and 600.

3 Forward or backward the value is the same.

4 The sum of my digits is a dozen.

5 The number formed by my last 2 digits is a square.

6 My middle digit is a special prime.
Who am I?
ANSWER: 525

RC 21 ROUND 4A: TRUE OR FALSE
In partial fractions

1 $(2x + 1)/(x - 1) = 2 + 3/(x - 1)$
ANSWER: TRUE

2 $(2x - 1)/(2x + 1) = 1 + 2/(2x + 1)$
ANSWER: FALSE

3 $(3x + 1)/(x + 1) = 3 - 2/(x + 1)$
ANSWER: TRUE

4 $(x + 3)/(x - 1) = 1 + 4/(x - 1)$
ANSWER: TRUE

RC 21 ROUND 4B: TRUE OR FALSE

1 $\log(x^2 - 1) - \log(x + 1) = \log(x - 1)$
ANSWER: TRUE

2 $\log(x^2 - 4) - \log(x - 2) = \log(x + 2)$
ANSWER: TRUE

3 $\log(x + 3) - \log(x^2 - 9) = \log(x - 3)$
ANSWER: FALSE

4 $\log(x^2 - 25) - \log(x + 5) = -\log(x - 5)$
ANSWER: FALSE

RC 22 ROUND 1A

Find the cubic polynomial P(x) such that,

1 $P(1) = P(-1) = P(0) = 0$, $P(2) = 6$ **ANSWER: $P(x) = x^3 - x$**

[$P(x) = ax(x - 1)(x + 1)$, $P(2) = a(2)(1)(3) = 6$, $a = 1$, $P(x) = x^3 - x$]

2 $P(2) = P(-1) = P(3) = 0$, $P(1) = 8$ **ANSWER: $2(x + 1)(x - 3)(x - 2)$**

[$P(x) = a(x + 1)(x - 3)(x - 2)$, $P(1) = a(2)(-2)(-1) = 8$, $a = 2$,
$P(x) = 2(x + 1)(x - 3)(x - 2)$]

3 $P(2) = P(-2) = P(0) = 0$, $P(1) = -9$ **ANSWER: $P(x) = 3x^3 - 12x$, or $3x(x^2 - 4)$**

[$P(x) = a(x - 2)(x + 2)x$, $P(1) = a(-1)(3)1 = -9$, $a = 3$, $P(x) = 3x(x^2 - 4)$]

4 $P(-1) = P(-2) = P(2) = 0$, $P(1) = -6$ **ANSWER: $(x + 1)(x + 2)(x - 2)$**

[$P(x) = a(x + 1)(x + 2)(x - 2)$, $P(1) = a(2)(3)(-1) = -6$, $a = 1$,
$P(x) = (x + 1)(x + 2)(x - 2)$]

RC 22 ROUND 1B

Find the sum of the coefficients (including x^0) in the expansion of

1 $(2 + x)^4$ **ANSWER: 81**

[$16 + 8(4)x + 4(6)x^2 + 2(4)x^3 + x^4$, $16 + 32 + 24 + 8 + 1 = 41 + 40 = 81$]

2 $(1 + 2x)^4$ **ANSWER: 81**

[$1 + 4(2x) + 6(2x)^2 + 4(2x)^3 + (2x)^4$, $1 + 8 + 24 + 32 + 16 = 41 + 40 = 81$]

3 $(3 + 2x)^3$ **ANSWER: 125**

[$27 + 3(9)(2x) + 3(3)(2x)^2 + (2x)^3$, $27 + 54 + 36 + 8 = 81 + 44 = 125$]

4 $(3 - 2x)^3$ **ANSWER: 1**

[$27 - 3(9)(2x) + 3(3(2x)^2 - (2x)^3$, $27 - 54 + 36 - 8 = 63 - 62 = 1$]

RC 22 ROUND 2: SPEED RACE

1 Differentiate $y = (2x^2 – 3)/(3x^2 + 2)$
ANSWER: $dy/dx = 26x/(3x^2 + 2)^2$
$[dy/dx = (4x(3x^2 + 2) - 6x(2x^2 – 3))/(3x^2 + 2)^2 = 26x/(3x^2 + 2)^2]$

2 Find the greatest and least values of $\sin^2 x + 8\sin x – 4$
ANSWER: greatest value = 5, least value = - 11.
$[(\sin x + 4)^2 – 20$, greatest $= 5^2 – 20 = 5$, least $= 3^2 – 20 = -11]$

3 Given that $f(x) = \sqrt{(x + 2)}$ and $g(x) = x^2$, find an expression for $g(f(x))$ and state its domain.
ANSWER: $(x + 2)$, domain = $\{x: x \geq -2\}$
$[g(f(x)) = (f(x))^2 = (\sqrt{(x + 2)})^2 = x + 2$, domain $= \{x: x \geq -2\}]$

RC 22 ROUND 5: RIDDLE

1 I am a binary operation.

2 My set of operation is the set of functions.

3 I am a combination of 2 functions.

4 As a binary operation I am not commutative.

5 I am associative.

6 I produce a function of a function, or a composite function.
Who am I?
ANSWER: COMPOSITION OF FUNCTIONS

RC 22 ROUND 4A: TRUE OR FALSE

The given value of x is a solution of the inequality $(x - 3)(x + 1) > 0$

1 $x = 2$
ANSWER: FALSE

2 $x = 0$
ANSWER: FALSE

3 $x = 3$
ANSWER: FALSE

4 $x = -3$
ANSWER: TRUE

RC 22 ROUND 4B: TRUE OR FALSE

1 If $7 + 8 = 13$, then $8 + 7 = -13$.
ANSWER: TRUE

2 If $8 \times 4 = 32$, then $4 \times 8 = 23$.
ANSWER: FALSE

3 If the sum of the angles of a triangle is 200°, then it is a quadrilateral.
ANSWER: TRUE

4 If a triangle is obtuse, then all its angles are obtuse.
ANSWER: FALSE

RC 23 ROUND 1A

Find an expression for the general term U_n of an exponential sequence given that,

1 the 3rd term is 12 and the 6th term is 96, **ANSWER: $U_n = 3(2^{n-1})$**

[$ar^2 = 12$, $ar^5 = 96$, $r^3 = 96/12 = 8$, $r = 2$, $a = 12/4 = 3$, $U_n = ar^{n-1} = 3(2^{n-1})$]

2 the 3rd term is 36, and the 6th term is 972,**ANSWER: $U_n = 4(3)^{n-1}$**

[$ar^2 = 36$, $ar^5 = 972$, $r^3 = 972/36 = 108/4 = 27$, $r = 3$, $a = 36/9 = 4$,
$U_n = ar^{n-1}$, $U_n = 4(3)^{n-1}$]

3 the 2nd term is 8, and the 5th term is 512, **ANSWER: $U_n = 2(4^{n-1})$**

[$ar = 8$, $ar^4 = 512$, $r^3 = 512/8 = 64$, $r = 4$, $a(4) = 8$, $a = 2$, $U_n = 2(4^{n-1})$]

4 the 3rd term is 10, the 6th term is 1250, **ANSWER: $U_n = 2(5^{n-2})$**

[$ar^2 = 2(5)$, $ar^5 = 2(5)^4$, $r^3 = 125$, $r = 5$, $a(25) = 10$, $a = (2/5)$, $U_n = 2(5^{n-2})$]

RC 23 ROUND 1B

Eliminate θ between the pair of equations and find an equation in x and y,

1 $x = 3\sin\theta + 4$, $y = 4\cos\theta - 3$ **ANSWER: $(x - 4)^2/9 + (y + 3)^2/16 = 1$**

[$\sin\theta = (x - 4)/3$, $\cos\theta = (y + 3)/4$, $\sin^2\theta + \cos^2\theta = 1$,
$(x - 4)^2/9 + (y + 3)^2/16 = 1$]

2 $x = 2\cos\theta - 3$, $y = 5\sin\theta + 2$ **ANSWER: $(x + 3)^2/4 + (y - 2)^2/25 = 1$**

[$\cos\theta = (x + 3)/2$, $\sin\theta = (y - 2)/5$, $(x + 3)^2/4 + (y - 2)^2/25 = 1$

3 $x = 4\cos\theta + 5$, $y = 6\sin\theta - 4$ **ANSWER: $(x - 5)^2/16 + (y + 4)^2/36 = 1$**

[$\cos\theta = (x - 5)/4$, $\sin\theta = (y + 4)/6$, $(x - 5)^2/16 + (y + 4)^2/36 = 1$]

4 $x = 3\cos\theta - 2$, $y = 2\sin\theta + 3$, **ANSWER: $(x + 2)^2/9 + (y - 3)^2/4 = 1$**

[$\cos\theta = (x + 2)/3$, $\sin\theta = (y - 3)/2$, $(x + 2)^2/9 + (y - 3)^2/4 = 1$]

RC 23 ROUND 2: SPEED RACE

1 A point A is reflected in the x-axis and the image point is B(-3, 5). Find the coordinates of the point A.
ANSWER: A(-3, -5)
[Reflect B in x – axis to get A, (-3, 5)→(-3, -5)]

2 Find the length of a side of a triangle if the altitude to that side is twice the length of that side, and the area is 25 cm^2.
ANSWER: 5 cm
[$(1/2)\ (2a)(a) = a^2 = 25$, $a = 5$]

3 Two fair dice are thrown. Find the probability that both scores are prime numbers given that the total score is 8.
ANSWER: 2/5
[S = {(2, 6), (6, 2), (3, 5), (5, 3), (4, 4)], A = {(3, 5), (5, 3)}, P(A) = 2/5]

RC 23 ROUND 4: RIDDLE

1 I am a type of a section.

2 I am a cross-section of a geometric figure.

3 I am a cross-section of a hollow circular pyramid.

4 I may be a circle or an ellipse.

5 I may be a hyperbola or a parabola.

6 My name suggests I have something to do with a cone.
Who am I?
ANSWER: CONIC-SECTION

RC 23 ROUND 3A: TRUE OR FALSE

1 The sequence 1, 3, 5, 7, . . . is a linear series.
ANSWER: FALSE

2 The sequence 3, 6, 12, 24, . . . is an exponential series.
ANSWER: FALSE

3 The series $1 + 2 + 3 + 4 + \ldots$ is a linear series.
ANSWER: TRUE

4 The series $2 + 4 + 6 + 8 + \ldots$ is an exponential series.
ANSWER: FALSE

RC 23 ROUND 3B: TRUE OR FALSE

1 $\sin 123° = \cos 33°$
ANSWER: TRUE

2 $\tan 123° = \cot 33°$
ANSWER: FALSE

3 $\cos 143° = -\sin 57°$
ANSWER: TRUE

4 $\sin 157° = \cos 67°$
ANSWER: TRUE

RC 24 ROUND 1A

From a point P outside a circle, 2 line segments are drawn to intersect the circle first at A and B, and a second at C and D. Find the length of

1 PB given PA = 12 cm, PC = 15cm and PD = 20 cm ANSWER: PB = 25 cm
[PA x PB = PC x PD, 12 x PB = 15 x 20 = 300, PB = 300/12 = 25 cm]

2 PD given PA = 9 cm, PB = 12 cm, and PC = 18 cm ANSWER: PD = 6 cm
[PA x PB = PC x PD, 9 x 12 = 18 x PD, PD = 108/18 = 6 cm]

3 PA given PB = 14 cm, PC =12 cm and PD = 21 cm ANSWER: PA = 18 cm
[PA x PB = PC x PD, 14 x PA = 12 x 21, PA = 12 x 21/14 = 6 x 3 = 18 cm]

4 PC given PD = 20 cm, PA = 12 cm, PB = 15 cm ANSWER: PC = 9 cm
[PA x PB = PC x PD, 15 x 12 = 20 x PC, PC = 180/20 = 9 cm]

RC 24 ROUND 1B

Solve for x and y given,

1 $5^{2x-y} = 125$, and $6^{x/y} = 36$ ANSWER: x = 2, y = 1
[$125 = 5^3$, 2x – y = 3, x/y = 2, x = 2y, 4y – y = 3, y = 1, x = 2y = 2]

2 $2^{3x+y} = 64$, and $10^{y/x} = 1000$ ANSWER: x = 1, y = 3
[$64 = 2^6$, 3x + y = 6, y/x = 3, y = 3x, 3x + 3x = 6, x = 1, y = 3x = 3]

3 $11^{3x+y} = 121$, and $9^{x/y} = 1/9$ ANSWER: x = 1, y = -1
[$121 = 11^2$, 3x + y = 2, x/y = -1, y = -x, 3x – x = 2x = 2, x = 1, y = -1]

4 $7^{x+y} = 343$, and $5^{y/x} = 25$ ANSWER: x = 1, y = 2
[$343 = 7^3$, x + y = 3, $5^{y/x} = 5^2$, y/x = 2, y = 2x, 3x = 3, x = 1, y = 2]

RC 24 ROUND 2: SPEED RACE

1 The gradient of a curve is $2x + k$. Find the value of k if the curve passes through the points A(1, 2) and B(-1, 4)

ANSWER: $k = -1$

[$y = x^2 + kx + C$, $2 = 1 + k + C$, $4 = 1 - k + C$, $2 + 2C = 2 + 4$, $C = 2$, $k = -1$]

2 A binary operation is defined on the set R of real numbers by $m*n = m + n\sqrt{3}$. Evaluate$(3*2)*4$

ANSWER: $3 + 6\sqrt{3}$

[$3*2 = 3 + 2\sqrt{3}$, $(3 + 2\sqrt{3})*4 = (3 + 2\sqrt{3}) + 4\sqrt{3} = 3 + 6\sqrt{3}$]

3 If α and β are the roots of the equation $x^2 + 5x + 3 = 0$, evaluate $(1/\alpha + 1/\beta)^2$

ANSWER: 25/9 [$(1/\alpha + 1/\beta)^2 = ((\alpha + \beta)/\alpha\beta)^2 = (-5)^2/3^2 = 25/9$]

RC 24 ROUND 4: RIDDLE

1 I am a rule in Mathematics.

2 I am applied to triangles.

3 I am one of 2 such rules.

4 Both are used in solving triangles.

5 I have nothing to do with Pythagoras theorem.

6 I am in the form of proportions.

Who am I?

ANSWER: SINE RULE

RC 24 ROUND 3A: TRUE OR FALSE
The given pair of lines

1 $x + 3y = 5$ and $3x + y = 10$ are perpendicular,
ANSWER: FALSE

2 $y = 2x + 5$ and $2x - y = 7$ are parallel,
ANSWER: TRUE

3 $2x + 3y = 4$, and $3x - 2y = 9$ are perpendicular,
ANSWER: TRUE

4 $5x + y = 9$ and $10x + 2y = 17$ are parallel.
ANSWER: TRUE

RC 24 ROUND 3B: TRUE OR FALSE
The given equation has exactly one solution.

1 $\sin x = \frac{1}{2}$.
ANSWER: FALSE

2 $\sin x = \frac{1}{2}$ in the interval $0 < x < \pi$.
ANSWER: FALSE

3 $\cos x = \frac{1}{2}$ in the interval $0 < x < \pi$.
ANSWER: TRUE

4 $\tan x = 1$ in the interval $0 < x < \pi$.
ANSWER: TRUE

RC 25 ROUND 1A

Find integer values of a, b, and c such that the solution set of the inequality,

1 $ax^2 + bx + c < 0$, is $\{x: -1/2 < x < 1/3\}$ ANSWER: a = 6, b = 1, c = -1

[$(2x + 1)(3x - 1) = 6x^2 + x - 1 < 0$, a = 6, b = 1, c = -1]

2 $ax^2 + bx + c > 0$, is $\{x: x < 2/3, \text{ or } x > 3/2\}$ ANSWER: a = 6, b = -13, c = 6

[$(3x - 2)(2x - 3) > 0$, $6x^2 - 13x + 6 > 0$, a = 6, b = -13, c = 6]

3 $ax^2 + bx + c < 0$, is $\{x: -2/5 < x < ¾\}$ ANSWER: a = 20, b = -7, c = -6

[$(5x + 2)(4x - 3) < 0$, $20x^2 - 7x - 6 < 0$, a = 20, b = -7, c = -6]

4 $ax^2 + bx + c > 0$, is $\{x: x < -3/4, \text{ or } x > -1/2\}$ ANSWER: a = 8, b = 10, c = 3

[$(4x + 3)(2x + 1) = 8x^2 + 10x + 3 > 0$, a = 8, b = 10, c = 3]

RC 25 ROUND 1B

Find the coordinates of the image point after the point,

1 A(5, 3) is reflected in the point P(3, -2) ANSWER: (1, - 7)

[(5, 3)→(2(3) – 5, 2(-2) – 3) = (1, -7)]

2 B(-2, 4) is reflected in the point P(4, 3) ANSWER: (10, 2)

[(-2, 4)→(2(4) + 2, 2(3) – 4) = (10, 2)]

3 C(4, -3) is reflected in the point P(1, 2) ANSWER: (-2, 7)

[(4, -3)→(2(1) – 4, 2(2) – (-3)) = (-2, 7)]

4 D(2, 3) is reflected in the point P(-3, -2), ANSWER: (-8, -7)

[(2, 3)→(2(-3) – 2, 2(-2) – 3) = (-8, -7)]

RC 25 ROUND 2: SPEED RACE

1 A straight line makes an angle of 150° with the positive direction of the x-axis. Find the slope of the line.
ANSWER: $-\sqrt{3}/3$
$[m = \tan 150 = -\tan 30 = - 1/\sqrt{3} = -\sqrt{3}/3]$

2 The area of a circle is increasing at a steady rate of (2π) cm^2/s. Find the rate of increase of the radius when the radius $R = 4$ cm
ANSWER: ¼ cm/s
$[A = \pi R^2$, $dA/dt = 2\pi R(dR/dt) = 2\pi$, $dR/dt = (2\pi)/(2\pi R) = 1/R =$ ¼ cm/s]

3 Rationalize the denominator of $(5 - 2\sqrt{3})/(5 + 2\sqrt{3})$
ANSWER: $(27 - 20\sqrt{3})/13$
$[(5 - 2\sqrt{3})^2/(25 – 12) = (25 + 12 - 20\sqrt{3})/13 = (37 - 20\sqrt{3})/13$

RC 25 ROUND 4: RIDDLE

1 I am a 3-digit number.

2 I am an odd number.

3 My last digit is a square.

4 I am myself a square between 500 and 600.

5 The sum of my digits is 16.

6 My 2nd digit is a special prime.
Who am I?
ANSWER: 529

RC 25 ROUND 3A: TRUE OR FALSE

1 The slope of the line segment joining the points A(-1, 2) and B(3, -4) is -2.
ANSWER: FALSE

2 The mid-point of the line joining A(3, 1) and B(-5, 3) is (-2, 4).
ANSWER: FALSE

3 The slope of a perpendicular to the line joining A(3, 1) and B(1, 3) is 1.
ANSWER: TRUE

4 The slope of any line parallel to the line $2x + 3y = 6$ is -3/2.
ANSWER: FALSE

RC 25 ROUND 3B: TRUE OR FALSE

1 $\sin(90° + A) = \sin A$
ANSWER: FALSE

2 $\cos(90° + A) = \sin A$
ANSWER: FALSE

3 $\tan(45° + A) = (1 + \tan A)/(1 - \tan A)$
ANSWER: TRUE

4 $\sin(45° + A) = (\cos A + \sin A)/\sqrt{2}$
ANSWER: TRUE

RC 26 ROUND 1A

Find a relation between h and k of the form $h^2 = \ldots$, or $k^2 = \ldots$
if the point P(h, k) is equidistant from

1 the point A(3, 0) and the y-axis, ANSWER: $k^2 = 6h - 9$
$[(h - 3)^2 + k^2 = h^2, h^2 - 6h + 9 + k^2 = h^2, k^2 = 6h - 9]$

2 the point B(0, 4) and the x-axis, ANSWER: $h^2 = 8k - 16$
$[h^2 + (k - 4)^2 = k^2, h^2 + k^2 - 8k + 16 = k^2, h^2 = 8k - 16]$

3 the point C(-2, 0) and the y-axis, ANSWER: $k^2 =- 4h - 4$
$[(h + 2)^2 + k^2 = h^2, h^2 + 4h + 4 + k^2 = h^2, k^2 + 4h + 4 = 0, k^2 = -4h - 4]$

4 the point D(0, 5) and the x-axis. ANSWER: $h^2 = 10k - 25$
$[h^2 + (k - 5)^2 = k^2, h^2 + k^2 - 10k + 25 = k^2, h^2 = 10k - 25]$

RC 26 ROUND 1B

Find the set of values of k such that the quadratic equation,

1 $x^2 + (k - 2)x + k - 3 = 0$ has distinct real roots ANSWER: $\{k: k \neq 4\}$
$[(k - 2)^2 - 4(k - 3) > 0, k^2 - 4k + 4 - 4k + 12 = k^2 - 8k + 16 = (k - 4)^2 > 0,$
$k \neq 4]$

2 $x^2 + kx + (k + 3) = 0$ has no real roots ANSWER: $\{k: -2 < k < 6\}$
$[k^2 - 4(k + 3) = k^2 - 4k - 12 = (k - 6)(k + 2) < 0, -2 < k < 6]$

3 $x^2 - 2kx + (2 - k) = 0$ has real roots. ANSWER: $\{k: k \geq 1, \text{ or } k \leq -2\}$
$[4k^2 - 4(2 - k) = 4k^2 + 4k - 8 = 4(k^2 + k - 2) = 4(k + 2)(k - 1) \geq 0,$
$k \geq 1, \text{ or } k \leq -2]$

4 $x^2 + 2kx + (k + 2) = 0$ has no real roots. ANSWER: $\{k: -1 < k < 2\}$
$[4k^2 - 4(k + 2) = 4(k^2 - k - 2) < 0, (k - 2)(k + 1) < 0, -1 < k < 2]$

RC 26 ROUND 2: SPEED RACE

1 Find the value of the gradient of the curve $y = x^3 - 3x^2 + 4x - 2$ at the point $x = 2$ on the curve.
ANSWER: 4
[$dy/dx = 3x^2 - 6x + 4$, at $x = 2$, $dy/dx = 12 - 12 + 4 = 4$]

2 Given that $\sin A = 2/3$, find $\cos 2A$.
ANSWER: 1/9
[$\cos 2A = 1 - 2\sin^2 A = 1 - 2(2/3)^2 = 1 - 8/9 = 1/9$]

3 Find the inverse of the linear transformation $(x, y) \rightarrow (2x - y, 3x - y)$
ANSWER: $(x, y) \rightarrow (-x + y, -3x + 2y)$
[$A = \begin{pmatrix} 2 & -1 \\ 3 & -1 \end{pmatrix}$ $\det A = -2 + 3 = 1$, $A^{-1} = \begin{pmatrix} -1 & 1 \\ -3 & 2 \end{pmatrix}$, $(x, y) \rightarrow (-x + y, -3x + 2y)$]

RC 26 ROUND 4: RIDDLE

1 I am a compound statement.

2 I am formed from 2 simple statements.

3 I am false only in one instant.

4 I am true when any one of the 2 statements is true.

5 I am false only when both statements are false.

6 Part of me is a junction.
Who am I?
ANSWER: DISJUNCTION

RC 26 ROUND 3A: TRUE OR FALSE

Given that $f(x) = \sin^{-1}x$, then

1 $-1 \leq f(x) \leq 1$

ANSWER: FALSE

2 $0 \leq f(x) \leq \pi/2$

ANSWER: FALSE

3 $-\pi/2 \leq f(x) \leq \pi/2$

ANSWER: TRUE

4 $-1 \leq x \leq 1$

ANSWER: TRUE

RC 26 ROUND 3B: TRUE OR FALSE

1 The base angles of an inscribed trapezium are congruent.

ANSWER: TRUE

2 A parallelogram inscribed in a circle is a rectangle.

ANSWER: TRUE

3 In a circle, if a diameter bisects a chord, then it is perpendicular to the chord.

ANSWER: TRUE

4 In a circle, congruent chords are equidistant from the center.

ANSWER: TRUE

RC 27 ROUND 1A

Factorize the cubic polynomial completely,

1 $f(x) = 2x^3 + 7x^2 – 5x – 4$ **ANSWER: $f(x) = (x – 1)(x + 4)(2x + 1)$**

[$f(1) = 2 + 7 - 5 – 4 = 0$, $f(x) = (x – 1)(2x^2 + ax + 4)$, for x^2, $7 = -2 + a$, $a = 9$

$(2x^2 + 9x + 4) = (2x + 1)(x + 4)$, $f(x) = (x – 1)(x + 4)(2x + 1)$]

2 $f(x) = x^3 – x^2 – x – 2$ **ANSWER: $f(x) = (x – 2)(x^2 + x + 1)$**

[$f(2) = 8 – 4 – 2 – 2 = 0$, $f(x) = (x – 2)(x^2 + ax + 1)$, for x^2, $-2 + a = -1$, $a = 1$

$(x^2 + x + 1)$, $f(x) = (x – 2)(x^2 + x + 1)$]

3 $f(x) = x^3 - 7x^2 + 7x +15$ **ANSWER: $f(x) = (x + 1)(x – 3)(x – 5)$**

[$f(-1)= -1 – 7 – 7 + 15 = 0$, $f(x) = (x + 1)(x^2 + ax + 15)$, for x^2, $-7 = 1 + a$,

$a = -8$, $(x^2 – 8x + 15) = (x – 3)(x – 5)$, $f(x) = (x + 1)(x – 3)(x – 5)$]

4 $f(x) = x^3 + 2x^2 – x - 2$ **ANSWER: $(x – 1)(x + 1)(x + 2)$**

[$x^2(x + 2) – 1(x + 2) = (x^2 – 1)(x + 2) = (x – 1)(x + 1)(x + 2)$]

RC 27 ROUND 1B

Find the greatest and least values of

1 $\sin^2x – 4\sin x + 1$ **ANSWER: greatest = 6, least = -2**

[$(\sin x – 2)^2 – 4 + 1$, greatest = $9 – 3 = 6$, least = $1 – 3 = -2$]

2 $\cos^2x + 6\cos x + 13$ **ANSWER: greatest = 20, least = 8**

[$(\cos x + 3)^2 + 13 - 9$, greatest = $4^2 + 4 = 20$, least = $2^2 + 4 = 8$]

3 $\sin^2x + 8\sin x + 7$ **ANSWER: greatest = 16, least = 0**

[$(\sin x + 4)^2 – 16 + 7$, greatest = $5^2 – 9 = 16$, least = $3^2 – 9 = 0$]

4 $\cos^2x – 10\cos x + 18$ **ANSWER: greatest = 29, least = 9**

[$(\cos x – 5)^2 – 25 + 18$, greatest = $36 – 7 = 29$, least = $16 – 7 = 9$]

RC 27 ROUND 2: SPEED RACE

1 Given the vectors $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j}$ and $\mathbf{b} = \mathbf{i} + \mathbf{j}$, find the magnitude of the vector $\mathbf{b} - \mathbf{a}$.

ANSWER: $2\sqrt{5}$

[$b - a = (i + j) - (3i - 4j) = -2i + 4j$, $|b - a| = \sqrt{(4 + 16)} = \sqrt{20} = 2\sqrt{5}$]

2 The line $y = 2x - 1$ is a tangent to the curve $y = x^2 - 2x + 3$. Find the coordinates of the point of contact of the tangent with the curve.

ANSWER: (2, 3)

[$dy/dx = 2x - 2 = 2$, $x = 2$, $y = 3$, (2, 3)]

3 A linear sequence is given by -8, -5, - 2, 1, . . .

What term of the sequence is 34?

ANSWER: 15th term

[$-8 + 3(n - 1) = 34$, $3n = 45$, $n = 15$, hence 15th term]

RC 27 ROUND 4: RIDDLE

1 I am a 2-digit number.

2 One digit is a factor of the other digit.

3 The sum of my digits is a dozen.

4 My first digit is a factor of the second digit.

5 The absolute difference of my digits is a square.

6 I am an even number.

Who am I?

ANSWER: 48

RC 27 ROUND 3A: TRUE OR FALSE

Consider the function $y = -2\sin 3x$

1 The range of the function is $\{y: -2 \leq y \leq 2\}$
ANSWER: TRUE

2 The domain of the function is $-\pi \leq x \leq \pi$.
ANSWER: FALSE

3 The amplitude of the function is -2
ANSWER: FALSE

4 The period of the function is $T = 2\pi/3$
ANSWER: TRUE

RC 27 ROUND 3B: TRUE OR FALSE
For the curve $x^2/4 + y^2/9 = 1$,

1 the point (4, 0) is an x-intercept,
ANSWER: FALSE

2 the point (3, 0) is a y-intercept,
ANSWER: FALSE

3 the point (0, 3) is a y-intercept,
ANSWER: TRUE

4 the point (-2, 0) is an x-intercept.
ANSWER: TRUE

RC 28 ROUND 1A

A sequence is defined by the given recurrence relation. If $U_n = U_1$ for all positive integers n, find the value of U_1.

1 $U_{n+1} = 3U_n + 8$ ANSWER: $U_1 = -4$

$[U_1 = 3U_1 + 8, 2U_1 + 8 = 0, U_1 = -4]$

2 $U_{n+1} = 6 - 2U_n$ ANSWER: $U_1 = 2$

$[U_1 = 6 - 2U_1, 3U_1 = 6, U_1 = 2]$

3 $U_{n+1} = 5U_n - 16$ ANSWER: $U_1 = 4$

$[U_1 = 5U_1 - 16, 4U_1 = 16, U_1 = 4]$

4 $U_{n+1} = 5 - 4U_n$ ANSWER: $U_1 = 1$

$[U_1 = 5 - 4U_1, 5U_1 = 5, U_1 = 1]$

RC 28 ROUND 1B

Find the greatest value and the least value of the trigonometric expression.

1 $30/(3\cos x - 4\sin x + 10)$ ANSWER: greatest = 6, least = 2

$[30/(5\cos(x + \alpha) + 10)$, greatest $= 30/(-5 + 10) = 6$, least $= 30/(15) = 2]$

2 $28/(5\cos x + 12\sin x + 15)$ ANSWER: greatest = 14, least = 1

$[28/(13\cos(x + \alpha) + 15)$, greatest $= 28/(2) = 14$, least $= 28/(13 + 15) = 1]$

3 $15/(6\cos x - \sqrt{13}\sin x + 8)$ ANSWER: greatest = 15, least = 1

$[15/(7\cos(x + \alpha) + 8)$, greatest $= 15/(1) = 15$, least $= 15/(15) = 1]$

4 $36/(5\cos x + \sqrt{11}\sin x + 12)$ ANSWER: greatest = 6, least = 2

$[36/(6\cos(x + \alpha) + 12)$, greatest $= 36/6 = 6$, least $= 36/18 = 2]$

RC 28 ROUND 2: SPEED RACE

1 The radius of the wheel of a vehicle is 28 cm. If it travels a distance of 8.8 km, find the number of complete turns made by the wheel. Take $\pi = 22/7$.
ANSWER: 5,000
[$2\pi r = 2(22/7)28 = 176$ cm, $n = 8.8 \times 10^5/176 = 10^4/2 = 5{,}000$]

2 Find the length of an altitude of a triangle if its area is 24 cm^2 and the altitude is 2 cm shorter than the corresponding side.
ANSWER: 6 cm
[Area = $\frac{1}{2} h(h + 2) = 24$, $h^2 + 2h - 48 = (h + 8)(h - 6) = 0$, $h = 6$]

3 Solve for x and y from the equation $(7^{x+y})(3^{x-2y}) = 7 \times 81$
ANSWER: x = 2, y = -1
[$7^{x+y} = 7$, $x + y = 1$, $3^{x-2y} = 3^4$, $x - 2y = 4$, $3y = -3$, $y = -1$, $x - 1 = 1$, $x = 2$]

RC 28 ROUND 4: RIDDLE

1 I am a 3-digit odd number.

2 I am an exact square.

3 The number formed by my first 2 digits is an exact square.

4 I am between 300 and 400.

5 The sum of my digits is 10.

6 My last digit is an identity.
Who am I?
ANSWER: 361

RC 28 ROUND 3A: TRUE OR FALSE
In a circle, if 2 chords AB and CD intersect at E, then

1 AB x EB = CD x EC
ANSWER: FALSE

2 AE x BE = CE x CD
ANSWER: FALSE

3 AB x EA = CD x EC
ANSWER: FALSE

4 AE x BE = CE x DE
ANSWER: TRUE

RC 28 ROUND 3B: TRUE OR FALSE

1 A relation is a mapping.
ANSWER: FALSE

2 A many to one relation is a mapping.
ANSWER: TRUE

3 Some mappings are not relations.
ANSWER: FALSE

4 A many to many relation is a mapping.
ANSWER: FALSE

RC 29 ROUND 1A

A and B are events in a sample space S.

1 If $P(A) = 0.75$, $P(B) = 0.65$ and $P(A\cup B) = 0.90$, find $P(A\cap B)$

ANSWER: 0.50 $[P(A\cap B) = 0.75 + 0.65 - 0.90 = 1.40 - 0.90 = 0.50]$

2 If $P(A\cup B) = 0.85$, $P(A\cap B) = 0.65$ and $P(A) = 0.80$, find $P(B)$

ANSWER: 0.70 $[P(B) = P(A\cup B) + P(A\cap B) - P(A) = 0.85 + 0.65 - 0.80 = 0.70]$

3 If $P(A) = 0.74$, $P(B) = 0.65$ and $P(A\cap B) = 0.54$, find $P(A\cup B)$.

ANSWER: 0.85 $[P(A\cup B) = 0.74 + 0.65 - 0.54 = 1.39 - 0.54 = 0.85]$

4 If $P(B) = 0.72$, $P(A\cup B) = 0.87$, $P(A\cap B) = 0.58$, find $P(A)$.

ANSWER: 0.73 $[P(A) = 0.87 + 0.58 - 0.72 = 1.45 - 0.72 = 0.73]$

RC 29 ROUND 1B

Solve for x in the interval $0 < x < \pi/2$

1 $(\sin x)^{\cos x} = 1$ ANSWER: NO SOLUTION

[$\cos x = 0$, no solution in interval, $\sin x = 1$, no solution in the interval]

2 $(4\cos x - 1)^{(\tan x - 1)} = 1$ ANSWER: $x = \pi/4, \pi/3$

[$\tan x - 1 = 0$, $\tan x = 1$, $x = \pi/4$, $4\cos x - 1 = 1$, $4\cos x = 2$, $\cos x = ½$, $x = \pi/3$]

3 $(\tan x)^{(2\sin x - 1)} = 1$ ANSWER: $x = \pi/6, \pi/4$

[$2\sin x - 1 = 0$, $\sin x = ½$, $x = \pi/6$, $\tan x = 1$, $x = \pi/4$]

4 $(2\cos x - 1)^{\sin x - 1} = 1$ ANSWER: x = NO SOLUTION

[$\sin x - 1 = 0$, $\sin x = 1$, no solution, $2\cos x - 1 = 1$, $\cos x = 1$, no solution]

RC 29 ROUND 2: SPEED RACE

1 Evaluate the limit $\lim_{x \to -1} \frac{x^2 - 2x - 3}{x + 1}$

ANSWER; -4

[(x – 3)(x + 1)/(x +1) = x - 3 , limit = -1 – 3 = -4]

2 Expand $(1 + x - x^2)^5$ in ascending powers up to x^2

ANSWER: $1 + 5x + 5x^2$

$(1 + 5(x - x^2) + 10(x - x^2)^2 + \ldots = 1 + 5x + (10 - 5)x^2 = 1 + 5x + 5x^2]$

3 A curve has gradient $(3x^2 + ax - 5)$ and has a stationary point at (1, -2). Find the equation of the curve.

ANSWER: $y = x^3 + x^2 - 5x + 1$

$[y = x^3 + ax^2/2 - 5x + C$, $3 + a - 5 = 0$, $a = 2$, $y = x^3 + x^2 - 5x + C$,
$-2 = 1 + 1 - 5 + C$, $C = 1$, $y = x^3 + x^2 - 5x + 1]$

RC 29: ROUND 4: RIDDLE

1 I am a binary operation on the set of real numbers.

2 I am not one of the 2 basic binary operations on the real numbers.

3 I am not commutative.

4 Neither am I associative.

5 With polynomials I give rise to the Remainder Theorem.

6 I also give rise to quotients.

Who am I?

ANSWER: DIVISION

RC 29 ROUND 3A: TRUE OR FALSE

1 $x^3 - 64 = (x - 4)(x^2 - 4x + 16)$
ANSWER: FALSE

2 $x^3 - 64 = (x - 4)(x^2 - 4x - 16)$
ANSWER: FALSE

3 $x^3 + 64 = (x + 4)(x^2 - 4x - 16)$
ANSWER: FALSE

4 $x^3 + 64 = (x + 4)(x^2 - 4x + 16)$
ANSWER: TRUE

RC 29 ROUND 3B: TRUE OR FALSE
In a rhombus,

1 the diagonals are perpendicular or congruent,
ANSWER: TRUE

2 the diagonals bisect each other, and bisect the vertex angles,
ANSWER: TRUE

3 opposite angles are congruent, and adjacent angles are supplementary,
ANSWER: TRUE

4 there are 2 axes of symmetry, and the diagonals are congruent.
ANSWER: FALSE

RC 30 ROUND 1A

Find the equation of the circle $(x - 2)^2 + (y + 2)^2 = 9$ after a reflection

1 in the line $y = x$ ANSWER: $(x + 2)^2 + (y - 2)^2 = 9$

$[(x, y) \rightarrow (y, x), (y - 2)^2 + (x + 2)^2 = 9$, or $(x + 2)^2 + (y - 2)^2 = 9]$

2 in the origin ANSWER: $(x + 2)^2 + (y - 2)^2 = 9$

$[(x, y)\rightarrow(-x, -y), (-x - 2)^2 + (-y + 2)^2 = (x + 2)^2 + (y - 2)^2 = 9]$

3 in the x-axis ANSWER: $(x - 2)^2 + (y - 2)^2 = 9$

$[(x, y)\rightarrow(x, -y), (x - 2)^2 + (-y + 2)^2 = (x - 2)^2 + (y - 2)^2 = 9]$

4 in the y-axis ANSWER: $(x + 2)^2 + (y + 2)^2 = 9$

$[(x, y)\rightarrow(-x, y)\ (-x - 2)^2 + (y + 2)^2 = 9, (x + 2)^2 + (y + 2)^2 = 9]$

RC 30 ROUND 1B

Express the repeating decimal as a rational number in simplest form.

1 2.313131…. ANSWER: 229/99

$[x = 2.313131, 100x = 231.313131.., 99x = 229, x = 229/99]$

2 3.121212… ANSWER:103/33

$[x = 3.121212..., 100x = 312.1212..., 99x = 309/99 = 103/33]$

3 2.515151… ANSWER: 83/33

$[x = 2.515151..., 100x = 251.515151..., 99x = 249, x = 83/33]$

4 5.232323 . . . ANSWER: $x = 518/99$

$[x = 5.232323. . ., 100x = 523.23232, 99x = 518, x = 518/99]$

RC 30 ROUND 2: SPEED RACE

1 The sides of a triangle lie on the lines $x + y = 6$, $x - y = 4$, $x - 2y = 12$. .
Find the coordinates of the vertices of the triangle.
ANSWER: (5, 1), (-4, -8), (8, -2)
[$x + y = 6$, $x - y = 4$, $2x = 10$, $x = 5$, $y = 1$(5, 1), $x - y = 4$, $x - 2y = 12$, $y = -8$
$x = -4$, (-4, -8), $x + y = 6$, $x - 2y = 12$, $3y = -6$, $y = -2$, $x = 8$, (8 , -2)]

2 Simplify $6/(\sqrt{(x + 3)} - \sqrt{(x - 3)})$ by rationalizing the denominator .
ANSWER: $\sqrt{(x + 3)} + \sqrt{(x - 3)}$ [$6(\sqrt{(x + 3)} + \sqrt{(x - 3)})/((x + 3) - (x - 3)) = 6(\sqrt{(x + 3)} + \sqrt{(x - 3)})/6 = \sqrt{(x + 3)} + \sqrt{(x - 3)}$]

3 In a triangle find the length of an altitude if the altitude is 3 cm longer than the corresponding side and the area is 27 cm^2 ANSWER: 9 cm
[$(1/2)h(h - 3) = 27$, $h^2 - 3h - 54 = (h - 9)(h + 6) = 0$, $h = 9$ cm]

RC 30 ROUND 4: RIDDLE

1 I am a physical quantity.

2 I am a vector quantity.

3 I am used in defining a force.

4 Force is my rate of change with respect to time.

5 In collision problems, my sum is always conserved.

6 Normally for a particle, I am proportional to the velocity.
Who am I?
ANSWER: LINEAR MOMENTUM (accept MOMENTUM)

RC 30 ROUND 3A: TRUE OR FALSE

1 All isosceles triangles are similar.
ANSWER: FALSE

2 All equilateral triangles are similar.
ANSWER: TRUE

3 All rectangles are similar.
ANSWER: FALSE

4 All regular polygons are similar.
ANSWER: FALSE

RC 30 ROUND 3B: TRUE OR FALSE

1 $\int x^{-1/2} dx = x^{1/2} + C$
ANSWER: FALSE

2 $\int x^{1/2} dx = (3/2)x^{3/2} + C$
ANSWER: FALSE

3 $\int (3/2)x^{1/2} dx = x^{3/2} + C$
ANSWER: TRUE

4 $\int x^{3/2} dx = (2/5)x^{5/2} + C$
ANSWER: TRUE

RC 31 ROUND 1A

Find the remainder when the polynomial $x^3 - x^2 - 5x + 6$ is divided by

1 $(x^2 - x - 2)$ ANSWER: **$(-3x + 6)$** [$x^3 - x^2 - 5x + 6 = (x - 2)(x + 1)q(x) + ax + b$, for $x = -1$, $9 = -a + b$, for $x = 2$, $0 = 2a + b$, $3a = -9$, $a = -3$, $b = 6$]

2 $x^2 - 4$ ANSWER: **$(-x + 2)$** [$x^3 - x^2 - 5x + 6 = (x - 2)(x + 2)q(x) + ax + b$, for $x = 2$, $2a + b = 0$, for $x = -2$, $4 = -2a + b$, $2b = 4$, $b = 2$, $a = -1$]

3 $(x^2 - 3x + 2)$ ANSWER: **$(-x + 2)$** [$x^3 - x^2 - 5x + 6 = (x - 2)(x - 1)q(x) + ax + b$, for $x = 1$, $a + b = 1$, for $x = 2$, $2a + b = 0$, $a = -1$, $b = 2$]

4 $(x^2 + x - 2)$, ANSWER: **$(-x + 2)$** [$x^3 - x^2 - 5x + 6 = (x - 1)(x + 2)q(x) + ax + b$, for $x = 1$, $1 = a + b$, for $x = -2$, $4 = -2a + b$, $3a = -3$, $a = -1$, $b = 2$]

RC 31 ROUND 1B

Find the coordinates of the point D(x, y) such that ABCD is a parallelogram given

1 A(1, 3), B(-2, 5) and C(3, -2) ANSWER: **D(6, - 4)**
[$\overrightarrow{AB} = \overrightarrow{DC}$, $(-3, 2) = (3 - x, -2 - y)$, $3 - x = -3$, $x = 6$, $-2 - y = 2$, $y = -4$, D(6, -4)]

2 A(3, -2), B(-1, 4), C(5, - 3) ANSWER: **D(9, -9)**
[$\overrightarrow{AB} = \overrightarrow{DC}$, $(-4, 6) = (5 - x, -3 - y)$, $5 - x = -4$, $x = 9$, $-3 - y = 6$, $y = -9$, D(9, -9)]

3 A(-2, 4), B(3, 1) and C(4, -4) ANSWER: **D(-1, -1)**
[$\overrightarrow{AB} = \overrightarrow{DC}$, $(5, -3) = (4 - x, -4 - y)$, $4 - x = 5$, $x = -1$, $-4 - y = -3$, $y = -1$, D(-1, -1)]

4 A(2, 5), B(-2, 4), C(3, - 4) ANSWER: **D(7, -3)**
[$\overrightarrow{AB} = \overrightarrow{DC}$, $(-4, -1) = (3 - x, -4 - y)$, $3 - x = -4$, $x = 7$, $-4 - y = -1$, $y = -3$, D(7, -3)]

RC 31 ROUND 2: SPEED RACE

1 Solve for x given $\log_2(2x + 1) = \log_2(x - 1) + 2$
ANSWER: $x = 5/2$
[$\log_2(2x + 1)/(x - 1) = 2$, $(2x + 1)= 4(x - 1)$, $2x = 5$, $x = 5/2$]

2 Find the gradient of the implicit curve $x^2 + 2xy - y^2 = 15$
ANSWER: $dy/dx = (x + y)/(y - x)$
[$2x + 2(y + x(dy/dx) - 2y(dy/dx) = 0$, $(dy/dx)(2y - 2x) = 2x + 2y$,
$dy/dx = (x + y)/(y - x)$]

3 Evaluate $(\sin47°\cos15° - \cos47°\sin15°)/\cos58°$
ANSWER: 1
[$\sin(47 - 15)/\cos58 = \sin32/\sin32 = 1$]

RC 31 ROUND 4: RIDDLE
1 I am a 2-digit number.

2 My digits are powers of the same prime number.

3 The sum of my digits is a dozen.

4 My digits are a square and a cube.

5 My last digit is a square.

6 I am an even number.
Who am I?
ANSWER: 84

RC 31 ROUND 3A: TRUE OR FALSE
If $a/b = c/d$, then

1$(a + b)/(c + d) = a/b$
ANSWER: FALSE

2 $(a + c)/(b + d) = c/d$
ANSWER: TRUE

3 $(a - c)/(b - d) = a/b$
ANSWER: TRUE

4 $ad/bc = 1$
ANSWER: TRUE

RC 31 ROUND 3B: TRUE OR FALSE

1 A parallelogram has no axis of symmetry.
ANSWER: TRUE

2 A parallelogram has no center of symmetry.
ANSWER: FALSE

3 The order of rotational symmetry of a parallelogram is 1.
ANSWER: FALSE

4 A parallelogram has a fixed center of rotation.
ANSWER: TRUE

RC 32 ROUND 1A

Factorise completely

1...$(2x + y - 2)^2 - (x - 2y + 2)^2$ ANSWER: $(3x - y)(x + 3y - 4)$

$[(3x - y)(x + 3y - 4)]$

2 $(x + 2y + 5)^2 - (2x + y - 5)^2$ ANSWER: $3(x + y)(y - x + 10)$

$[(3x + 3y)(y - x + 10) = 3(x + y)(y - x + 10)]$

3 $(3x - 2y + 3)^2 - (2x - 3y - 3)^2$ ANSWER: $5(x - y)(x + y + 6)$

$[(5x - 5y)(x + y + 6) = 5(x - y)(x + y + 6)]$

4 $(5x - 4y - 6)^2 - (4x + 5y + 6)^2$ ANSWER: $(9x + y)(x - 9y - 12)$

$[(9x + y)(x - 9y - 12)]$ (NB: $a^2 - b^2 = (a + b)(a - b)$)

RC 32 ROUND 1B

Determine the constant term in the expansion of

1 $(x^2 + 2/x)^6$ ANSWER: 240

$[6C_r (x^2)^r(2/x)^{6 - r} = 6C_r(2)^{6 - r}(x^{3r - 6})$, $3r - 6 = 0$, $r = 2$, $6C_2 2^4 = 15(16) = 240]$

2 $(x^3 + 3/x^2)^5$ ANSWER: 270

$[5C_r(x^3)^r(3/x^2)^{5 - r} = 5C_r(3)^{5 - r}(x^{5r - 10})$, $5r - 10 = 0$, $r = 2$, $10(3^3) = 270]$

3 $(x^2 + 4/x^2)^4$ ANSWER: 96

$[4C_r(x^2)^r(4/x^2)^{4 - r} = 4C_r(x^{4r - 8})(4^{4 - r})$, $4r - 8 = 0$, $r = 2$, $4C_2(4^2) = 6(16) = 96]$

4 $(x^2 + 1/x)^9$ ANSWER: 84

$[9C_r(x^2)^r(1/x)^{9 - r} = 9C_r x^{3r - 9}$, $3r - 9 = 0$, $r = 3$, $9C_3 = 9(8)7/6 = 12 \times 7 = 84]$

RC 32 ROUND: 2 SPEED RACE

1 Find the radius R and the coordinates of the center C of the circle
$x^2 + y^2 - 4x + 8y = 16$
ANSWER: R = 6, C(2, -4)
$[(x - 2)^2 + (y + 4)^2 = 4 + 16 + 16 = 36$, R = 6, C(2, -4)]

2 The volume of a sphere is increasing at a constant rate of 40π cm^3/s. Find the rate of increase of the radius R when the radius R = 10 cm
ANSWER: (1/10) cm/s, or 0.1 cm/s
$V = 4\pi R^3/3$, $dV/dt = 4\pi R^2 dR/dt = 40\pi$, $dR/dt = 40\pi/4\pi 100 = 1/10$ cm/s]

3 Evaluate $\lim_{x \to -2} \frac{x^3 + 8}{x^2 - 4}$ ANSWER: -3
$[(x + 2)(x^2 - 2x + 4)/(x + 2)(x - 2) = (x^2 - 2x + 4)/(x - 2)$,
limit = (4 + 4 + 4)/-4 = - 3]

RC 32 ROUND 4: RIDDLE

1 I am an operation.

2 I obtain a statement from a given statement.

3 My effect is contrary to the original statement.

4 I am noted for my negativity.

5 I am not a binary operation.

6 The statement I produce and the original statement are at loggerheads.
Who am I?
ANSWER: NEGATION OF A STATEMENT (accept NEGATION)

RC 32 ROUND 3A: TRUE OR FALSE
As x increases in the interval $90° < x < 180°$, the function

1 $y = \sin x$ increases,
ANSWER: FALSE

2 $y = \cos x$ decreases,
ANSWER: TRUE

3 $y = \tan x$ increases,
ANSWER: TRUE

4 $y = \operatorname{cosec} x$ increases.
ANSWER: TRUE

RC 32 ROUND 3B: TRUE OR FALSE
If a set A maps onto a set B, then

1 some elements of B are not images of elements of A,
ANSWER: FALSE

2 the mapping is one to one,
ANSWER: FALSE

3 some elements in A are not elements of B,
ANSWER: FALSE

4 every element of B is the image of at least one element of A.
ANSWER: TRUE

RC 33 ROUND 1A

Given that $f(x) = \sqrt{(x-1)}$ and $g(x) = x^2$, find an expression and a domain for

1 $f(g(x))$ **ANSWER: $\sqrt{(x^2 - 1)}$, $\{x: x \geq 1\}$**

[$f(g(x)) = \sqrt{(g(x) - 1)} = \sqrt{(x^2 - 1)}$, Domain = $\{x: x \geq 1\}$

2 $g(f(x))$ **ANSWER: $x - 1$, $\{x: x \geq 1\}$**

[$g(f(x)) = (f(x))^2 = (\sqrt{(x - 1)})^2 = x - 1$, domain = $\{x: x \geq 1\}$

3 $f(1/g(x))$ **ANSWER: $\sqrt{(1 - x^2)}/x$ domain = $\{x: -1 \leq x \leq 1, x \neq 0\}$**

[$f(1/g(x)) = f(1/x^2) = \sqrt{(1/x^2 - 1)} = \sqrt{(1 - x^2)}/x$, D = $\{x: -1 \leq x \leq 1, x \neq 0\}$]

4 $g(1/f(x))$ **ANSWER: $1/(x - 1)$ domain = $\{x: x > 1\}$**

[$g(1/\sqrt{(x - 1)}) = 1/(x - 1)$, domain = $\{x: x > 1\}$]

RC 33 ROUND 1B

Find a cartesian equation in x and y from the given set of equations.

1 $x = 3\cos t + 4$, $y = 4\sin t + 3$ **ANSWER: $(x - 4)^2/9 + (y - 3)^2/16 = 1$**

[$\cos t = (x - 4)/3$, $\sin t = (y - 3)/4$, $(x - 4)^2/9 + (y - 3)^2/16 = 1$]

2 $x = 2\sin t + 5$, $y = 5\cos t - 4$ **ANSWER: $(x - 5)^2/4 + (y + 4)^2/25 = 1$**

[$\sin t = (x - 5)/2$, $\cos t = (y + 4)/5$, $(x - 5)^2/4 + (y + 4)^2/25 = 1$]

3 $x = 4\cos t - 3$, $y = 3\sin t - 5$ **ANSWER: $(x + 3)^2/16 + (y + 5)^2/9 = 1$**

[$\cos t = (x + 3)/4$, $\sin t = (y + 5)/3$, $(x + 3)^2/16 + (y + 5)^2/9 = 1$]

4 $x = 5\sin t - 3$, $y = 4\cos t - 6$ **ANSWER: $(x + 3)^2/25 + (y + 6)^2/16 = 1$**

[$\sin t = (x + 3)/5$, $\cos t = (y + 6)/4$, $(x + 3)^2/25 + (y + 6)^2/16 = 1$]

RC 33 ROUND 2: SPEED RACE

1 Evaluate the integral $\int_1^2 (3x^2 - 4x + 3)dx$

ANSWER: 6

$[x^3 - 2x^2 + 5x|_1^2 = (8 - 8 + 10) - (1 - 2 + 5) = 10 - 4 = 6]$

2 Find the equation of a line through the point A(2, -5) and perpendicular to the line $2x + 5y = 8$

ANSWER: $5x - 2y = 20$

$[5x - 2y = k = 10 + 10, 5x - 2y = 20]$

3 A circle has center C(-5, 3) and radius 5. P(x, y) is a point inside the circle. State the condition satisfied by the coordinates of P.

ANSWER: $(x + 5)^2 + (y - 3)^2 < 25$

RC 33 ROUND 4: RIDDLE

1 I am a function associated with a given function.

2 Not every function can have me.

3 I literally reverse a function.

4 The composite of a given function and me is an identity function.

5 A linear function has me.

6 A quadratic function generally does not have me.

Who am I?

ANSWER: INVERSE FUNCTION

RC 33 ROUND 3A: TRUE OR FALSE

1 A sheared rectangle is a parallelogram.
ANSWER: TRUE

2 A sheared square is a rhombus.
ANSWER: TRUE

3 A rectangle can be sheared into a square.
ANSWER: FALSE

4 A sheared isosceles triangle is an equilateral triangle.
ANSWER: FALSE

RC 33 ROUND 3B: TRUE OR FALSE

If P(x, y) is a point in the cartesian plane, then under a reflection in

1 the origin, $(x, y) \rightarrow (-y, -x)$
ANSWER: FALSE

2 the x-axis, $(x, y) \rightarrow (x, -y)$
ANSWER: TRUE

3 the y-axis, $(x, y) \rightarrow (-x, y)$
ANSWER: TRUE

4 the line $y = x$, $(x, y) \rightarrow (-y, x)$
ANSWER: FALSE

RC 34 ROUND 1A

Find the equation of the locus of the point P(x, y) if it moves in the x-y plane such that it is equidistant from the given point and the given axis,
(Express answer in the form $y^2 = \ldots$ or $x^2 = \ldots$)

1 the point A(5, 0) and the y-axis **ANSWER: $y^2 = 10x - 25$**
[$(x - 5)^2 + y^2 = x^2$, $x^2 - 10x + 25 + y^2 = x^2$, $y^2 - 10x + 25 = 0$, $y^2 = 10x - 25$]

2 the point B(0, 6) and the x-axis, **ANSWER: $x^2 = 12y - 36$**
[$x^2 + (y - 6)^2 = y^2$, $x^2 + y^2 - 12y + 36 = y^2$, $x^2 = 12y - 36$]

3 the point C(-4, 0) and the y-axis **ANSWER: $y^2 = - 8x - 16$**
[$(x + 4)^2 + y^2 = x^2$, $x^2 + 8x + 16 + y^2 = x^2$, $y^2 = - 8x - 16$]

4 the point D(0, 5) and the x-axis **ANSWER: $x^2 = 10y - 25$**
[$x^2 + (y - 5)^2 = y^2$, $x^2 - 10y + 25 = 0$, $x^2 = 10y - 25$]

RC 34 ROUND 1B

Find the remainder $ax + b$ when the polynomial $f(x) = x^3 - 3x^2 + 2x + 5$ is divided by,

1 $(x - 2)(x + 1)$ **ANSWER: $2x + 1$** [$f(x) = (x - 2)(x + 1)q(x) + ax + b$, $f(2) = 5 = 2a + b$, $f(-1) = -1 = -a + b$, $2a + b = 5$, $3a = 6$, $a = 2$, $b = 1$]

2 $(x + 2)(x - 1)$ **ANSWER: $8x - 3$** [$f(x) = (x + 2)(x - 1)q(x) + ax + b$, $f(1) = 5 = a + b$, $f(-2) = -8 - 12 - 4 + 5= -19 = -2a + b$, $3a = 24$, $a = 8$, $b = -3$]

3 $(x - 1)(x + 1)$ **ANSWER: $3x + 2$** [$f(x) = (x - 1)(x + 1)q(x) + ax + b$, $f(1) = 5, = a + b$, $f(-1) = -1 -3 - 2 + 5 = -1 = -a + b$, $2a = 6$, $a = 3$, $b = 2$]

4 $(x - 2)(x + 2)$ **ANSWER: $6x - 7$** [$f(x) = (x -2)(x +2)q(x) + ax + b$,[$f(2) = 5 = 2a + b$, $f(-2) = -8 -12 - 4 +5= -19 = -2a + b$, $2b = -14$, $b = -7$, $a = 6$]
[NB: The remainder can also be obtained by long division]

RC 34 ROUND 2: SPEED RACE

1 Find the coordinates of the final image point after the point A(-2, 3) is reflected in the origin and the image reflected in the point P(2, -1)
ANSWER: (2, 1)
[(-2, 3)→(2, -3)→(4 – 2, -2 –(-3)) = (2, 1)]

2 Solve for x given $4^{2x} = 3^{3x-1}$
ANSWER: x = log3/(3log3 – 2log4), or log3/(3log3 – 4log2)
[2xlog4 = (3x – 1)log3, x(3log3 – 2log4) = log3, x = log3/(3log3 – 2log4)]

3 Find the stationary point of the curve y = (x + 1)/x
ANSWER: NO STATIONARY POINT
[y = 1 + 1/x, dy/dx = - $1/x^2 \neq 0$, no stationary point]

RC 34 ROUND 4: RIDDLE

1 I am a 2-digit odd number.

2 When my digits are reversed, the number formed is an even number.

3 My 2 digits are prime numbers.

4 The sum of my digits is an odd square digit.

5 I am an exact cube.

6 My digits differ by 5.
Who am I?
ANSWER: 27

RC 34 ROUND 3A: TRUE OR FALSE

1 For an acute triangle, the circum-center is inside the triangle.
ANSWER: TRUE

2 For a right-angle triangle, the circum-center is outside the triangle.
ANSWER: FALSE

3 For a right-angle triangle, the circum-center is inside the triangle.
ANSWER: FALSE

4 For an obtuse triangle, the circum-center is outside the triangle.
ANSWER: TRUE

RC 34 ROUND 3B: TRUE OR FALSE

1 For any real number a, $y = a^x$ is an exponential function
ANSWER: FALSE (only if $a \neq 1$)

2 The exponential function $y = a^x$ is an increasing function.
ANSWER: FALSE

3 The exponential function is a decreasing function.
ANSWER: FALSE

4 The inverse of $y = a^x$, is $y = \log_a x$.
ANSWER: TRUE

RC 35 ROUND 1A

Use implicit differentiation to find the gradient (dy/dx) for the given curve.

1 $2x^3 + 3x^2y^2 - 2y^3 = 17$ **ANSWER: $(dy/dx) = (x^2 + xy^2)/(y^2 - x^2y)$**
$[6x^2 + 6xy^2 + 6x^2y(dy/dx) - 6y^2(dy/dx) = 0, (dy/dx) = (x^2 + xy^2)/(y^2 - x^2y)]$

2 $x^4 - x^3y^2 + y^3 = 19$ **ANSWER: $(dy/dx) = (4x^3 - 3x^2y^2)/(2x^3y - 3y^2)$**
$[4x^3 - 3x^2y^2 - 2x^3y dy/dx + 3y^2 dy/dx = 0, dy/dx = (4x^3 - 3x^2y^2)/(2x^3y - 3y^2)]$

3 $x^4 + 2x^2y^2 - 4y^3 = 15$ **ANSWER: $(dy/dx) = (x^3 + xy^2)/(3y^2 - x^2y)$**
$[4x^3 + 4xy^2 + 4x^2y dy/dx - 12y^2 dy/dx = 0, (dy/dx) = (x^3 + xy^2)/(3y^2 - x^2y)]$

4 $x^3 - 3xy^2 + y^3 = 19$ **ANSWER: $dy/dx = (x^2 - y^2)/(2xy - y^2)$**
$[3x^2 - 3y^2 - 6xy dy/dx + 3y^2 dy/dx = 0, dy/dx = (x^2 - y^2)/(2xy - y^2)]$

RC 35 ROUND 1B

Expand the given expression in ascending powers of x up to x^3.

1 $(1 + x + x^2)^5$ **ANSWER: $1 + 5x + 15x^2 + 30x^3$** $[1 + 5(x + x^2) + 10(x + x^2)^2 + 10(x + x^2)^3 = 1 + 5x + 15x^2 + 30x^3]$

2 $(1 + x - x^2)^6$ **ANSWER: $1 + 6x + 9x^2 - 10x^3$** $[1 + 6(x - x^2) + 15(x - x^2)^2 + 20(x - x^2)^3 = 1 + 6x + (-6 + 15)x^2 + (15(-2) + 20)x^3 = 1 + 6x + 9x^2 - 10x^3]$

3 $(1 - x + x^2)^5$ **ANSWER: $1 - 5x + 15x^2 - 30x^3$** $[1 + 5(-x + x^2) + 10(-x + x^2)^2 + 10(-x + x^2)^3 = 1 - 5x + (5 + 10)x^2 + (10(-2) - 10)x^3 = 1 - 5x + 15x^2 - 30x^3]$

4 $(1 - x - x^2)^4$ **ANSWER: $1 - 4x + 2x^2 + 8x^3$** $[1 - 4(x + x^2) + 6(x + x^2)^2 - 4(x + x^2)^3 = 1 - 4x + x^2(-4 + 6) + x^3(12 - 4) = 1 - 4x + 2x^2 + 8x^3]$

RC 35 ROUND 2: SPEED RACE

1 The points A(2, 3) and B(-4, - 5) are at the ends of a diameter of a circle. Find the equation of the circle.
ANSWER: $(x + 1)^2 + (y + 1)^2 = 25$
[center C(-1, -1), $R^2 = 3^2 + 4^2 = 25$, $(x + 1)^2 + (y + 1)^2 = 25$]

2 If $f(x) = x^2 – 3x + 2$, and $g(x) = 2x – 1$, evaluate $f(g(2))$
ANSWER: 2
[$g(2) = 3$, $f(3) = 9 – 9 + 2 = 2$]

3 Solve $\tan^2 x = 1$ for x in the interval $0 < x < \pi$
ANSWER: $x =\pi/4, 3\pi/4$
[$\tan x = \pm 1$, $\tan x = 1$, $x = \pi/4$, $\tan x = -1$, $x = 3\pi/4$]

RC 35 ROUND 4: RIDDLE

1 I am a point of concurrency of a triangle.

2 I am a point of intersection of some 3 lines associated with the triangle.

3 For an acute triangle, I am inside the triangle.

4 For an obtuse triangle, I am outside the triangle.

5 For a right-angle triangle, I am at the vertex of the right angle.

6 I am not associated with any circle concerning the triangle.
Who am I?
ANSWER: ORTHO-CENTER

RC 35 ROUND 3A: TRUE OR FALSE

1 $5^{-3} < 5^{-4}$
ANSWER: FALSE

2 $4^3 > 3^4$
ANSWER: FALSE

3 $5^3 < 3^5$
ANSWER: TRUE

4 $4^{-2} > 2^{-4}$
ANSWER: FALSE

RC 35 ROUND 3B: TRUE OR FALSE

1 The degree of the expression $x^3 + y^4 – 3x^2y^3$ is 6.
ANSWER: FALSE

2 The degree of the expression $3x^3 + 2x^2y^2 – xy^4$ is 5
ANSWER: TRUE

3 The degree of the expression $x^3(x^2 + y^4)$ is 9.
ANSWER: FALSE

4 The degree of the expression $(x^2 + y)^3$ is 6
ANSWER: TRUE

RC 36 ROUND 1A

The functions f, g and h are defined on the set R of real numbers by $f(x) = 2x + 1$, $g(x) = x^2 + 1$ and $h(x) = 1/(x^2 + 1)$. Find

1 $f(g(x))$ ANSWER: $2x^2 + 3$
[$f(g(x)) = 2g(x) + 1 = 2(x^2 + 1) + 1 = 2x^2 + 3$]

2 $g(f(x))$ ANSWER: $(2x + 1)^2 + 1$, or $4x^2 + 4x + 2$
[$g(f(x)) = f(x)^2 + 1 = (2x + 1)^2 + 1$]

3 $h(f(x))$ ANSWER: $1/((2x + 1)^2 + 1)$ or $1/(4x^2 + 4x + 2)$
[$h(f(x)) = 1/(f(x)^2 + 1) = 1/((2x + 1)^2 + 1) = 1/(4x^2 + 4x + 2)$

4 $h(g(x))$ ANSWER: $1/((x^2 + 1)^2 + 1)$, or $1/(x^4 + 2x^2 + 2)$
[$h(g(x)) = 1/(g(x)^2 + 1) = 1/((x^2 + 1)^2 + 1) = 1/(x^4 + 2x^2 + 2)$]

RC 36 ROUND 1B

Find the values of the constants a and b such that the inequality

1 $ax^2 + bx + 3 < 0$ has solution set $\{x: x > 1, \text{ or } x < -3\}$ANSWER: $a = -1, b = 2,$
[$(x – 1)(x + 3) > 0$, $x^2 – 2x – 3 > 0$, $-x^2 + 2x + 3 < 0$, $a = -1$, $b = 2$]

2 $ax^2 + bx + 10 > 0$ has solution set $\{x: -2 < x < 5\}$ ANSWER: $a = -1, b = 3$
[$(x + 2)(x – 5) = x^2 – 3x – 10 < 0$, $-x^2 + 3x + 10 > 0$, $a = -1$, $b = 3$]

3 $ax^2 + bx - 15 < 0$ has solution set $\{x: x > 5, \text{ or } x < 3\}$ ANSWER: $a = -1, b = 8$
[$(x – 5)(x – 3) > 0$, $x^2 – 8x + 15 > 0$, $-x^2 + 8x – 15 < 0$, $a = -1$, $b = 8$]

4 $ax^2 + bx + 10 > 0$ has solution set $\{x: -2 < x < 5\}$ ANSWER: $a = -1, b = 3$
[$(x + 2)(x – 5) = x^2 – 3x - 10 < 0$, $-x^2 + 3x + 10 > 0$, $a = -1$, $b = 3$]

RC 36 ROUND 2: SPEED RACE

1 Express $\log(x^3y^2/z^4) - \log(x^2y^3/z^2)$ as a single logarithm in simplest form.
ANSWER: $\log(x/yz^2)$
[$\log(x^3y^2z^2/x^2y^3z^4) = \log(x/yz^2)$]

2 Find dy/dx from the implicit relation $x^3 - 3xy^2 + y^3 = 10$
ANSWER: $dy/dx = (x^2 - y^2)/(2xy - y^2)$
[$3x^2 - 3y^2 - 6xydy/dx + 3y^2dy/dx = 0$, $dy/dx(6xy - 3y^2) = 3x^2 - 3y^2$,
$dy/dx = (x^2 - y^2)/(2xy - y^2)$]

3 Factorize completely $8x^3 + 27y^3$
ANSWER: $(2x + 3y)(4x^2 - 6xy + 9y^2)$
[$(2x)^3 + (3y)^3 = (2x + 3y)(4x^2 - 6xy + 9y^2)$]

RC 36 ROUND 4: RIDDLE

1 I am a polygon.

2 I am a quadrilateral.

3 I have 2 pairs of congruent adjacent angles.

4 I have a pair of congruent opposite sides.

5 I have a pair of parallel opposite sides.

6 My congruent sides are not parallel.
Who am I?
ANSWER: ISOSCELES TRAPEZIUM

RC 36 ROUND 3A: TRUE OR FALSE

At a point of inflexion, the tangent

1 is above the curve,
ANSWER: FALSE

2 is below the curve,
ANSWER: FALSE

3 is either horizontal or vertical,
ANSWER: FALSE

4 cuts the curve.
ANSWER: TRUE

RC 36 ROUND 3B: TRUE OR FALSE
In the cartesian plane, the transformation

1 $(x, y)\rightarrow(x, -y)$ is a reflection in the x- axis.
ANSWER: TRUE

2 $(x, y)\rightarrow(-x, -y)$ is a reflection in the line $y = -x$.
ANSWER: FALSE

3 $(x, y)\rightarrow(2 - x, y - 2)$ is a translation.
ANSWER: FALSE

4 $(x + 2, y - 2)$ is a translation.
ANSWER: TRUE

RC 37 ROUND 1A

Find the common ratio of an exponential sequence if the following are successive terms.

1 $x + 2, x + 5, x + 7$ **ANSWER: 2/3**

[$(x + 2)(x + 7) = (x + 5)^2$, $9x + 14 = 10x + 25$, $x = -11$, $r = -6/-9 = 2/3$]

2 $x, x - 3, x - 7$ **ANSWER: 4/3**

[$(x - 3)^2 = x(x - 7)$, $-6x + 9 = -7x$, $x = -9$, $r = -12/-9 = 4/3$]

3 $x - 2, x + 4, x + 6$ **ANSWER: 1/3**

[$(x + 4)^2 = (x - 2)(x + 6)$, $8x + 16 = 4x - 12$, $4x = -28$, $x = -7$, $r = -3/-9 = 1/3$]

4 $x + 1, x + 3, x + 7$ **ANSWER: 2**

[$(x + 3)^2 = (x + 1)(x + 7)$, $6x + 9 = 8x + 7$, $2x = 2$, $x = 1$, $r = 4/2= 2$]

RC 37 ROUND 1B

Find constants A, B and C such that

1 $8 + 6x - x^2 = A(x + B)^2 + C$ **ANSWER: A = -1, B = -3, C = 17**

[$-1(x^2 - 6x) + 8 = -1(x - 3)^2 + 9 + 8 = -1(x - 3)^2 + 17$, A = -1, B = -3, C = 17]

2 $15 - 10x - x^2 = A(x + B)^2 + C$ **ANSWER: A = -1, B = 5, C = 40**

[$-1(x^2 + 10x) + 15 = -1(x + 5)^2 + 25 + 15$, A = -1, B = 5, C = 40]

3 $7 + 8x - x^2 = A(x + B)^2 + C$ **ANSWER: A = -1, B = 4, C = 23**

[$-1(x^2 - 8x) + 7 = -1(x - 4)^2 +16 + 7= -1(x - 4)^2 + 23$, A = -1, B = -4, C =23]

4 $3 - 12x - x^2 = A(x + B)^2 + C$ **ANSWER: A = -1, B = 6, C = 39**

[$-1(x^2 + 12x) + 3 = -1(x + 6)^2 + 3 + 36$, A = -1, B = 6, C = 39]

RC 37 ROUND 2: SPEED RACE

1 Evaluate $\displaystyle \lim_{x \to -3} \frac{x^2 + 2x - 3}{x + 3}$

ANSWER: - 4

[$(x^2 + 2x - 3)/(x + 3) = (x + 3)(x - 1)/(x + 3) = x - 1$, limit = -3 - 1 = -4]

2 Evaluate and simplify $(\tan 45° - \tan 30°)/(1 + \tan 45° \tan 30°)$

ANSWER: $2 - \sqrt{3}$

[$(1 - 1/\sqrt{3})(1 + 1/\sqrt{3}) = (\sqrt{3} - 1)/(\sqrt{3} + 1) = (\sqrt{3} - 1)^2/2 = (4 - 2\sqrt{3})/2 = 2 - \sqrt{3}$]

3 Find the coordinates of the point of inflexion of the curve,
$y = x^3 + 3x^2 + 3x + 5$

ANSWER: (-1, 4) [$dy/dx = 3x^2 + 6x + 3$, $d^2y/dx^2 = 6x + 6 = 0$, $x = -1$, $y = -1 + 3 - 3 + 5 = 8 - 4 = 4$, (-1, 4)]

RC 37 ROUND 4: RIDDLE

1 I am a 3 digit even number.

2 My first digit is an unusual prime.

3 I am an exact square.

4 I am a square of a square.

5 The number formed by my first 2 digits is a square.

6 The sum of my digits is 13.
Who am I?

ANSWER: 256

RC 37 ROUND 3A: TRUE OR FALSE

1 Zero is an even number.
ANSWER: TRUE

2 The product of two odd numbers is an odd number.
ANSWER: TRUE

3 The difference of 2 odd numbers is an odd number.
ANSWER: FALSE

4 The sum of an odd number and an even number is an odd number.
ANSWER: TRUE

RC 37 ROUND 3B: TRUE OR FALSE
The graph of the curve

1 $y = (x + 3)^2 - 5$ has vertex (3, -5)
ANSWER: FALSE

2 $y = (x - 4)^2 + 3$ has vertex (4, 3)
ANSWER: TRUE

3 $y = -(x + 5)^2 - 6$ has vertex (-5, -6)
ANSWER: TRUE

4 $y = 2(x - 3)^2 + 7$ has vertex (3, 7)
ANSWER: TRUE

RC 38 ROUND 1A

Find the coordinates of the final image point after the point,

1 A(-2, 3) is reflected in the origin and the image reflected in the point P(2, -1)
ANSWER: (2, 1) [(-2, 3)→(2, -3)→(4 – 2, -2 –(-3)) = (2, 1)]

2 B(4, 5) is reflected in the x-axis and the image reflected in the point P(3, 2),
ANSWER: (2, 9) [(4, 5)→(4, -5)→(6 – 4, 4 – (-5)) = (2, 9)]

3 C(3, - 5) is reflected in the line y = x, and the image reflected in the point P(4, 3).
ANSWER: (13, 3) [(3, -5)→(-5, 3)→(8 – (-5), 6 – 3) = (13, 3)]

4 D(4, -5) is reflected in the x-axis and the image reflected in the point P(3, -2)
ANSWER: (2, -9) [((4, -5)→(4, 5)→(6 – 4, -4 - 5) = (2, -9)]

RC 38 ROUND 1B

Two chords AB and CD of a circle intersect at a point E inside the circle. Find

1 the length of BE given AE = 10 cm, CE = 15 cm and DE = 8 cm,
ANSWER: 12 cm [AE x BE = CE x DE, BE = 15 x 8/10 = 12]

2 the length of DE given AE = 9 cm, BE= 12 cm, CE = 18 cm,
ANSWER: 6 cm [AE x BE = CE x DE, DE = 9 x 12/18 = 6]

3 the length of AE, given CE = 8 cm, DE = 12 cm, and BE = 6 cm,
ANSWER: 16 cm [AE x BE = CE x DE, AE = 8 x 12/6 = 16]

4 the length of CE, given AE = 5 cm, BE = 8 cm, and DE = 10 cm,
ANSWER: 4 cm [AE x BE = CE x DE, CE = 5 x 8/10 = 4]

RC 38 ROUND 2: SPEED RACE

1 Triangle ABC has sides of lengths a = 12 cm, b = 8 cm and c = 10 cm. Find cosB

ANSWER: 3/4

[$\cos B = (a^2 + c^2 - b^2)/2ac = (144 + 100 - 64)/2(12)10 = 180/2(120) = 3/4$]

2 Given $y = (x^2 - 4)/(x^2 + 4)$, find dy/dx.

ANSWER: $dy/dx = 16x/(x^2 + 4)^2$

[$dy/dx = (2x(x^2 + 4) - 2x(x^2 - 4))/(x^2 + 4)^2 = 16x/(x^2 + 4)^2$]

3 Solve for x given $\log_3(2x - 2) = \log_3(x - 4) + 1$

ANSWER: $x = 10$

[$2x - 2 = 3(x - 4)$, $x = 12 - 2 = 10$]

RC 38 ROUND 4: RIDDLE

1 I am a rigid body motion.

2 I preserve distance as well as angles.

3 I am in general not a linear transformation.

4 I am characterized by a fixed point.

5 I may be clockwise or anti-clockwise.

6 I have something to do with a bearing.
Who am I?

ANSWER: ROTATION

RC 38 ROUND 3A: TRUE OR FALSE

1 An arc of a circle is a part of the circle.
ANSWER: TRUE

2 A sector of a circle is a part of the circle.
ANSWER: FALSE

3 A center of a circle is a part of the circle.
ANSWER: FALSE

4 A segment of a circle is a part of the circle.
ANSWER: FALSE

RC 38 ROUND 3B: TRUE OR FALSE

1If $(x + 1)(x - 3) < 0$, then $-3 < x < 1$
ANSWER: FALSE

2 If $(x + 2)(3 - x) < 0$, then $x > 3$ or $x < -2$
ANSWER: TRUE

3 If $(x - 1)(5 - x) > 0$, then $1 < x < 5$
ANSWER: TRUE

4 If $(x + 1)(x + 3) > 0$, then $x > -1$, or $x < -3$
ANSWER: TRUE

RC 39 ROUND 1A

Express the given number as the difference of 2 squares.

1 37 **ANSWER: $19^2 - 18^2$** [$37 = a^2 - b^2 = (a + b)(a – b) = 37(1)$, $a + b = 37$, $a – b = 1$, $2a = 38$, $a = 19$, $b = 18$]

2 41 **ANSWER: $21^2 - 20^2$** [$41 = a^2 – b^2 = (a + b)(a – b) = 41(1)$, $a + b = 41$, $a – b = 1$, $2a = 42$, $a = 21$, $b = 20$]

3 17 **ANSWER: $9^2 – 8^2$** [$17 = a^2 – b^2 = (a + b)(a – b) = 17(1)$, $a + b = 17$, $a - b = 1$, $2a = 18$, $a = 9$, $b = 8$]

4 23 **ANSWER: $12^2 – 11^2$** [$23 = a^2 – b^2 = (a + b)(a – b) = 23(1)$, $a + b = 23$, $a – b = 1$, $2a = 24$, $a = 12$, $b = 11$]

RC 39 ROUND 1B

Find constants A and B such that

1 $3/(x^2 – x – 2) = A/(x – 2) + B/(x + 1)$ **ANSWER: A = 1, B = -1**
[$3 = A(x + 1) + B(x – 2)$, $A + B = 0$, $A – 2B = 3$, $3B = -3$, $B = -1$, $A = 1$]

2 $(x + 3)/(x^2 – 1) = A/(x – 1) + B/(x + 1)$, **ANSWER: A = 2, B = -1**
[$x + 3 = A(x + 1) + B(x – 1)$, $2A = 4$, $A = 2$, $-2B = 2$, $B = -1$]

3 $14/(x^2 – 3x – 10) = A/(x – 5) + B/(x + 2)$ **ANSWER: A = 2, B = -2**
[$14 = A(x + 2) + B(x – 5)$, $7A = 14$, $A = 2$, $-7B = 14$, $B = -2$]

4 $(x + 9)/(x^2 + 3x – 4) = A/(x + 4) + B/(x – 1)$ **ANSWER: A = -1, B = 2**
[$(x + 9) = A(x – 1) + B(x + 4)$, $5B = 10$, $B = 2$, $-5A = 5$, $A = -1$]

RC 39 ROUND 2: SPEED RACE

1 Find dy/dx from the equation $2x^2 + xy - y^3 = 10$
ANSWER: $dy/dx = (4x + y)/(3y^2 - x)$
[$4x + y + x(dy/dx) - 3y^2(dy/dx) = 0$, $(dy/dx)(3y^2 - x) = (4x + y)$,
$dy/dx = (4x + y)/(3y^2 - x)$]

2 Find the equation of a circle with center (-4, 5) touching the y-axis.
ANSWER: $(x + 4)^2 + (y - 5)^2 = 16$
[radius = 4, $(x + 4)^2 + (y - 5)^2 = 16$]

3 Solve the equation $\cos x = -\sqrt{3}/2$ in the interval $\pi/2 < x < \pi$.
ANSWER: $x = 5\pi/6$
[$\cos x = \sqrt{3}/2$, $x = \pi/6$, $\cos x = -\sqrt{3}/2$, $x = \pi - \pi/6 = 5\pi/6$]

RC 39 ROUND 4: RIDDLE

1 I am a statistical measure from a given data.

2 I am a measure of centrality.

3 I am a measure of location.

4 I am one of the 3 commonest such measures.

5 I am not unique.

6 I represent a measurement with the largest frequency.
Who am I?
ANSWER: MODE

RC 39 ROUND 3A: TRUE OR FALSE

1 A kite has 2 pairs of congruent adjacent sides.
ANSWER: TRUE

2 A kite has 2 pairs of congruent opposite angles.
ANSWER: FALSE

3 A parallelogram has 2 pairs of congruent opposite angles.
ANSWER: TRUE

4 A trapezium has 2 pairs of adjacent supplementary angles.
ANSWER: TRUE

RC 39 ROUND 3B: TRUE OR FALSE

1 A relation is a function.
ANSWER: FALSE

2 A function is a relation.
ANSWER: TRUE

3 A many to one relation is a function.
ANSWER: TRUE

4 A relation has an inverse only if it is one to one.
ANSWER: FALSE

RC 40 ROUND 1A

The sides of a triangle lie on the given lines. Find the coordinates of the vertices of the triangle.

1 $x + y = 6$, $x - y = 2$, $y = 3$, ANSWER: (4, 2), (4, 3), (3, 3)

[$x + y = 6$, $x - y = 2$, $x = 4$, $y = 2$, (4, 2), $x - y = 1$, $y = 3$, $x = 4$, (4, 3)
$x + y = 6$, $y = 3$, $x = 3$, (3, 3)]

2 $x - 2y = -3$, $x = 5$, $x + y = 3$ ANSWER: (5, 4), (5, -2), (1, 2)

[$x - 2y = -3$, $x = 5$, $y = 4$, (5, 4), $x = 5$, $x + y = 3$, $y = -2$, (5, -2)
$x - 2y = -3$, $x + y = 3$, $3y = 6$, $y = 2$, $x = 1$, (1, 2)]

3 $2x + y = 3$, $x + y = 1$, $y = 5$ ANSWER: (2, -1), (-4, 5), (-1, 5)

[$2x + y = 3$, $x + y = 1$, $x = 2$, $y = -1$, (2, -1), $x + y = 1$, $y = 5$, $x = -4$, (-4, 5)
$2x + y = 3$, $y = 5$, $x = -1$, (-1, 5)]

4 $x - 2y = 6$, $x + 2y = 2$, $x = -2$ ANSWER: (4, 1), (-2, -4), (-2, 2)

[$x - 2y = 6$, $x + 2y = 2$, $x = 4$, $y = -1$, (4, -1), $x - 2y = 6$, $x = -2$, $y = -4$, (-2, -4)
$x + 2y = 2$, $x = -2$, $y = 2$, (-2, 2)

RC 40 ROUND 1B

Two fair dice are tossed, and the total score is 6. Find the probability that,

1 the 2 scores are both even, ANSWER: 2/5

[S = {(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)}, A = {(2, 4), (4, 2)}, P(A) = 2/5]

2 the scores are both odd ANSWER: 3/5

[S = {(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)}, B = {(1, 5), (3, 3), (5, 1)}, P(B) = 3/5]

3 the absolute difference of the scores is a square, ANSWER: 2/5

[S = {(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)}, C = {(1, 5), (5,1)}, P(C) = 2/5]

4 the first score divides the second score. ANSWER: 3/5

[S = {(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)}, D = {(1, 5), (2, 4), (3, 3)}, P(D) = 3/5]

RC 40 ROUND 2: SPEED RACE

1 Evaluate (dy/dx) from $x^2 + xy - y^2 = 4$ at P(2, 2).
ANSWER: 3
$2x + y + xdy/dx - 2ydy/dx = 0$, $dy/dx = (2x + y)/(2y - x)$, at (2, 2),
$dy/dx = (4 + 2)/(4 - 2) = 6/2 = 3$]

2 Express 43 as a difference of 2 squares
ANSWER: $43 = 22^2 - 21^2$
[$43 = a^2 - b^2 = (a + b)(a - b) = 43 \times 1$, $a + b = 43$, $a - b = 1$, $2a = 44$,
$a = 22$, $b = 21$]

3 The point A(2, 1) is reflected in the line $y = x$ and its image further reflected in the origin. Find the coordinates of the final image point.
ANSWER: (-1, 2)
[$(-2, 1) \rightarrow (1, -2) \rightarrow (-1, 2)$]

RC 40 ROUND 4: RIDDLE

1 I am a process in Algebra.

2 I am generally applied to a quadratic expression.

3 I can be used to solve any quadratic equation.

4 I am definitely not factorization.

5 I am used to derive the almighty formula.

6 I am in the completion business.
Who am I?
ANSWER: COMPLETING THE SQUARE

RC 40 ROUND 3A: TRUE OR FALSE

The graph of the inequality

1 $x - y < 5$ is the region below the line $x - y = 5$.
ANSWER: FALSE

2 $x + y > 6$ is the region below the line $x + y = 6$.
ANSWER: FALSE

3 $2x - y < 12$ is the region above the line $2x - y = 12$,
ANSWER: TRUE

4 $x + 2y > 7$ is the region above the line $x + 2y = 7$.
ANSWER: TRUE

RC 40 ROUND 3B: TRUE OR FALSE

1 $\log(x^2y) = 2(\log x + \log y)$
ANSWER: FALSE

2 $\log(x^2/y^3) = 2\log x - 3\log y$
ANSWER: TRUE

3 $\log(x^4y^2) = 2(2\log x + \log y)$
ANSWER: TRUE

4 $\log\sqrt{(x/y^3)} = \frac{1}{2}\log x - 3\log y$
ANSWER: FALSE

RC 41 ROUND 1A

Evaluate a and b if when $x^3 + ax^2 + bx - 2$ is divided by

1 $(x - 1)(x + 1)$, the remainder is $2x + 1$ **ANSWER: $a = 3, b = 1$**

[$x^3 + ax^2 + bx - 2 = (x^2 - 1)q(x) + 2x + 1$, $1 + a + b - 2 = 3$, $-1 + a - b - 2 = -1$
$a + b = 4$, $a - b = 2$, $2a = 6$, $a = 3$, $b = 4 - 3 = 1$]

2 $(x - 1)(x - 2)$, the remainder is $3x - 2$ **ANSWER: $a = -3, b = 5$**

[$x^3 + ax^2 + bx - 2 = (x - 1)(x - 2)q(x) + 3x - 2$, $1 + a + b - 2 = 1$, $a + b = 2$,
$8 + 4a + 2b - 2 = 4$, $4a + 2b = -2$, $2b = 10$, $b = 5$, $a + 5 = 2$, $a = -3$]

3 $(x + 1)(x - 2)$, the remainder is $5x - 4$ **ANSWER: $a = -2, b = 4$**

[$x^3 + ax^2 + bx - 2 = (x + 1)(x - 2)q(x) + 5x - 4$, $-1 + a - b - 2 = -9$, $a - b = -6$,
$8 + 4a + 2b - 2 = 6$, $4a + 2b = 0$, $6a = -12$, $a = -2$, $2b = -4a = 8$, $b = 4$]

4 $(x - 2)(x + 2)$, the remainder is $3x + 6$, **ANSWER: $a = 2, b = -1$**

[$x^3 + ax^2 + bx - 2 = (x - 2)(x + 2)q(x) + 3x + 6$, $-8 + 4a - 2b - 2 = 0$, $8 + 4a + 2b - 2 = 12$, $8a - 4 = 12$, $8a = 16$, $a = 2$, $4a + 2b = 6$, $2b = 6 - 8$, $b = -1$]

RC 41 ROUND 2: SPEED RACE

1 Express $\log(x^3y^2/z^4) + \log(z^2/xy)$ as a single logarithm in simplest form.

ANSWER: $\log(x^2y/z^2)$

[$\log(x^3y^2z^2/xyz^4) = \log(x^2y/z^2)$]

2 Evaluate $(\sin56°\sin64° - \cos56°\cos64°)/\sin30°$

ANSWER: 1

[$-\cos(56° + 64°)/\sin30° = -\cos120/\cos60 = \cos60/\cos60 = 1$]

3 Factorize completely $27x^3 - 8y^3$

ANSWER: $(3x - 2y)(9x^2 + 6xy + 4y^2)$

[$(3x)^3 - (2y)^3 = (3x - 2y)(9x^2 + 6xy + 4y^2)$, NB: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$]

RC 41 ROUND 1B
In an isosceles trapezium find the lengths of the bases if

1 each leg is 5cm, the altitude is 3cm and the area is $39cm^2$,
ANSWER: 9cm, 17cm
[Area = $\frac{1}{2}(2a + 8)3 = 39$, $(a + 4) = 13$, $a = 9$, base = 9 cm, 17 cm]

2 each leg is 17cm, the altitude is15 cm and the area is $300cm^2$
ANSWER: 12cm, 28 cm
[Area = $\frac{1}{2}(2a + 16)15 = 300$, $a + 8 = 20$, $a = 12$, base = 12, 28]

3 each leg is 13cm, the altitude is 12 cm and the area is 120 cm^2.
ANSWER: 5 cm, 15 cm
[Area = $\frac{1}{2}(2a + 10)12 = 120$, $a + 5 = 10$, $a = 5$, base = 5, 15]

4 each leg is 10 cm, the altitude is 8 cm and the area is 96 cm^2
ANSWER: 6 cm, 18 cm
[Area = $\frac{1}{2}(2a +12)8 = 96$, $a + 6 = 12$, $a = 6$, base = 6, 18]

RC 41 ROUND 4: RIDDLE

1 I am a quadrilateral.

2 I have a pair of congruent opposite sides.

3 I have 2 pairs of congruent adjacent angles.

4 I have a pair of parallel sides.

5 My congruent sides are not parallel.

6 I have a relation with an isosceles triangle.
Who am I?
ANSWER: ISOSCELES TRAPEZIUM

RC 41 ROUND 3A: TRUE OR FALSE
θ is an angle measure.

1 If $\cos\theta > 0$, then θ is in the 1st and 3rd quadrants.
ANSWER: FALSE

2 If $\sin\theta < 0$, then θ is in the 2nd and 3rd quadrants.
ANSWER: FALSE

3 If $\tan\theta < 0$, then θ is in the 2nd and 3rd quadrants.
ANSWER: FALSE

4 If $\cos\theta < 0$, then θ is in the 2nd and 4th quadrants.
ANSWER: FALSE

RC 41 ROUND 3B: TRUE OR FALSE

In the following, a is a real number.

1 $\sqrt{(a^2)} = a$
ANSWER: FALSE (only if $a \geq 0$)

2 $\sqrt[3]{a^3} = a$
ANSWER: TRUE

3 $\sqrt{(a^4)} = a^2$
ANSWER: TRUE

4 $\sqrt{(a^6)} = a^3$
ANSWER: FALSE (only if $a \geq 0$)

RC 42 ROUND 1Axxxx

Find the greatest value and the least value of the expression,

1 $(3\cos x - 4\sin x + 10)$ **ANSWER: Greatest value = 15, least value = 5**

[$(5\sin(x + \alpha) + 10)$, greatest = $(5 + 10) = 15$, least = $(-5 + 10) = 5$]

2 $(12\cos x + 5\sin x - 10)$ **ANSWER: greatest value = 3, least value = -23**

[$(13\cos(x + \alpha) - 10)$, greatest = $13 - 10 = 3$, least = $-13 - 10 = -23$]

3 $(5\cos x - \sqrt{11}\sin x + 4)$ **ANSWER: greatest = 10, least = -2**

[$6\cos(x + \alpha) + 4$, greatest = $6 + 4 = 10$, least = $-6 + 4 = -2$]

4 $(11\cos x + \sqrt{23}\sin x - 12)$ **ANSWER: greatest = 0, least = -24**

[$12\cos(x + \alpha) - 12)$, greatest = $12 - 12 = 0$, least = $-12 - 12 = -24$]

RC 42 ROUND 1B

Use implicit differentiation to find the gradient (dy/dx) for the given curve

1 $x^3 + 3x^2y - y^3 = 15$ **ANSWER: $(dy/dx) = (x^2 + 2xy)/(y^2 - x^2)$**

[$3x^2 + 6xy + 3x^2(dy/dx) - 3y^2(dy/dx) = 0$, $(dy/dx)(3y^2 - 3x^2) = 3x^2 + 6xy$

$dy/dx = (x^2 + 2xy)/(y^2 - x^2)$]

2 $3x^2 - 4xy + 2y^2 = 10$ **ANSWER: $(dy/dx) = (3x - 2y)/(2x - 2y)$**

[$6x - 4y - 4x(dy/dx) + 4y(dy/dx) = 0$, $(dy/dx)(4x - 4y) = (6x - 4y)$

$(dy/dx) = (3x - 2y)/(2x - 2y)$]

3 $2x^3 - xy^2 + y^3 = 20$ **ANSWER: $(dy/dx) = (6x^2 - y^2)/(2xy - 3y^2)$**

[$6x^2 - y^2 - 2xy(dy/dx) + 3y^2(dy/dx) = 0$, $(dy/dx)(2xy - 3y^2) = (6x^2 - y^2)$

$(dy/dx) = (6x^2 - y^2)/(2xy - 3y^2)$]

4 $x^3 + 2x^2y - 3y^2 = 25$ **ANSWER: $dy/dx = (3x^2 + 4xy)/(6y - 2x^2)$**

[$3x^2 + 4xy + 2x^2(dy/dx) - 6y(dy/dx) = 0$, $(dy/dx)(6y - 2x^2) = (3x^2 + 4xy)$

$(dy/dx) = (3x^2 + 4xy)/(6y - 2x^2)$]

RC 42 ROUND 2: SPEED RACE

1 Find the cosine of the angle between the vectors a = i + 5j and b = 3i – j
ANSWER:-$\sqrt{65}/65$
[$\cos\theta = a.b/|a||b| = -2/\sqrt{26}\sqrt{10} = -2\sqrt{260}/260 = -4\sqrt{65}/260 = -\sqrt{65}/65$]

2 Two fair dice are tossed. Find the probability the total score is either 6 or 7.
ANSWER : 11/36
[A = {(1, 5), (2, 4), (3, 3), (4, 2), (5, 1), (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}
P(A) = 11/36]

3 Solve the logarithmic equation $\log_2(x + 3) = \log_2(x - 3) + 1$
ANSWER: $x = 9$
[$x + 3 = 2(x - 3) = 2x - 6, x = 3 + 6 = 9$]

RC 42 ROUND 4: RIDDLE

1 I am a solid geometric figure.

2 I sit on a polygonal base.

3 I am formed when vertices of the base polygon are linked to a point above the base.

4 This makes me a pyramid.

5 I have 5 faces, five vertices and 8 edges.

6 My faces are 4 triangles and a regular quadrilateral.
Who am I?
ANSWER: SQUARE PYRAMID

RC 42 ROUND 3A: TRUE OR FALSE

The point P(x, y) is in the coordinate x-y plane. If the product

1 $xy > 0$, then P is in the 1st quadrant.
ANSWER: FALSE

2 $xy > 0$, then P is in the 3rd quadrant.
ANSWER: FALSE

3 $xy < 0$, then P is in the 2nd quadrant.
ANSWER: FALSE

4 $xy < 0$, then P is in the 4th quadrant.
ANSWER: FALSE

RC 42 ROUND 3B: TRUE OR FALSE

1 A linear transformation maps the origin into the origin.
ANSWER: TRUE

2 Under a linear transformation, only the origin is mapped onto the origin.
ANSWER: FALSE

3 Every linear transformation has an inverse.
ANSWER: FALSE

4 A linear transformation has an inverse if the determinant of its matrix is positive.
ANSWER: TRUE

RC 43 ROUND 1A

Find the range of the trigonometric function.

1 $y = 5\cos x - 12\sin x + 9$ ANSWER: $\{y: -4 \le y \le 22\}$
[$y = 13\cos(x + \alpha) + 9$, hence $\{y: -4 \le y \le 22\}$]

2 $y = 15\cos x + 8\sin x - 12$ ANSWER: $\{y: -29 \le y \le 5\}$
[$y = 17\cos(x + \alpha) - 12$, hence $\{y: -29 \le y \le 5\}$]

3 $y = 9\cos x - 12\sin x + 15$ ANSWER: $\{y: 0 \le y \le 30\}$
[$y = 15\cos(x + \alpha) + 15$, hence $\{y: 0 \le y \le 30\}$]

4 $y = 5\cos x + \sqrt{11}\sin x - 3$ ANSWER: $\{y: -9 \le y \le 3\}$
[$y = 6\cos(x + \alpha) - 3$, hence $\{y: -9 \le y \le 3\}$]

RC 43 ROUND 1B

Find the quadratic polynomial $Q(x)$ such that

1 $Q(1) = 3$, $Q(-1) = 5$, $Q(0) = 2$ ANSWER: $Q(x) = 2x^2 - x + 2$
[$Q(x) = ax^2 + bx + c$, $Q(1) = a + b + c = 3$, $Q(-1) = a - b + c = 5$, $Q(0) = c = 2$
$a + b = 1$, $a - b = 3$, $2a = 4$, $a = 2$, $b = -1$, $Q(x) = 2x^2 - x + 2$]

2 $Q(1) = 6$, $Q(-1) = 2$, $Q(0) = 3$ ANSWER: $Q(x) = x^2 + 2x + 3$
[$Q(x) = ax^2 + bx + c$, $Q(1) = a + b + c = 6$, $Q(-1) = a - b + c = 2$, $Q(0) = c = 3$
$a + b = 3$, $a - b = -1$, $2a = 2$, $a = 1$, $b = 2$, $Q(x) = x^2 + 2x + 3$]

3 $Q(-1) = 0$, $Q(0) = -3$, $Q(1) = 4$ ANSWER: $Q(x) = 5x^2 + 2x - 3$
[$Q(x) = ax^2 + bx + c$, $Q(-1) = a - b + c = 0$, $Q(1) = a + b + c = 4$
$Q(0) = c = -3$, $2b = 4$, $b = 2$, $a + 2 - 3 = 4$, $a = 5$, $Q(x) = 5x^2 + 2x - 3$]

4 $Q(1) = 3$, $Q(0) = 5$, $Q(-1) = 3$ ANSWER: $Q(x) = -5x^2 + 5$
[$a + b + c = 3$, $c = 5$, $a - b + c = 3$, $2b = 0$, $b = 0$, $a - b + c = a - 0 + 5 = 3$,
$a = -2$, $Q(x) = -2x^2 + 5$]

RC 43 ROUND 2: SPEED RACE

1 Rationalize the denominator and simplify $(5\sqrt{2} - 2\sqrt{5})/(5\sqrt{2} + 2\sqrt{5})$
ANSWER: $7/3 - (2/3)\sqrt{10}$
$[(50 + 20 - 20\sqrt{10})/(50 - 20) = 7/3 - (2/3)\sqrt{10}]$

2 Find the sum of the first 10 terms of the series
$-7 - 2 + 3 + 8 + \ldots$
ANSWER: 155
[Linear series $a = -7$, $d = 5$, $S_{10} = (10/2)(-14 + 5(9)) = 5(31) = 155$]

3 Differentiate the function $y = \sqrt{(x^3 + 2x^2 - 3x + 1)}$.
ANSWER: $dy/dx = (3x^2 + 4x - 3)/\sqrt{(x^3 + 2x^2 - 3x + 1)}$
$[dy/dx = (3x^2 + 4x - 3)(x^3 + 2x^2 - 3x + 1)^{-1/2} =$
$(3x^2 + 4x - 3)/\sqrt{(x^3 + 2x^2 - 3x + 1)}]$

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RC 43 ROUND 4: RIDDLE

1 I am used to evaluate a definite integral.

2 This involves finding the area under a curve.

3 I am an approximate method for numerical integration.

4 I am not the substitution method.

5 I am a rule for integration.

6 I am associated with a plane geometrical figure.
Who am I?
ANSWER: TRAPEZIUM RULE

RC 43 ROUND 3A: TRUE OR FALSE

1 $\sqrt{(1/4)} + \sqrt{(1/25)} = \sqrt{(49/100)}$
ANSWER: TRUE

2 $\sqrt{(125)}/5 = \sqrt{(25)}/5$
ANSWER: FALSE

3 $\sqrt{(-36)}$ is not a real number.
ANSWER: TRUE

4 $\sqrt{36} + \sqrt{64} = \sqrt{100}$
ANSWER: FALSE

RC 43 ROUND 3B: TRUE OR FALSE
The solution to the inequality $(x + 1)(3 - x) < 0$ is

1 $\{x: -3 < x < 1\}$
ANSWER: FALSE

2 $\{x: -1 < x < 3\}$
ANSWER: FALSE

3 $\{x: 3 < x < -1\}$
ANSWER: FALSE

4 $\{x: x > 3, \text{ or } x < -1\}$
ANSWER: TRUE

RC 44 ROUND 1A

Solve for x from the exponential equation.

1 $6^{(x - 5)} = 5^{(x + 6)}$ ANSWER: $x = (6\log5 + 5\log6)/(\log6 – \log5)$
[$(x - 5)\log6 = (x + 6)\log5$, $x(\log6 – \log5) = 6\log5 + 5\log6$,
$x = (6\log5 + 5\log6)/(\log6 – \log5)$]

2 $8^{(x – 3)} = 6^{(x + 3)}$ ANSWER: $x = 3(\log6 + \log8)/\log(8 – \log6)$
[$(x – 3)\log8 = (x + 3)\log6$, $x(\log8 – \log6) = 3(\log6 + \log8)$,
$x = 3(\log6 + \log8)/(\log8 – \log6)$

3 $9^{(x – 2)} = 4^{(x + 7)}$ ANSWER: $x = (7\log4 + 2\log9)/(\log9 – \log4)$
[$(x - 2)\log9 = (x + 7)\log4$, $x = (7\log4 + 2\log9)/(\log9 – \log4)$]

4 $7^{x – 5} = 5^{x + 3}$ ANSWER: $x = (5\log7 + 3\log5)/(\log7 – \log5)$
[$(x – 5)\log7 = (x + 3)\log5$, $x = (5\log7 + 3\log5)/(\log7 – \log5)$]

RC 44 ROUND 1B

Find the values of constants A, B and C such that,

1 $A(x + 1)(x + 2)) + (Bx + C)x = 4x^2 + 6x + 8$ ANSWER: $A = 4$, $B = 0$, $C = -6$
[For $x = 0$, $2A = 8$, $A = 4$, for x^2, $A + B = 4$, $B = 0$, for $x = -1$, $-C = 4 – 6 + 8 = 6$]

2 $A(x - 2)(x – 1) + (Bx + C)x = 3x^2 - 5x + 4$ ANSWER: $A = 2$, $B = 1$, $C = 1$
[For $x = 0$, $2A = 4$, $A = 2$, for x^2, $A + B = 3$, $B = 1$, for $x = 1$, $B + C = 2$, $C = 1$]

3 $A(x^2 + x – 2) + (Bx + C)x = x^2 – 3x + 4$ ANSWER: $A = -2$, $B = 3$, $C = -1$
[For $x = 0$, $-2A = 4$, $A = -2$, for x^2, $A + B = 1$, $B = 3$, for x, $A + C = -3$, $C = -1$]

4 $A(x^2 – 2x – 3) + (Bx + C)x = 2x^2 - 5x + 6$ ANSWER: $A = -3$, $B = B$, $C = 1$
[For $x = 0$, $-2A = 6$, $A = -3$, for x^2, $A + B = 2$, $B = 5$, for x, $-2A + C = -5$, $C = 1$]

RC 44 ROUND 2: SPEED RACE

1 Find the stationary points of the curve $y = x + 1/x$
ANSWER: (1, 2), (-1, -2)
[$dy/dx = 1 - 1/x^2$, $x^2 = 1$, $x = \pm 1$, $x = 1$, $y = 2$, (1, 2), $x = -1$, $y = -2$, (-1, -2)]

2 If $a = 2b/5$, and $c = 2b/3$, find the ratio a : b : c in simplest form
ANSWER: a : b : c = 6 : 15 : 10
[a : b : c = 2b/5 : b : 2b/3 = 2/5 : 1 : 2/3 = 6: 15 : 10]

3 Find the determinant of the matrix of the linear transformation T: (x, y)→(2x - y, 3x - y) and determine if T has an inverse or not.
ANSWER: det = 1, T has an inverse
[$A = \begin{pmatrix} 2 & -1 \\ 3 & -1 \end{pmatrix}$, $\det A = -2 + 3 = 1$, T has an inverse]

RC 44 ROUND 4: RIDDLE

1 I am a triangle.

2 I am an isosceles triangle.

3 I have an axis of symmetry.

4 However I am not strictly a geometric figure.

5 I am a triangle of numbers.

6 The numbers are binomial coefficients.
Who am I?
ANSWER: PASCAL'S TRIANGLE

RC 44 ROUND 3A: TRUE OR FALSE

1 If a number is divisible by 3 and 5, then it is divisible by 15.
ANSWER: TRUE

2 If a number is divisible by 6 and 4, then it is divisible by 24,
ANSWER: FALSE

3 If a number is divisible by7, then the sum of its digits is divisible by 7.
ANSWER: FALSE

4 If a number is divisible by 9, then the sum of its digits is divisible by 9.
ANSWER: TRUE

RC 44 ROUND 3B: TRUE OR FALSE
For a linear sequence with 1^{st} term a, and common difference d,

1 the general term is $U_n = a + (n - 1)d$
ANSWER: TRUE

2 the sum of the first n terms is $S_N = n(2a + (n - 1)d)$
ANSWER: FALSE

For an exponential series with 1^{st} term a, and common ratio r,

3 the general term is $U_n = ar^{n+1}$
ANSWER: FALSE

4 the sum of the 1^{st} n terms is $S_n = a(1 - r^n)/(1 - r)$.
ANSWER: TRUE

RC 45 ROUND 1A

Find the value of a if

1 $(\log_4 x)^2 = (\log_2 x)(\log_a x)$, **ANSWER: a = 16** $[(\log_2 x/\log_2 4)^2 = (\log_2 x)(\log_2 x)/4) = (\log_2 x)(\log_a x)$, $\log_a x = (\log_2 x)/4$, $\log_2 x/\log_2 a = (\log_2 x)/4$, $\log_2 a = 4$, $a = 2^4 = 16]$

2 $(\log_3 x)^2 = (\log_9 x)(\log_a x\ )$ **ANSWER: a = $\sqrt{3}$**

$[(\log_3 x)(\log_3 x) = (\log_3 x/\log_3 9)\log_a x$, $(\log_3 x)(\log_3 x) = (\log_3 x)(\log_a x)/2$,

$\log_a x = 2\log_3 x$, $\log_3 x/\log_3 a = 2\log_3 x$, $\log_3 a = ½$, $a = \sqrt{3}]$

3 $(\log_9 x)^2 = (\log_3 x)(\log_a x)$ **ANSWER: a = 81** $[(\log_3 x)(\log_3 x)/4 = (\log_3 x)(\log_a x)$, $\log_a x = (\log_3 x)/4$, $(\log_3 x)/(\log_3 a) = (\log_3 x)/4$, $\log_3 a = 4$, $a = 3^4 = 81]$

4 $(\log_5 x)^2 = (\log_{25} x)(\log_a x)$ ANSWER: $a = \sqrt{5}$

$[(\log_5 x)(\log_5 x) = ½(\log_5 x)(\log_a x), \log_5 x = ½\log_a x = ½\log_5 x/\log_5 a$, $2\log_5 a = 1$, $a = 5^{1/2} = \sqrt{5}]$

RC 45 ROUND 1B

Given the functions $f(x) = 2x + 3$, $g(x) = x^2 + 1$ and $h(x) = 1/(x^2 + 1)$, find expressions for the composite function

1 $f(g(x))$ ANSWER: $2x^2 + 5$

$[\ f(g(x)) = 2g(x) + 3 = 2(x^2 + 1) + 3 = 2x^2 + 5]$

2 $f(h(x))$ ANSWER: $2/(x^2 + 1) + 3$, or $(3x^2 + 5)/(x^2 + 1)$

$[f(h(x)) = 2(h(x) + 3 = 2/(x^2 + 1) + 3 = (3x^2 + 5)/(x^2 + 1)]$

3 $g(f(x))$ ANSWER: $(2x + 3)^2 + 1$

$[g(f(x)) = f(x)^2 + 1 = (2x + 3)^2 + 1]$

4 $g(1/f(x))$ ANSWER: $(x^2 + 1)^2 + 1$, or $(x^4 + 2x^2 + 2)$

$[g(1/f(x)) = g(x^2 + 1) = (x^2 + 1)^2 + 1$, or $x^4 + 2x^2 + 2$

RC 45 ROUND 2: SPEED RACE

1 Solve the equation $\log_3 x - 3\log_x 3 = 2$
ANSWER: $x = 27, 1/3$
[$\log_3 x - 3/\log_3 x = 2$, $(\log_3 x)^2 - 2\log_3 x - 3 = (\log_3 x - 3)(\log_3 x + 1) = 0$,
$\log_3 x = 3$, $x = 3^3 = 27$, $\log_3 x = -1$, $x = 3^{-1} = 1/3$]

2 Find the gradient dy/dx of the curve with equation $x^3 - 3xy^2 + y^3 = 25$
ANSWER: $(dy/dx) = (x^2 - y^2)/(2xy - y^2)$
[$3x^2 - 3y^2 - 6xy\,dy/dx + 3y^2 dy/dx = 0$, $(dy/dx)(6xy - 3y^3) = (3x^2 - 3y^2)$
$(dy/dx) = (x^2 - y^2)/(2xy - y^2)$

3 Find the solution set of the inequality $x^2 - 5x - 6 < 0$
ANSWER: $\{x: -1 < x < 6\}$
[$(x - 6)(x + 1) < 0$, $-1 < x < 6$]

RC 45 ROUND 4: RIDDLE

1I am a closed geometric figure.

2 I am all line segments and angles in a plane.

3 This makes me a polygon.

4 I may be equilateral, equiangular, scalene or isosceles.

5 I am the simplest of my type.

6 The sum of my interior angles is a straight angle.
Who am I?
ANSWER: TRIANGLE

RC 45 ROUND 3A: TRUE OR FALSE
The quadratic equation with roots

1 -3 and 5 is $x^2 + 2x - 15 = 0$
ANSWER: FALSE

2 4 and 7 is $x^2 - 11x + 28 = 0$
ANSWER: TRUE

3 -2 and – 6 is $x^2 + 8x + 12 = 0$
ANSWER: TRUE

4 7 and -5 is $x^2 + 2x - 35 = 0$
ANSWER: FALSE

RC 45 ROUND 3B: TRUE OR FALSE

1 A radius is a chord passing through the center of a circle.
ANSWER: FALSE

2 A chord of a circle passing through the center is a diameter.
ANSWER: TRUE

3 A chord of a circle is a line segment connecting any 2 points of the circle.
ANSWER: TRUE

4 A secant is a line in the plane of the circle containing 2 points of the circle.
ANSWER: TRUE

RC 46 ROUND 1A

Find the interval or set on which the given function is decreasing.

1 $y = x^3 - 6x^2 - 36x + 5$

ANSWER: $\{x: -2 < x < 6\}$

$[dy/dx = 3x^2 – 12x - 36 = 3(x^2 – 4x – 12) < 0, (x – 6)(x + 2) < 0, -2< x < 6]$

2 $y = x^3 + 2x^2 – 4x + 7$

ANSWER: $\{x: -2 < x < 2/3\}$

$[dy/dx = 3x^2 + 4x – 4 = (3x – 2)(x + 2) < 0, -2 < x < 2/3]$

3 $y = -x^3 - x^2 + 5x + 10$

ANSWER: $\{x: x > 1, \text{ or } x < -5/3 \}$ $[dy/dx = -3x^2 – 2x + 5 < 0, 3x^2 + 2x – 5 > 0,$
$(3x + 5)(x - 1) > 0, x > 1, \text{ or } x < -5/3]$

4 $y = -x^3 + 3x^2 + 45x + 8$

ANSWER: $\{x: x > 5, \text{ or } x < - 3\}$ $[dy/dx = -3x^2 + 6x + 45 < 0, x^2 – 2x – 15 > 0,$
$(x – 5)(x + 3) > 0, x > 5, \text{ or } x < - 3]$

RC 46 ROUND 1B

Find an expression for f(x) given,

1 $f(x + 2) = (x + 1)/(x – 3)$

ANSWER: **$f(x) = (x – 1)/(x – 5)$** $[a = x + 2, x = a – 2,$
$f(a) = (a – 2 + 1)/(a – 2 – 3) = (a – 1)/(a – 5), f(x) = (x – 1)/(x – 5)]$

2 $f(x - 3) = (x + 5)/(x – 4)$

ANSWER: $f(x) = (x + 8)/(x – 1)$ $[a = (x – 3, x = a + 3,$
$f(a) = (a + 3 + 5)/(a + 3 – 4) = (a + 8)/(a – 1), f(x) = (x + 8)/(x – 1)]$

3 $f(x + 3) = (x + 5)/(x + 6)$

ANSWER: $f(x) = (x + 2)/(x + 3)$ $[a = x + 3, x = (a – 3),$
$f(a)= (a – 3 + 5)/(a - 3 + 6) = (a + 2)/(a + 3), f(x) = (x + 2)/(x + 3)]$

4 $f(x – 2) = (x + 3)/(x – 4)$

ANSWER: $f(x) = (x + 5)/(x – 2)$

$[a = x – 2, x = a + 2, f(a) = (x + 2 + 3)/(x + 2 – 4) = (x + 5)/(x – 2)]$

RC 46 ROUND 2: SPEED RACE

1 Find the least common multiple of $x^2 - 7x + 12$, $x^2 - 4x$, and $x^2 - 3x$
ANSWER: $x(x - 4)(x - 3)$ or $x^3 - 7x^2 + 12x$
[$x^2 - 7x + 12 = (x - 3)(x - 4)$, $x^2 - 4x = x(x - 4)$, $x^2 - 3x = x(x - 3)$, hence lcm is $x(x - 3)(x - 4) = x^3 - 7x^2 + 12x$]

2 Integrate $\int (x^3+8)/(x+2)dx$
ANSWER: $x^3/3 - x^2 + 4x + C$
[$(x^3 + 8)/(x + 2) = x^2 - 2x + 4$, Integrate to give $x^3/3 - x^2 + 4x + C$]

3 If $\tan x = 2\sqrt{6}$, find possible value(s) for sec x.
ANSWER: ± 5
[$\sec^2 x = \tan^2 x + 1 = 24 + 1 = 25$, $\tan x = \pm 5$]

RC 46 ROUND 4: RIDDLE

1 I am a type of motion.

2 I am a transformation in the plane.

3 I preserve distance between points as well as angles.

4 There is something rigid about me.

5 Examples of me are rotation and reflection.

6 I am not a linear transformation.
Who am I?
ANSWER: RIGID BODY MOTION

RC 46 ROUND 3A: TRUE OR FALSE

In the interval $0 < x < \pi$,

1 cos x is decreasing,
ANSWER: TRUE

2 sin x is increasing,
ANSWER: FALSE

3 tan x is increasing,
ANSWER: FALSE

4 sin x increases and decreases.
ANSWER: TRUE

RC 46 ROUND 3B: TRUE OR FALSE

1 $(x + 1)$ is a factor of $x^3 - 3x^2 + 4x + 8$
ANSWER: TRUE

2 $(x - 1)$ is a factor of $2x^3 - 4x^2 + 7x - 5$
ANSWER: TRUE

3 $(x - 2)$ is a factor of $x^3 - 4x^2 + 3x + 3$
ANSWER: FALSE

4 $(x + 2)$ is a factor of $x^3 + 3x^2 + 3x - 2$
ANSWER: TRUE

EXTRA EXTRA

Speed race

1 If the point A(1, -3) is a stationary point of the curve $y = ax^2 + bx - 5$, find a and b.

ANSWER: $a = 2, b = -4$

[$dy/dx = 2ax + b$, $2a + b = 0$, $a + b - 5 = -3$, $a + b = -2$, $a = 2$, $b = -4$]

RC 47 ROUND 1A national Q/4 or S/2

Speed race

Find the equation of the curve with gradient $3ax^2 + 6x$ and passing the points A(1, 10) and B(-1, 4)

ANSWER: $y = 3x^3 + 3x^2 + 4$ [$y = ax^3 + 3x^2 + C$, $10 = a + 3 + C$, $a + C = 7$, $-a + 3 + C = 4$, $-a + C = 1$, $2C = 8$, $C = 4$, $a = 7 - 4 = 3$, $y = 3x^3 + 3x^2 + 4$]

TRUE OR FALSE

1 If a number is divisible by 12, it is divisible by 18

ANSWER: FALSE

2 If a number is divisible by 9, the sum of its digits is divisible by 3.

ANSWER: TRUE

3 If a number is divisible by 4, the number formed by the last 2 digits is divisible by 4

ANSWER: TRUE

4 If a number is divisible by 6, then it is divisible by 3.xxxx

ANSWER: TRUE

5 If a number is divisible by 5, then its last digit is 5.xxxxx
ANSWER: FALSE